



ECOLE
POLYTECHNIQUE
DE BRUXELLES

03. Mathematical Foundations II

INFO-H502

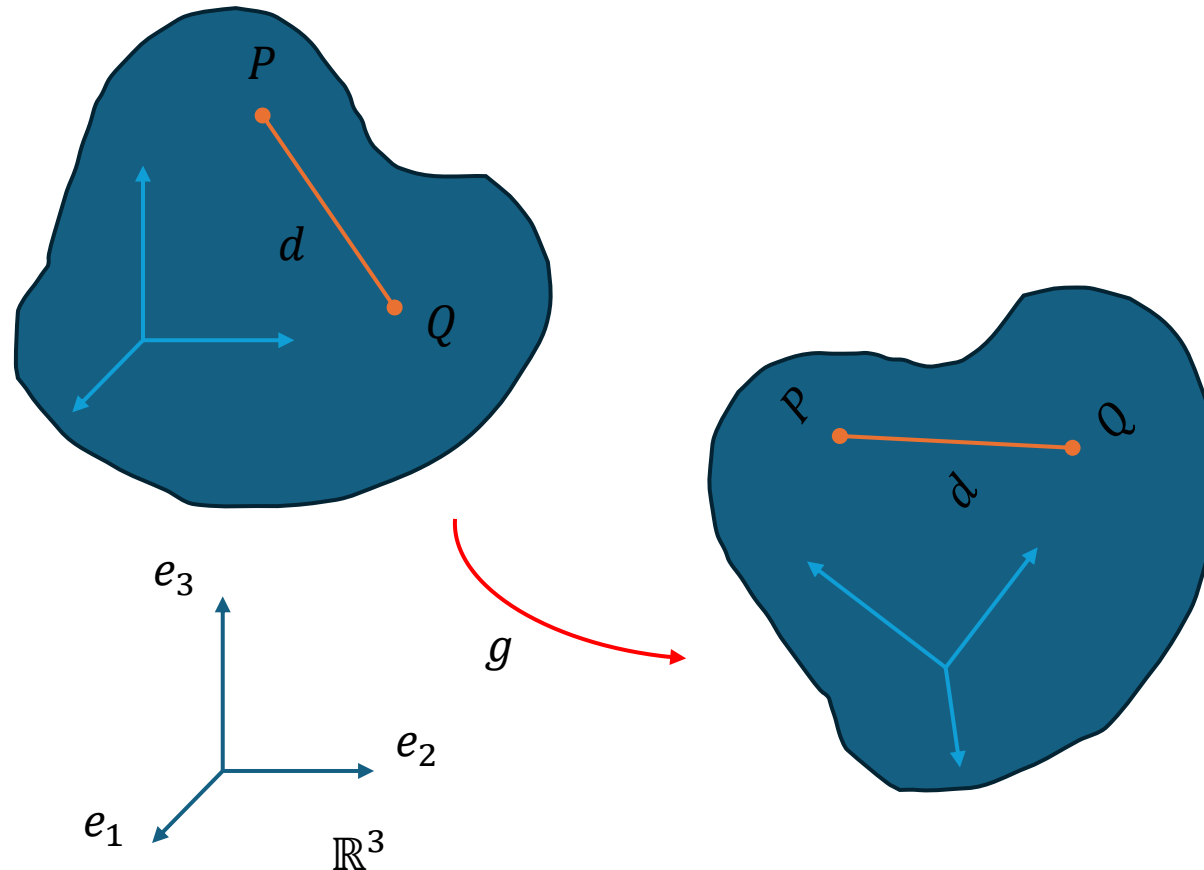
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LISA-VR

2024

Previously

- We constructed the main mathematical structures needed in computer graphics
- We defined rigid bodies
 - But they are not in matrix form.



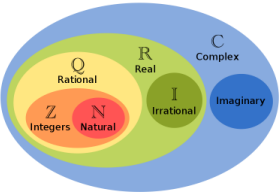
Summary

Add a law
 $+: G \times G \rightarrow G$

Add a law $\cdot: G \times G \rightarrow G$
 And $\cdot$ distribute $+$.

Objects = vectors
 Add 2 laws + inherit field ops
 $\oplus: V \times V \rightarrow V$
 $\odot: K \times V \rightarrow V$

Forget the origin $0 \in V$

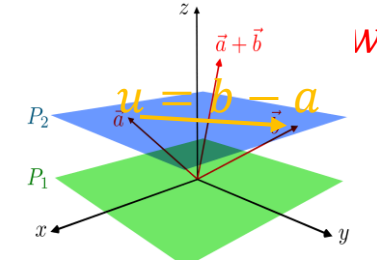
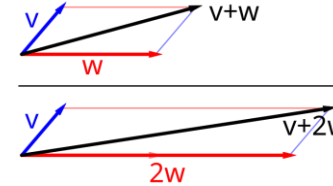


+	0	1	2	3
0	0	1	2	3
1	1	0	3	2
2	2	3	0	1
3	3	2	1	0

Addition Table

x	0	1	2	3
0	0	0	0	0
1	0	1	2	3
2	0	2	4	6
3	0	3	6	5

Multiplication Table



Sets

Groups $(G, \cdot)$

Field $(K, +, \cdot)$

Vector space $(V, \oplus, \odot)$

Affine spaces

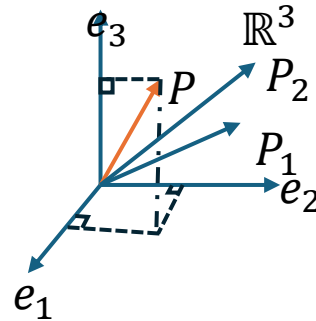
Elements = points

Add basis vectors
 Scalar product

Remove origin

Euclidean Space $\mathbb{E}^3$
 A set of points

3D Space $(\mathbb{R}^3, +, \cdot)$



Dot product
 cross-product

$d(v_1, v_2)$

$\mu(v)$

$proj_Q(v)$

lines

Ops: point-line, line-line

Planes

Ops: line-plane, normal

Rigid bodies $SE(3)$

Ops: (R, t)

Identify
 $\longleftrightarrow$

We have only topology

We cannot measure

We are stuck to origin

We have no matrices

- Our rigid transform is not in a matrix form $g = (R, T)$

Rigid-body motion $SE(3)$ – (R,T) to matrix

- *Definition (refined)*: Rigid-body motion, or Special Euclidean group $SE(3)$:

$$SE(3) = \{ g = (R, T) \mid R \in SO(3), T \in \mathbb{R}^3 \}$$

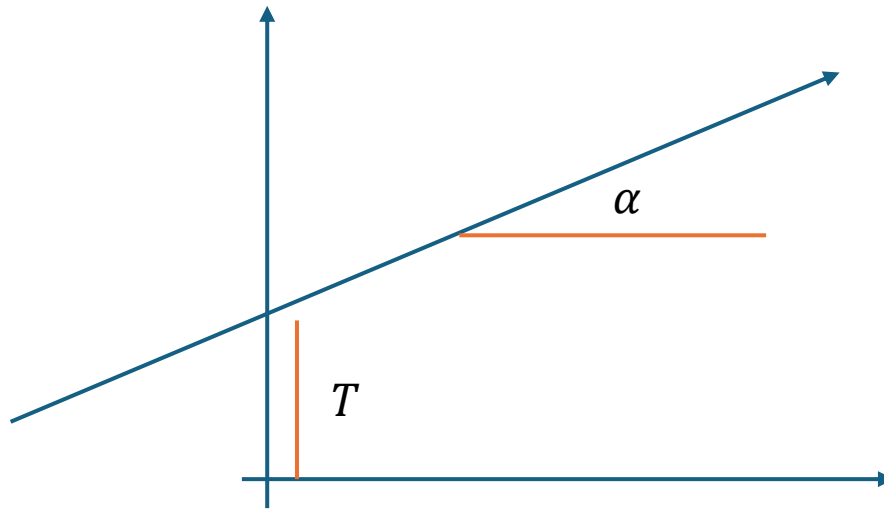
- However, $g = (R, T)$ is **not** a matrix, we want matrices in computer graphics.
- We introduce the real projective space and homogeneous coordinates for this purpose

Projective Geometry

- For rigid bodies, we have an affine transformation:

$$SE(3) = \{ g = (R, T) \mid R \in SO(3), T \in \mathbb{R}^3 \}$$

- If we transform a point p , we have $p' = Rp + T$
- This is an affine line



T is the offset of the line

- Affine transformations cannot be written as matrix products
- What could we do?

Why ?

Projective Geometry – $\mathbb{R}P^2$

- Let's focus on the operations in 2D

$$p' = Rp + T \Leftrightarrow \begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} t_1 \\ t_2 \end{pmatrix}$$

- We have:

$$\begin{cases} x' = ax + by + t_1 \\ y' = cx + dy + t_2 \end{cases}$$

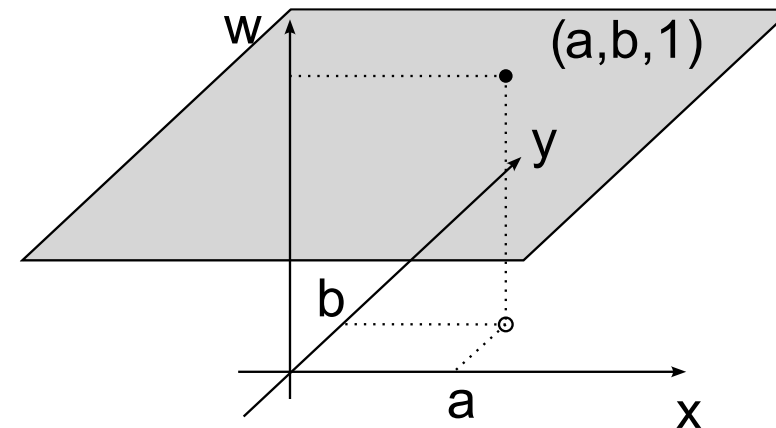
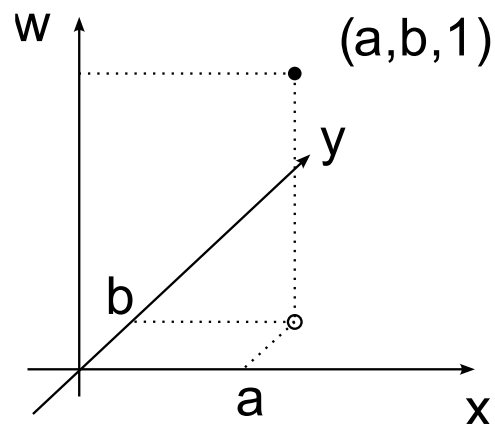
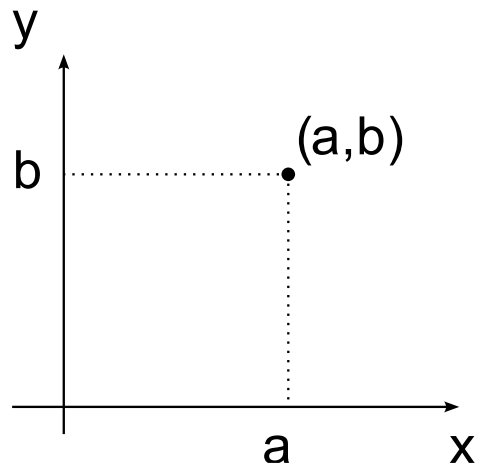
- We could extract a matrix:

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \overbrace{a \quad b}^{R+T} & t_1 \\ c & d & t_2 \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

- But we were forced to add a dimension to the point p
- What is the meaning of adding and removing new dimensions?

Projective Geometry

- What is the meaning of adding and removing new dimensions?
- We translated all our vector space to a plane $w = 1$

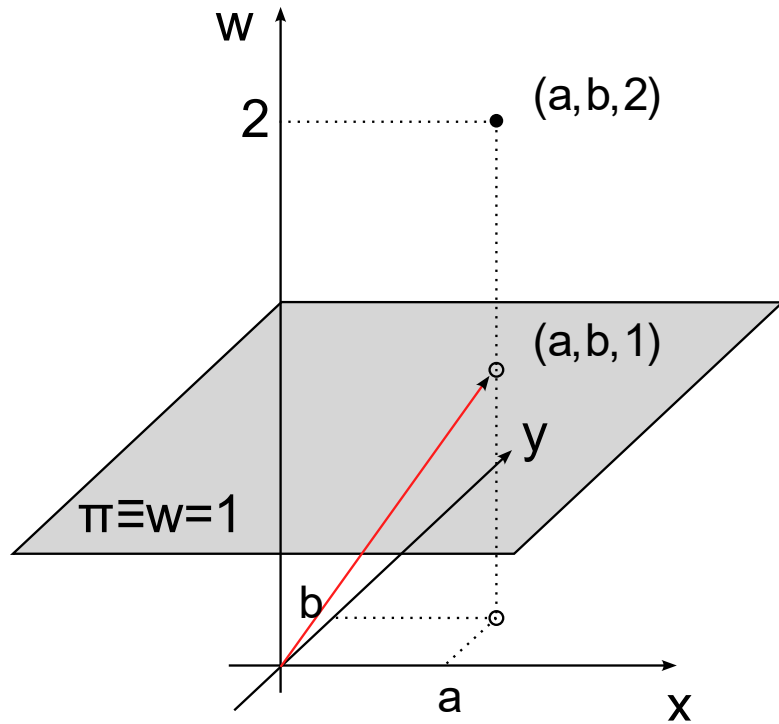


- Let's call this space the **projective space** $\mathbb{R}P^2$
- We can go back and forth between $\mathbb{R}^2$ and $\mathbb{R}P^2$ using
$$p: \mathbb{R}^2 \rightarrow \mathbb{R}P^2: (x, y) \mapsto (x, y, 1)$$
$$p^{-1}: \mathbb{R}P^2 \rightarrow \mathbb{R}^2: (x, y, 1) \mapsto (x, y)$$

But.. What happens if we change the w -coordinate to something else ?

Projective Geometry

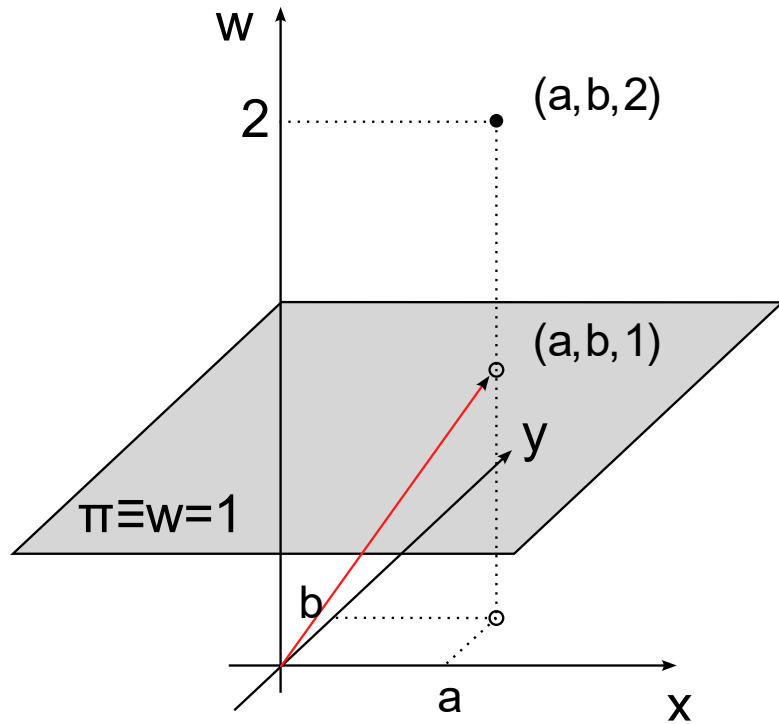
- But.. What happens if we change the w -coordinate to something else ?
- Say, $(x, y) \mapsto (x, y, 2)$



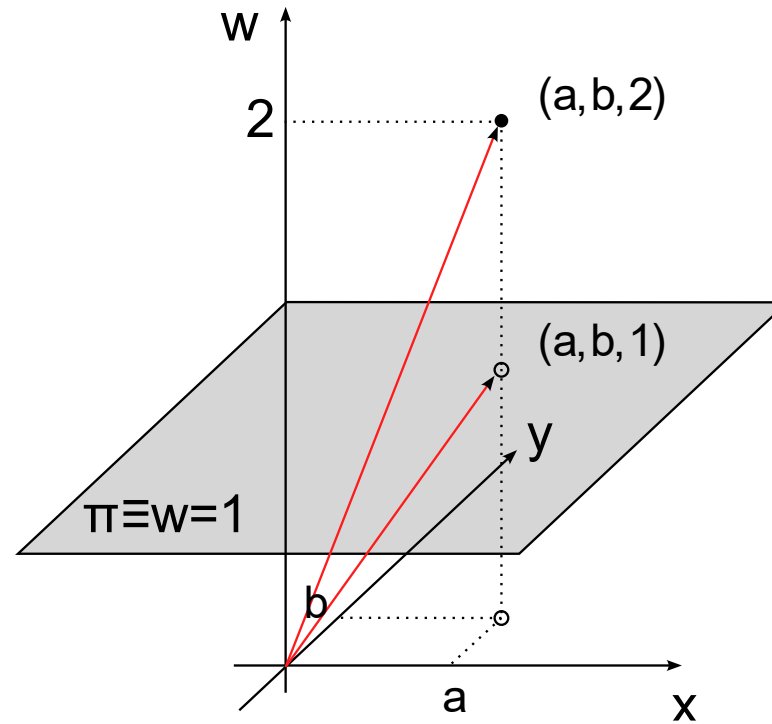
$$1) \mathbb{R}^2 \rightarrow \mathbb{RP}^2: (a, b) \mapsto (a, b, 2)$$

Projective Geometry

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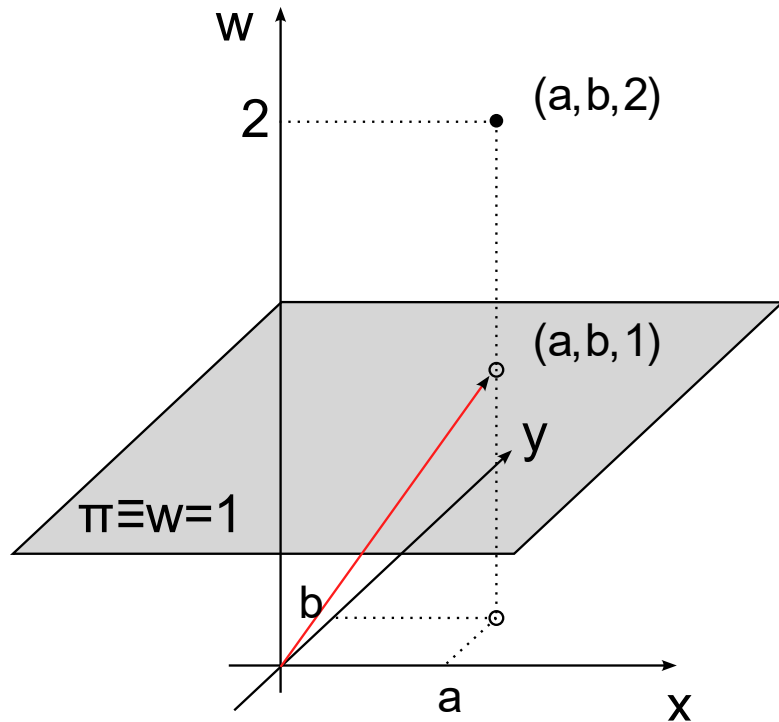
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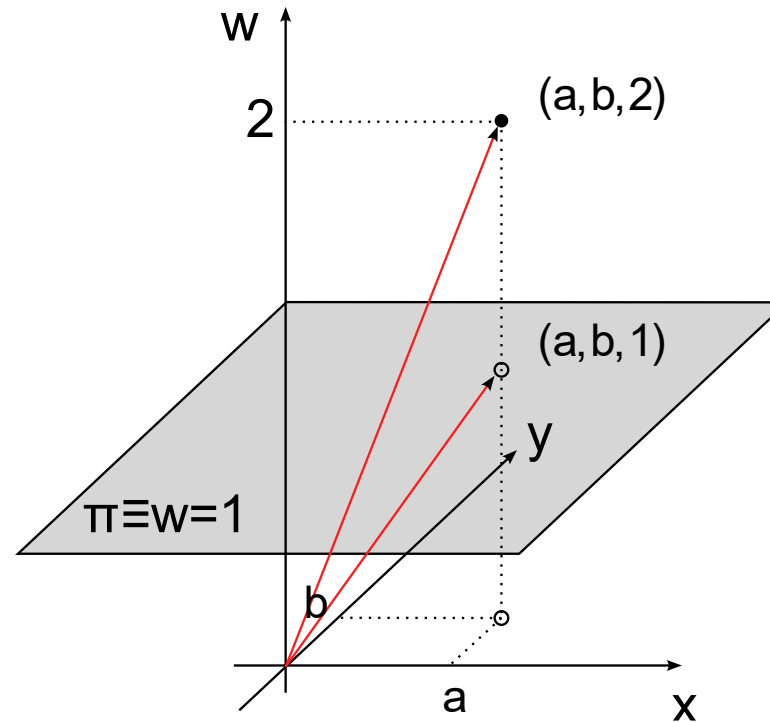
2) As a vector

Projective Geometry

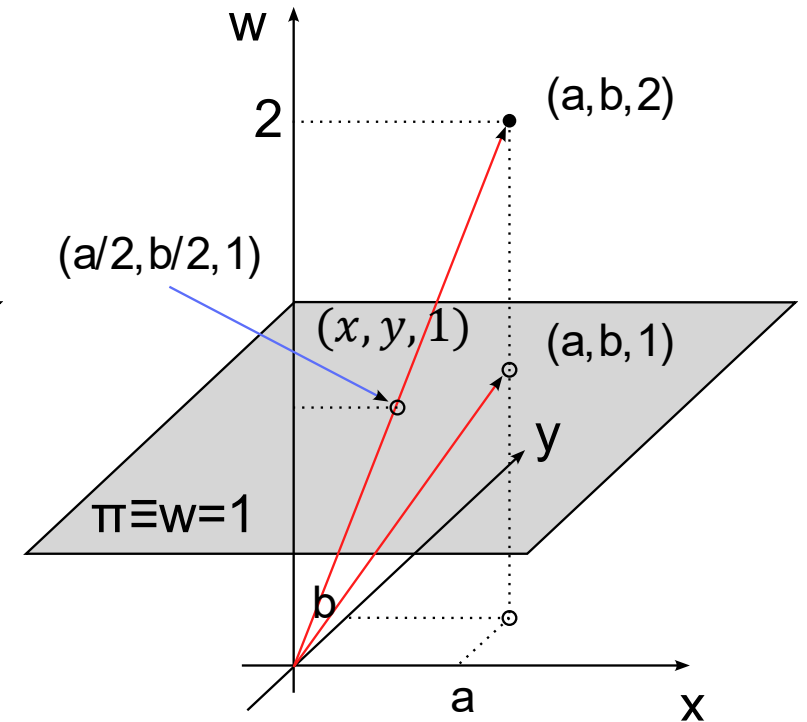
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1) $\mathbb{R}^2 \rightarrow \mathbb{RP}^2: (a, b) \mapsto (a, b, 2)$



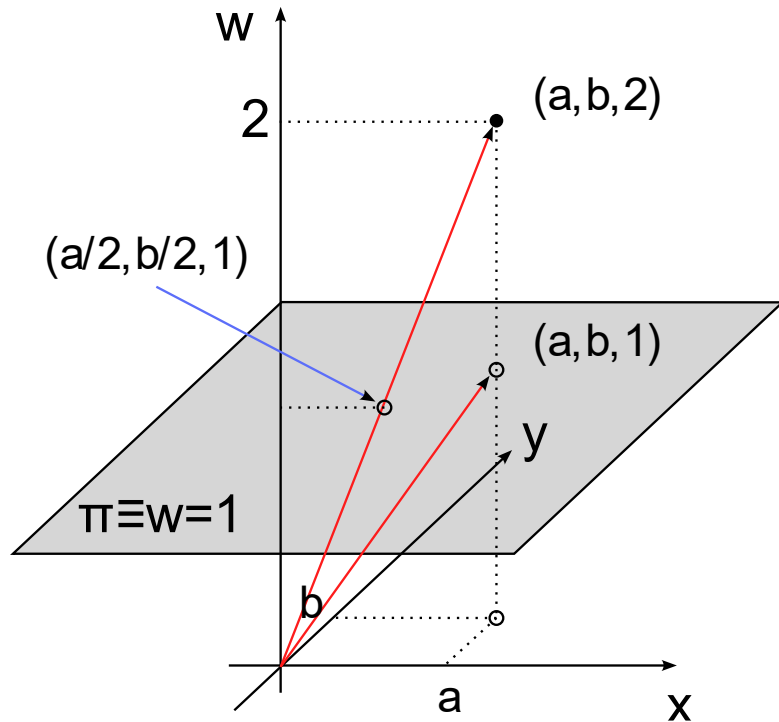
2) As a vector



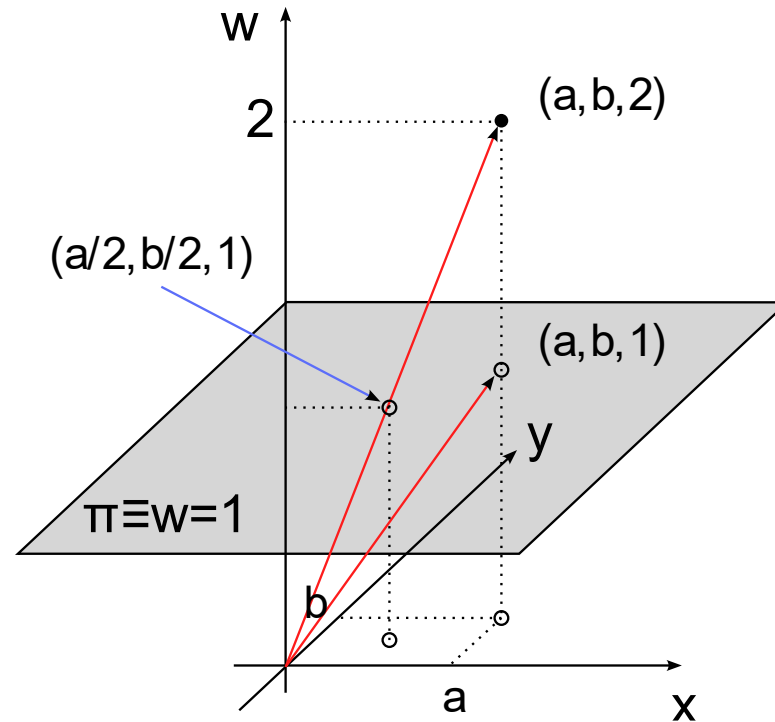
3) Find intersection with $w = 1$

Projective Geometry

- But.. What happens if we change the w -coordinate to something else ?
- Say, $(x, y) \mapsto (x, y, 2)$



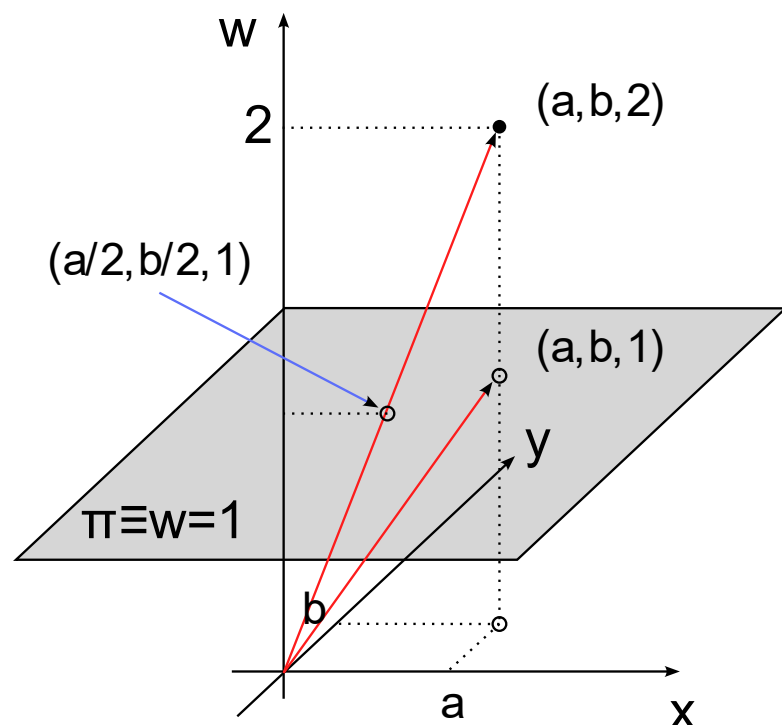
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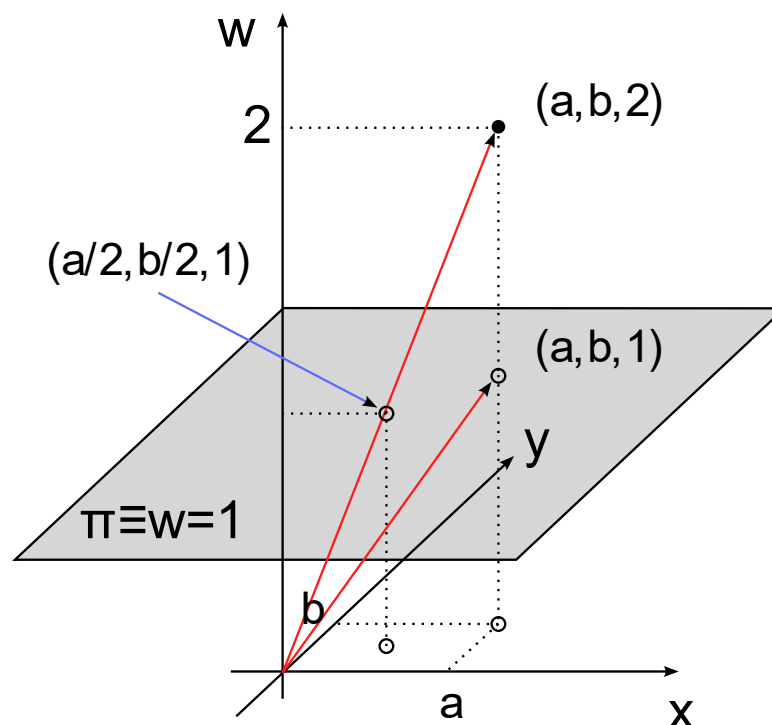
$$4) \mathbb{R}P^2 \rightarrow \mathbb{R}^2: \left(\frac{a}{2}, \frac{b}{2}, 1\right) \mapsto \left(\frac{a}{2}, \frac{b}{2}\right)$$

Projective Geometry

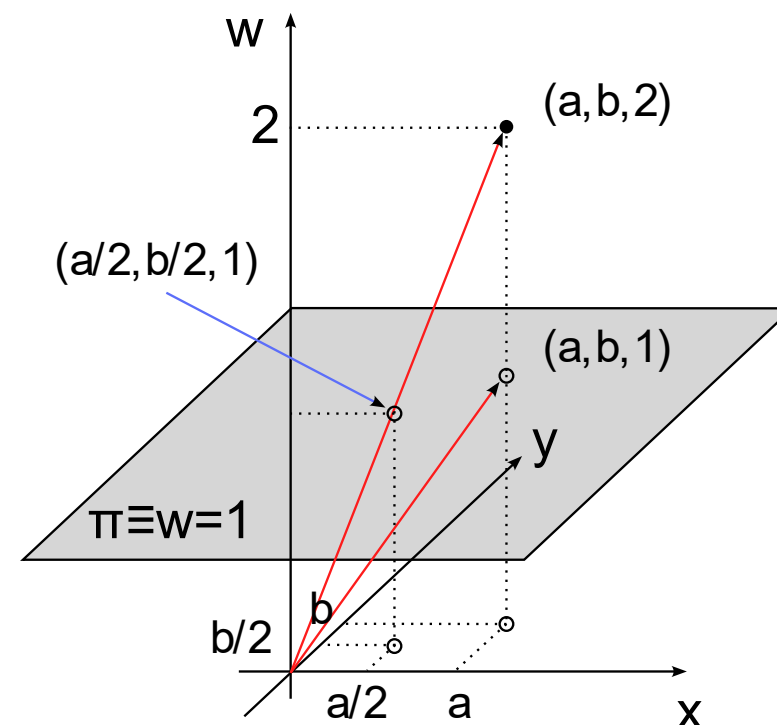
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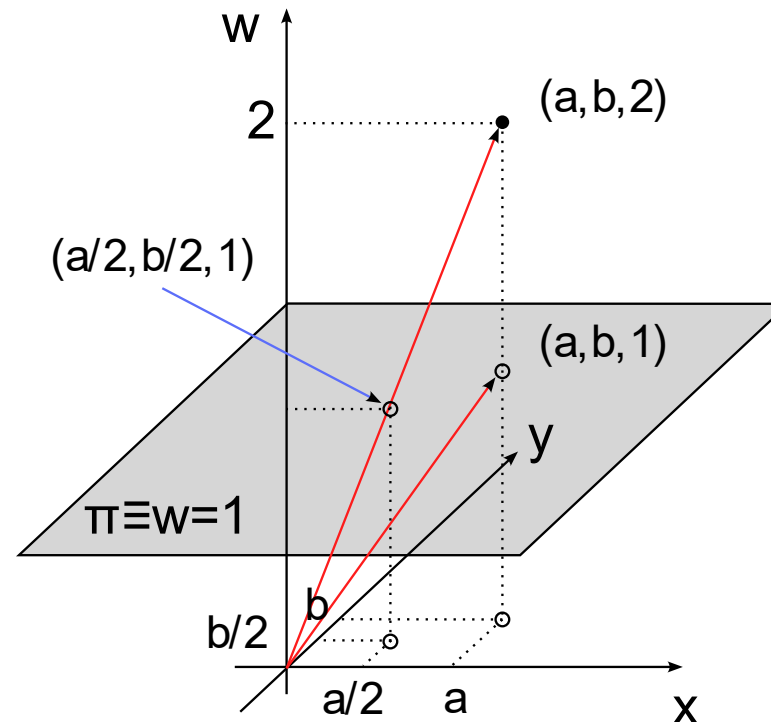
4) $\mathbb{R}P^2 \rightarrow \mathbb{R}^2: \left(\frac{a}{2}, \frac{b}{2}, 1\right) \mapsto \left(\frac{a}{2}, \frac{b}{2}\right)$



4) hum.. $\left(\frac{a}{2}, \frac{b}{2}\right) \neq (a, b)$

Projective Geometry

- But.. What happens if we change the w -coordinate to something else ?
- Say, $(x, y) \mapsto (x, y, 2)$
- We have a problem.. It does not represent the same point..



- We would like to change the last coordinate, but without changing the point

Projective Geometry

- Notice that $(a, b, 2)$ was transformed into $(\frac{a}{2}, \frac{b}{2}, 1)$
- What if we pre-multiply by the w -coord. the point in $\mathbb{R}^2$?

$$p: \mathbb{R}^2 \rightarrow \mathbb{RP}^2: (x, y) \mapsto (\textcolor{red}{k}x, \textcolor{red}{k}y, \textcolor{red}{k})$$

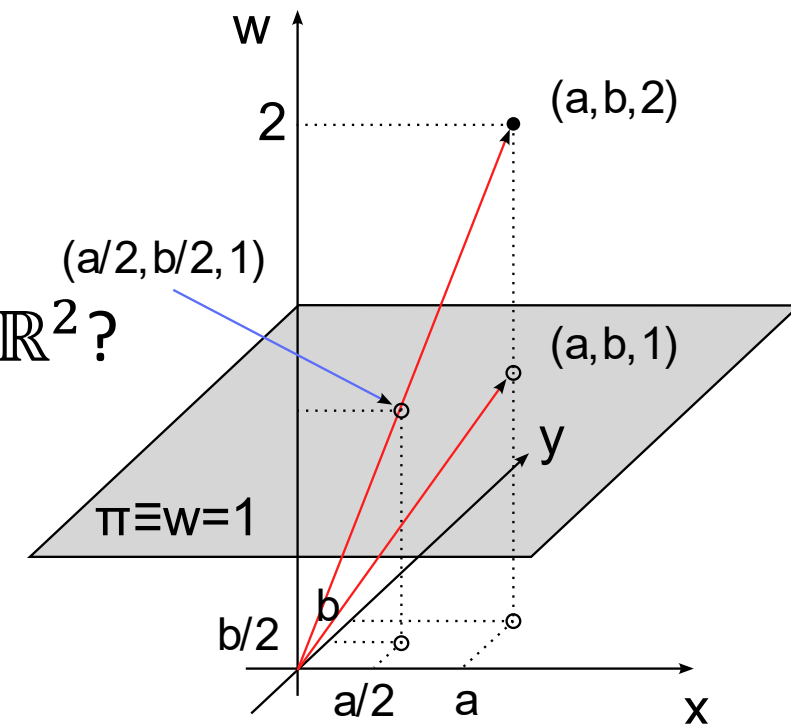
- Then when we find the projection on $w = 1$, we have:

$$\bullet \left(\frac{\textcolor{red}{k}x}{\textcolor{red}{k}}, \frac{\textcolor{red}{k}y}{\textcolor{red}{k}}, \frac{\textcolor{red}{k}}{\textcolor{red}{k}} \right) = (x, y, 1)$$

- Way better!
- Let's redefine our mappings:

$$p: \mathbb{R}^2 \rightarrow \mathbb{RP}^2: (x, y) \mapsto (kx, ky, k)$$

$$p^{-1}: \mathbb{RP}^2 \rightarrow \mathbb{R}^2: (kx, ky, k) \mapsto \left(\frac{kx}{k}, \frac{ky}{k} \right) = (x, y)$$

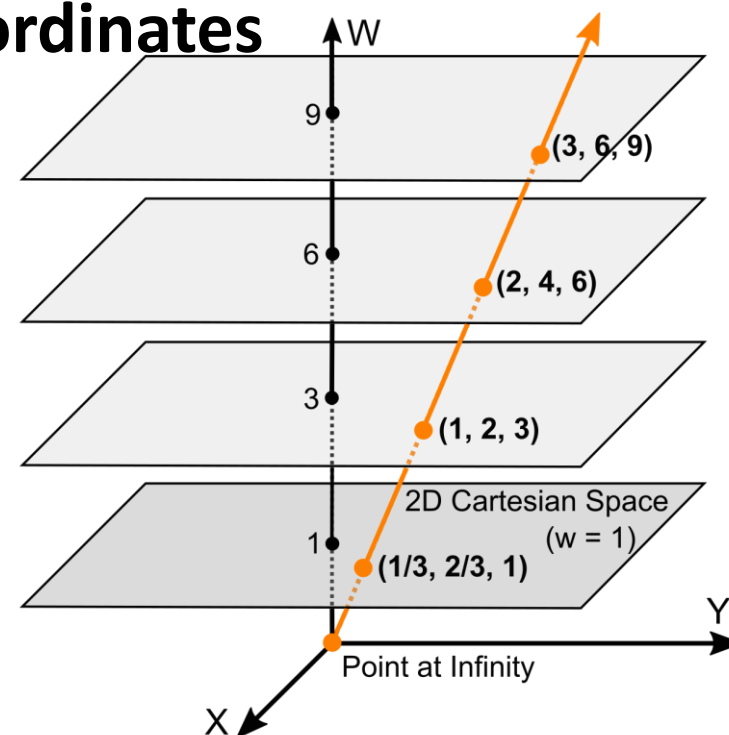
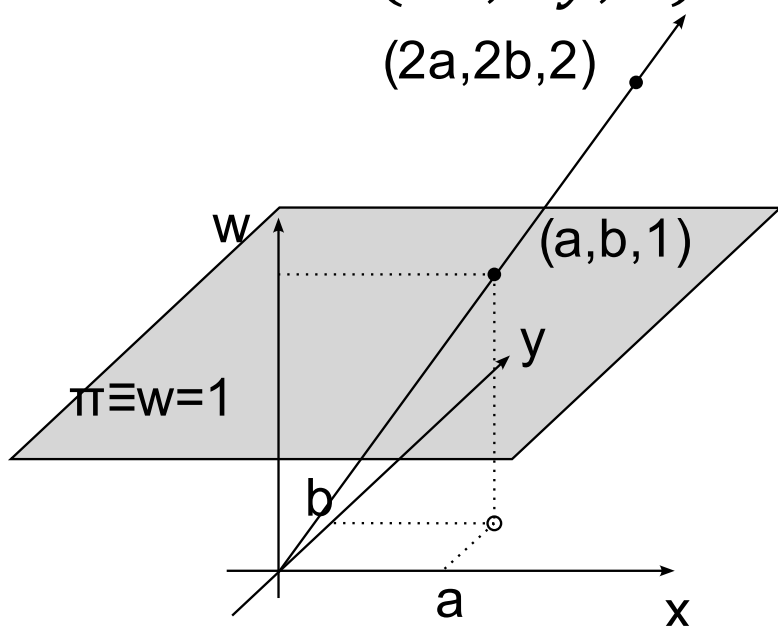


Projective Geometry

$$p: \mathbb{R}^2 \rightarrow \mathbb{RP}^2: (x, y) \mapsto (kx, ky, k)$$

$$p^{-1}: \mathbb{RP}^2 \rightarrow \mathbb{R}^2: (kx, ky, k) \mapsto \left(\frac{kx}{k}, \frac{ky}{k}\right) = (x, y)$$

- This is valid for any $k \neq 0$
- And for any k , the point (kx, ky, k) is the same as the point $(x, y, 1)$ by p^{-1}
- Points such as (kx, ky, k) are in **homogeneous coordinates**



Going back to the rigid transformations

- Now we have $\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} a & b & t_1 \\ c & d & t_2 \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$
- And we know that the point $(x, y, 1)^T$ is in homogeneous coordinates
- But $(x', y')^T$ is not !
- What happens if we change the w -coordinate?

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} a & b & t_1 \\ c & d & t_2 \end{pmatrix} \begin{pmatrix} kx \\ ky \\ k \end{pmatrix} = \begin{cases} (ka)x + (kb)y + kt_1 \\ (kc)x + (kd)y + kt_2 \end{cases}$$

- We have

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{cases} kax + kby + kt_1 \\ kcx + kdy + kt_2 \end{cases}$$

- This is not right at all ! We do not want this k in the new position
- what to do?

Going back to the rigid transformations

- This is not right at all ! We do not want this k in the new position
- We need to stay in homogeneous coordinates
 - I.e., to keep track of the k , let's save it !

$$\begin{pmatrix} kx' \\ ky' \\ k \end{pmatrix} = \begin{pmatrix} a & b & t_1 \\ c & d & t_2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} kx \\ ky \\ k \end{pmatrix} = \begin{cases} (ka)x + (kb)y + kt_1 \\ (kc)x + (kd)y + kt_2 \\ k \end{cases}$$

- Now, when we use p^{-1} to $(x', y', k)^T$ we correctly have:

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{cases} \frac{kax + kby + kt_1}{k} \\ \frac{kcx + kdy + kt_2}{k} \end{cases} = \begin{cases} ax + by + t_1 \\ cx + dy + t_2 \end{cases}$$

Correct transformation!
 $p' = Rp + T$

Going back to the rigid transformations

- So, if our rigid transformation is

$$SE(3) = \{ g = (R, T) \mid R \in SO(3), T \in \mathbb{R}^3 \}$$

- We can express it in homogeneous coordinates as

$$G = \begin{pmatrix} R & T \\ 0 & 1 \end{pmatrix}$$

- And our equation becomes

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{pmatrix} R & T \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

Our new *definition*:

$$SE(3) = \left\{ g = \begin{pmatrix} R & T \\ 0 & 1 \end{pmatrix} \mid R \in SO(3), T \in \mathbb{R}^3 \right\} \subset \mathbb{R}^{4 \times 4}$$

Rigid transforms – Homogeneous coordinates

$$SE(3) = \left\{ g = \begin{pmatrix} R & T \\ 0 & 1 \end{pmatrix} \middle| R \in SO(3), T \in \mathbb{R}^3 \right\} \subset \mathbb{R}^{4 \times 4}$$

- Note:

- $g_1 g_2 = \begin{pmatrix} R_1 & T_1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} R_2 & T_2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} R_1 R_2 & R_1 T_2 + T_1 \\ 0 & 1 \end{pmatrix} \in SE(3)$

- So, composition works correctly, check it !

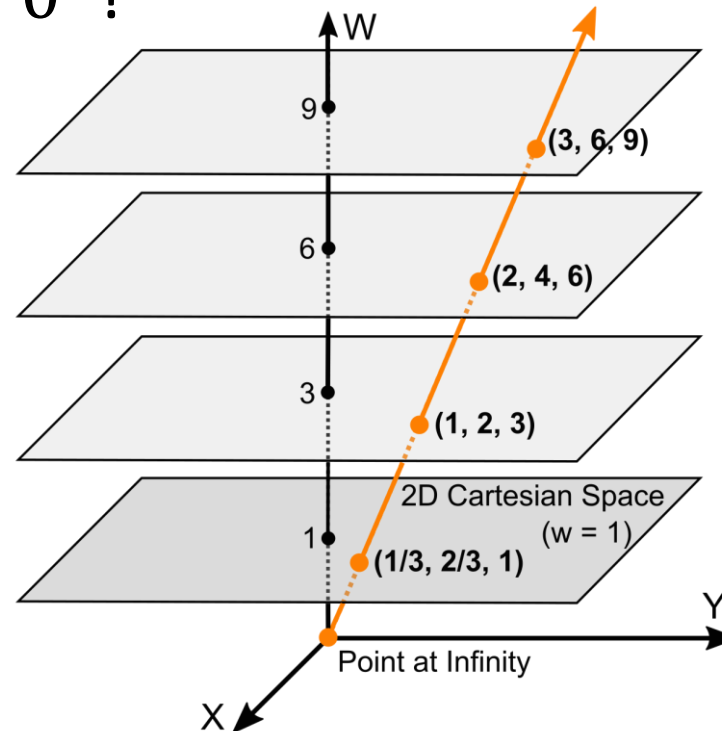
- $g^{-1} = \begin{pmatrix} R & T \\ 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} R^T & -R^T T \\ 0 & 1 \end{pmatrix} \in SE(3)$

- So, this is the inverse transformation of a rigid movement, check it !

Projective Geometry - Interpretations

- Why is it called projective space ?
- What does that $k = 1$ that we added mean ?
- What happens for objects in $0 < k < 1$?
- What happens for objects with negative k ?
- What happens when $k = 0$?

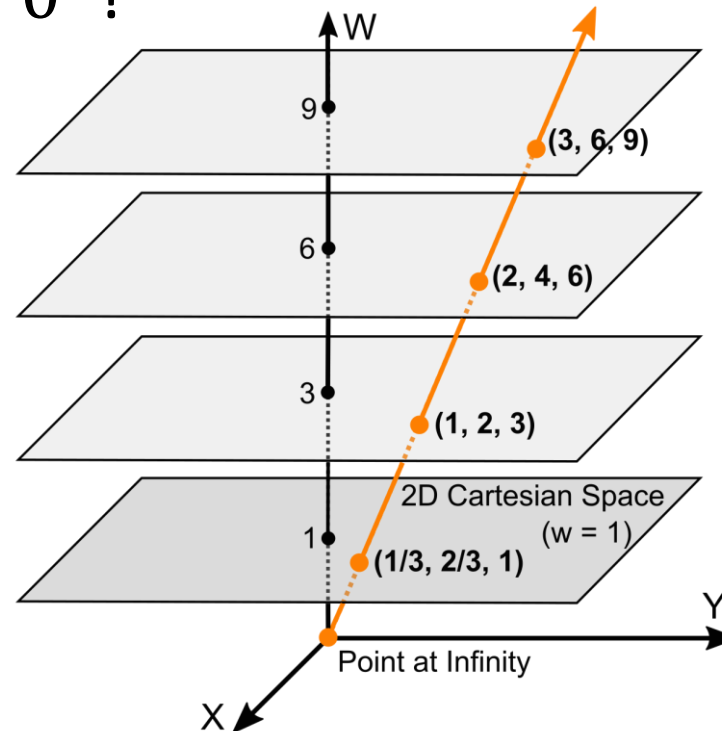
(elements such as $(1,2,0)$)



Projective Geometry - Interpretations

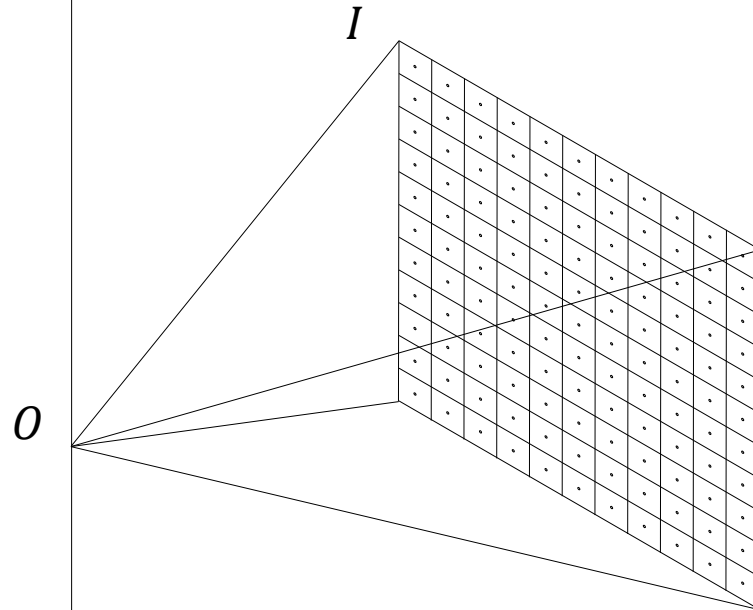
- Why is it called **projective space** ?
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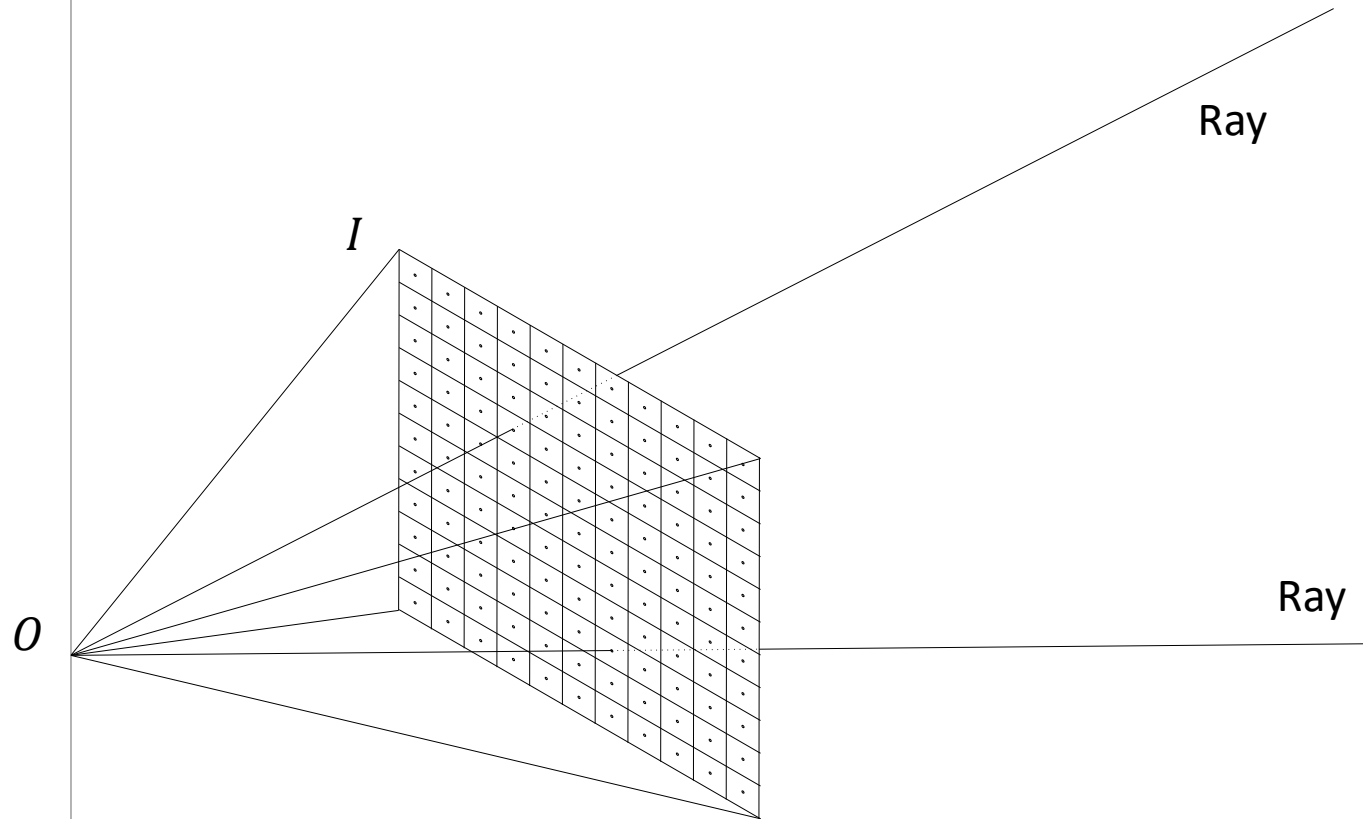
The Pinhole Camera Model

- A camera has a optical center O and an image plane I



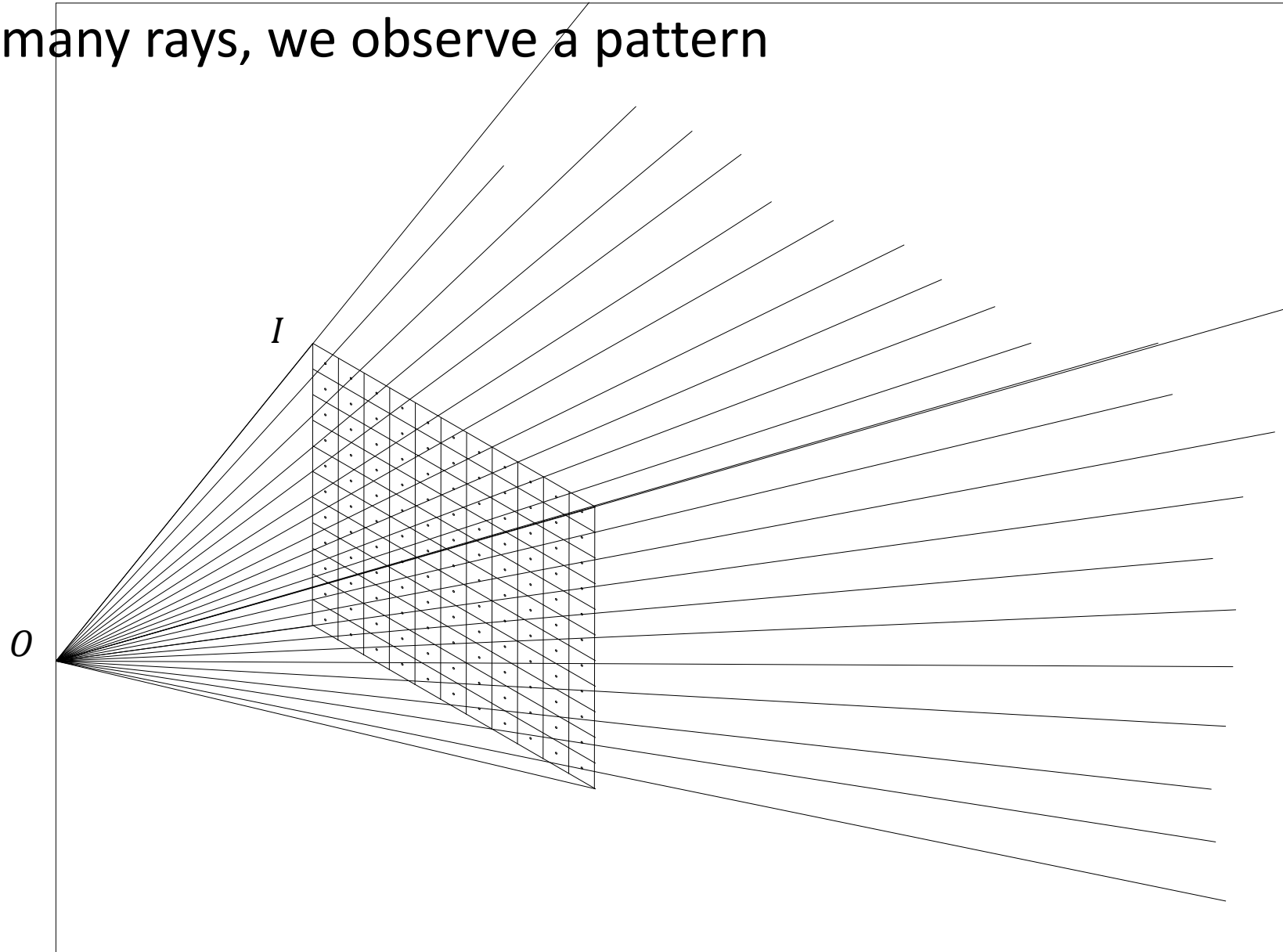
The Pinhole Camera Model

- Rays from the scene are projected into the image plane



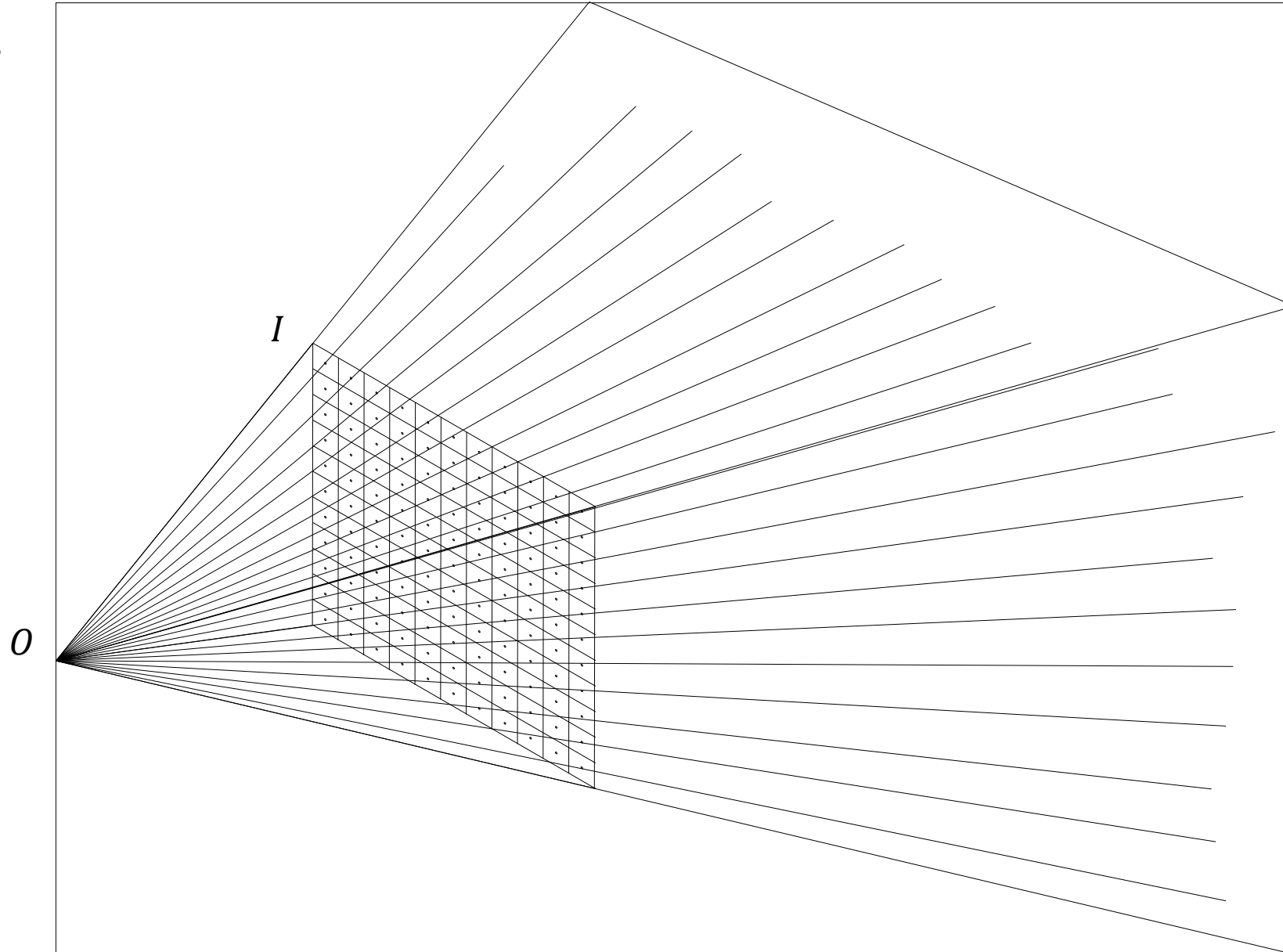
The Pinhole Camera Model

- If we show many rays, we observe a pattern



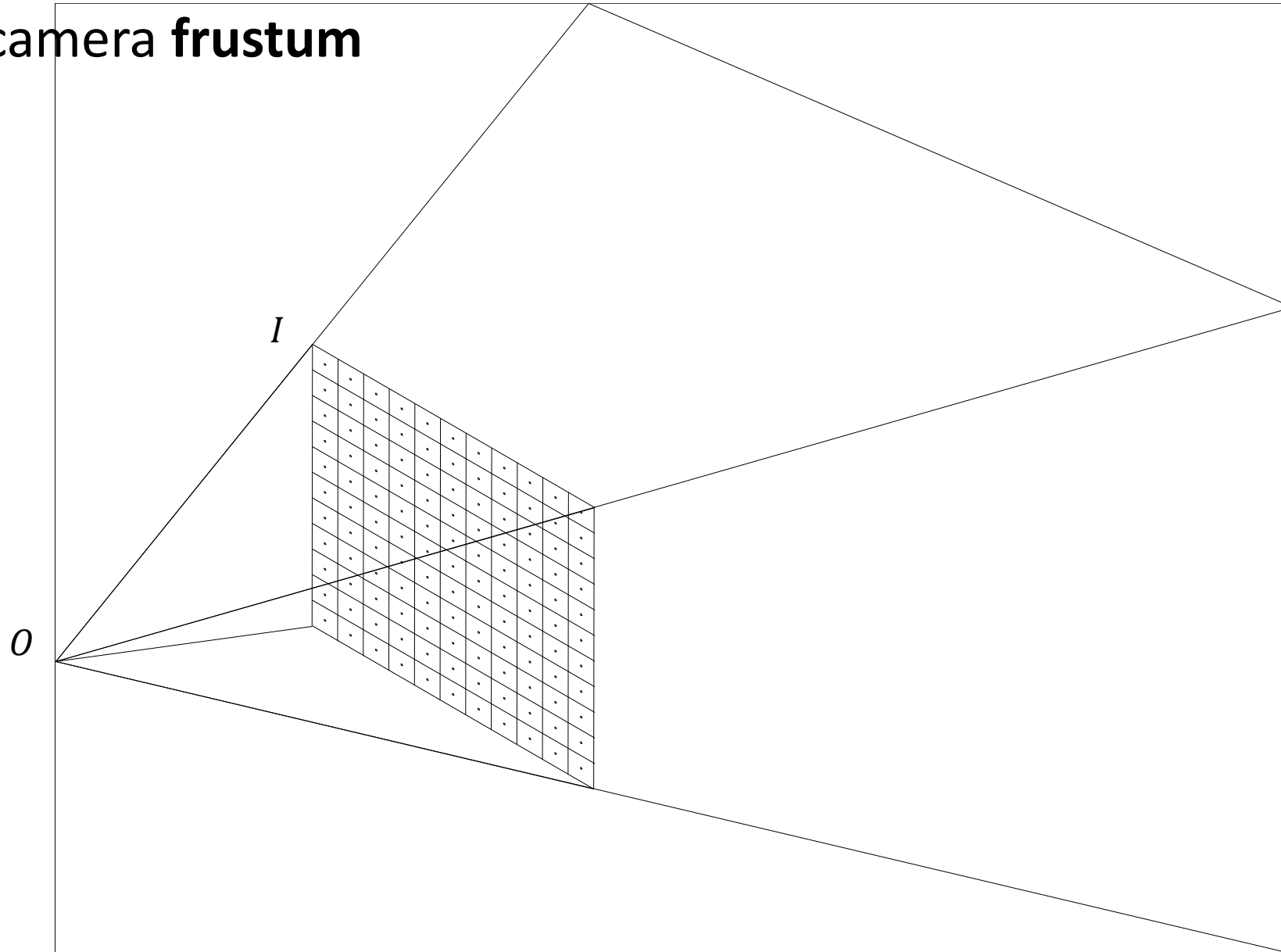
The Pinhole Camera Model

- A pyramid !



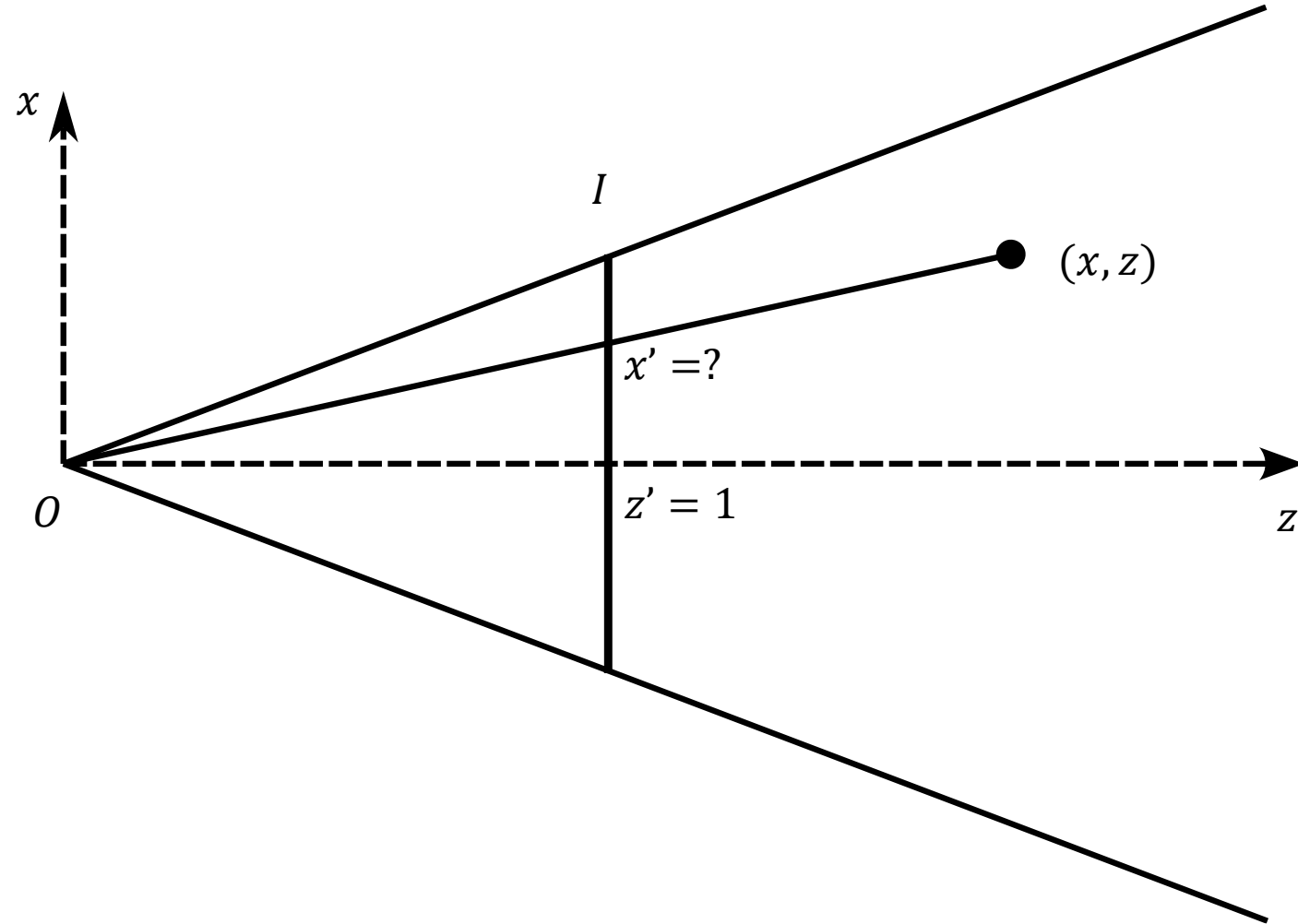
The Pinhole Camera Model

- This is the camera **frustum**



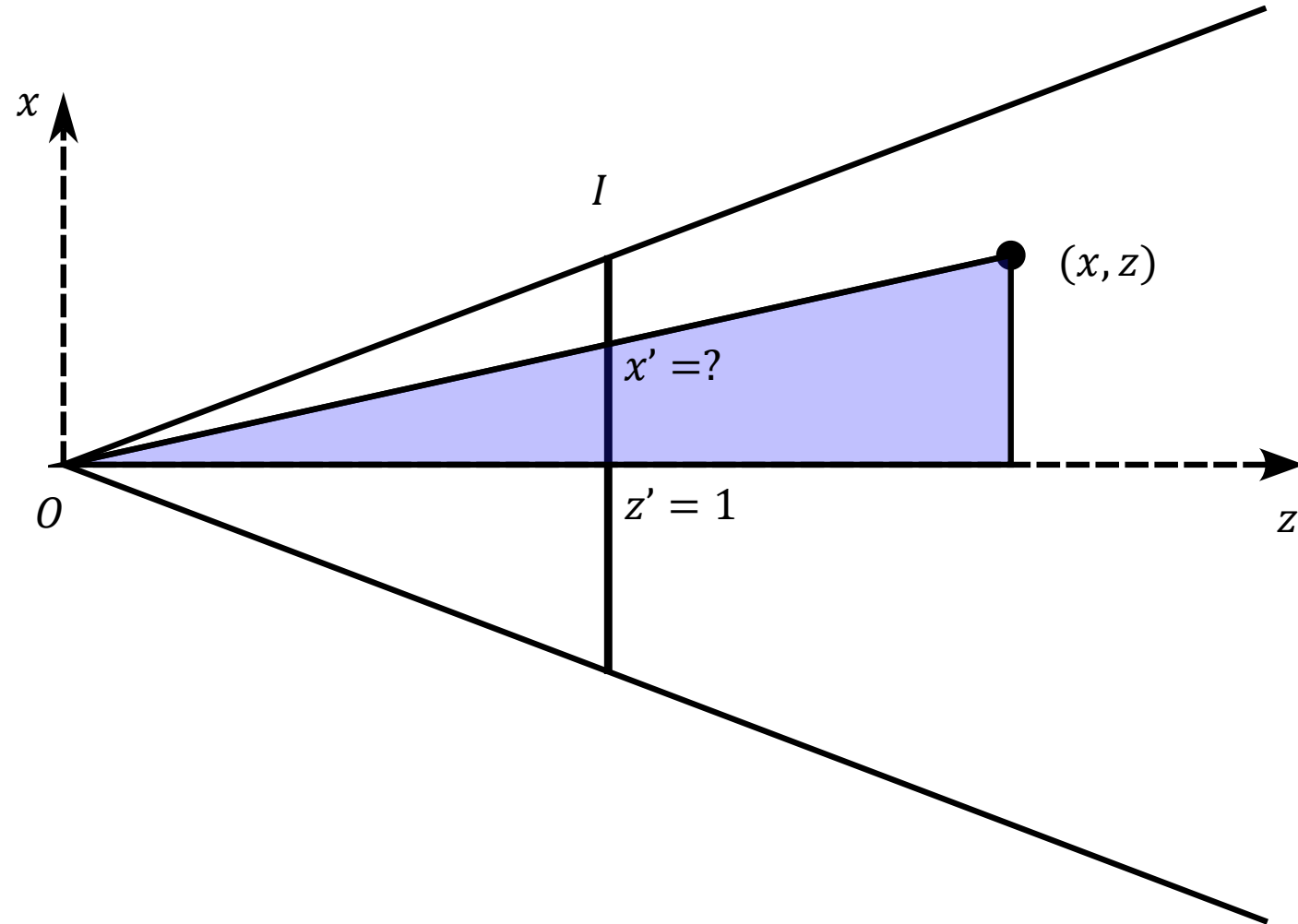
The Pinhole Camera Model

- In 2D, objects are *projected* to the image plane (at distance $z' = 1$)
- What's x' ?



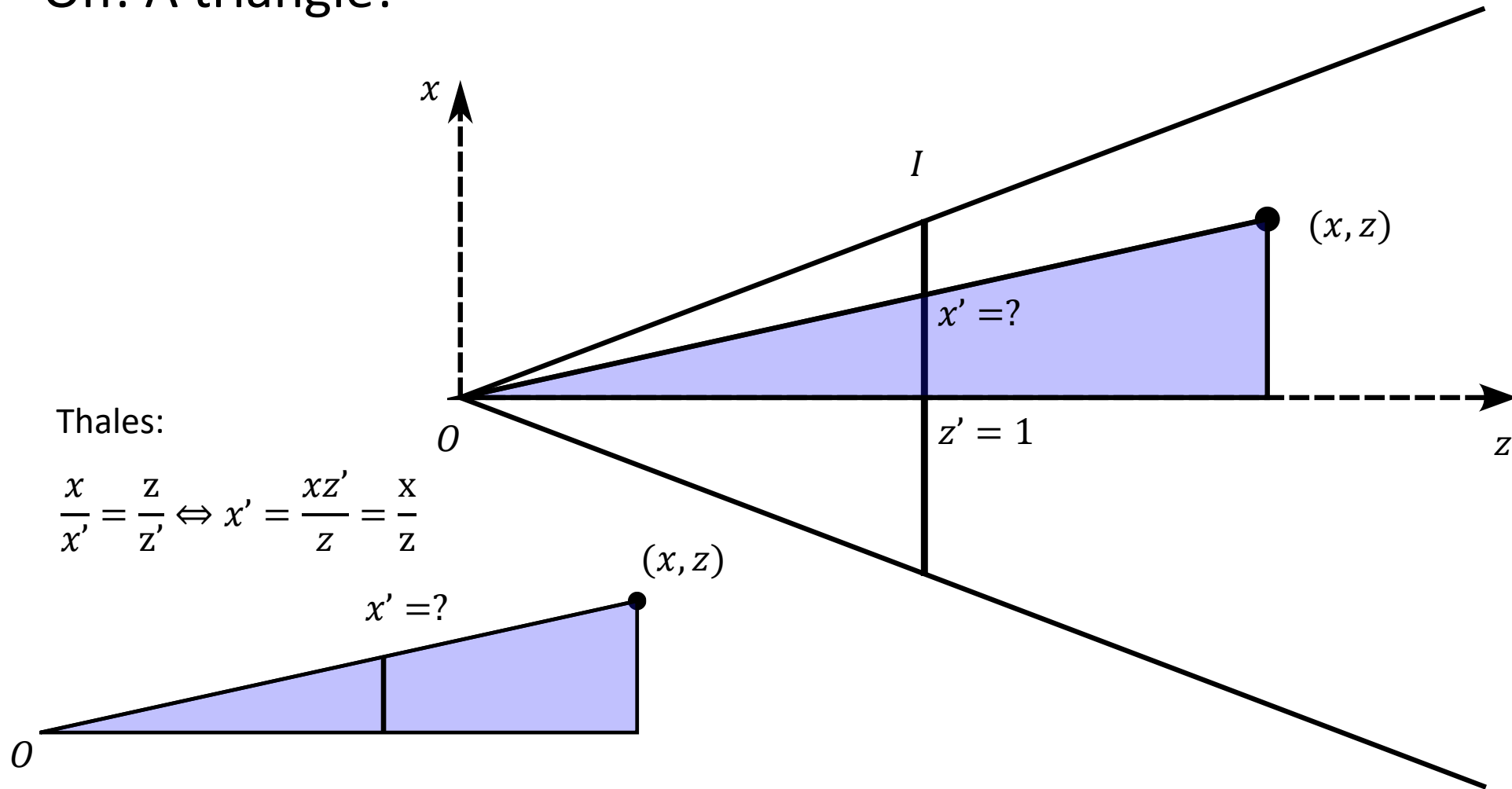
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- In 2D, objects are *projected* to the image plane (at distance $z' = 1$)
- Oh! A triangle!



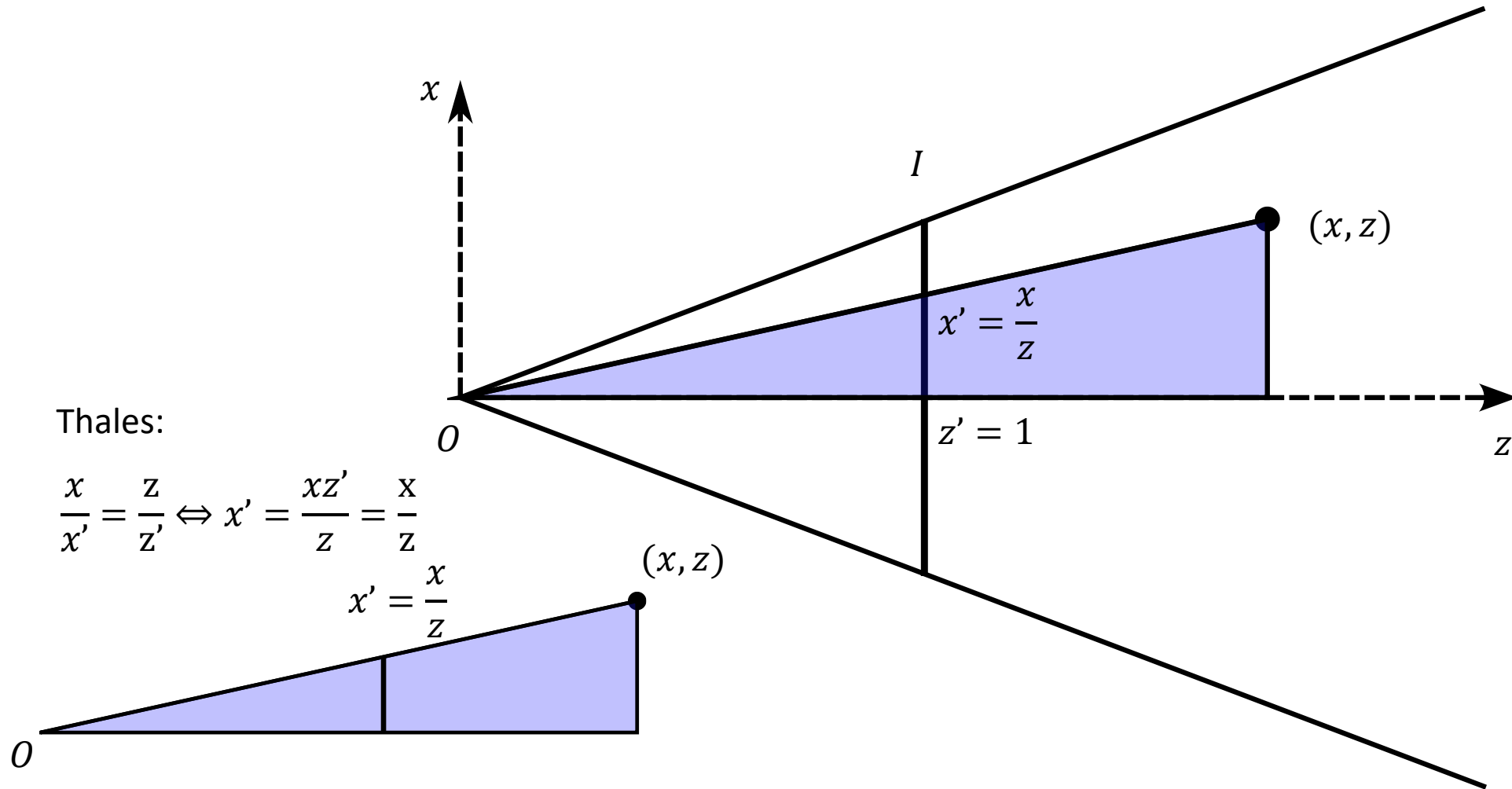
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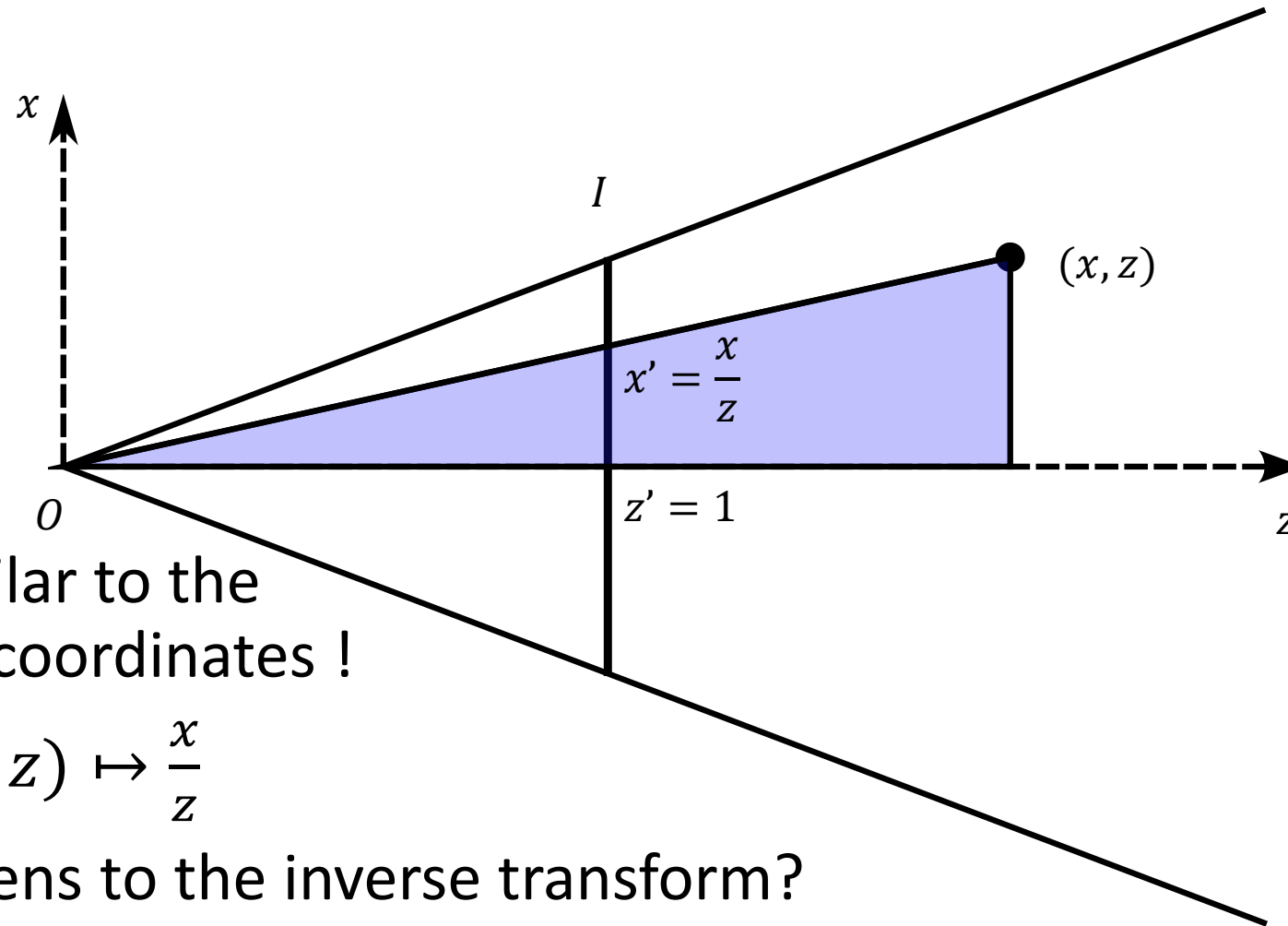
The Pinhole Camera Model

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The Pinhole Camera Model

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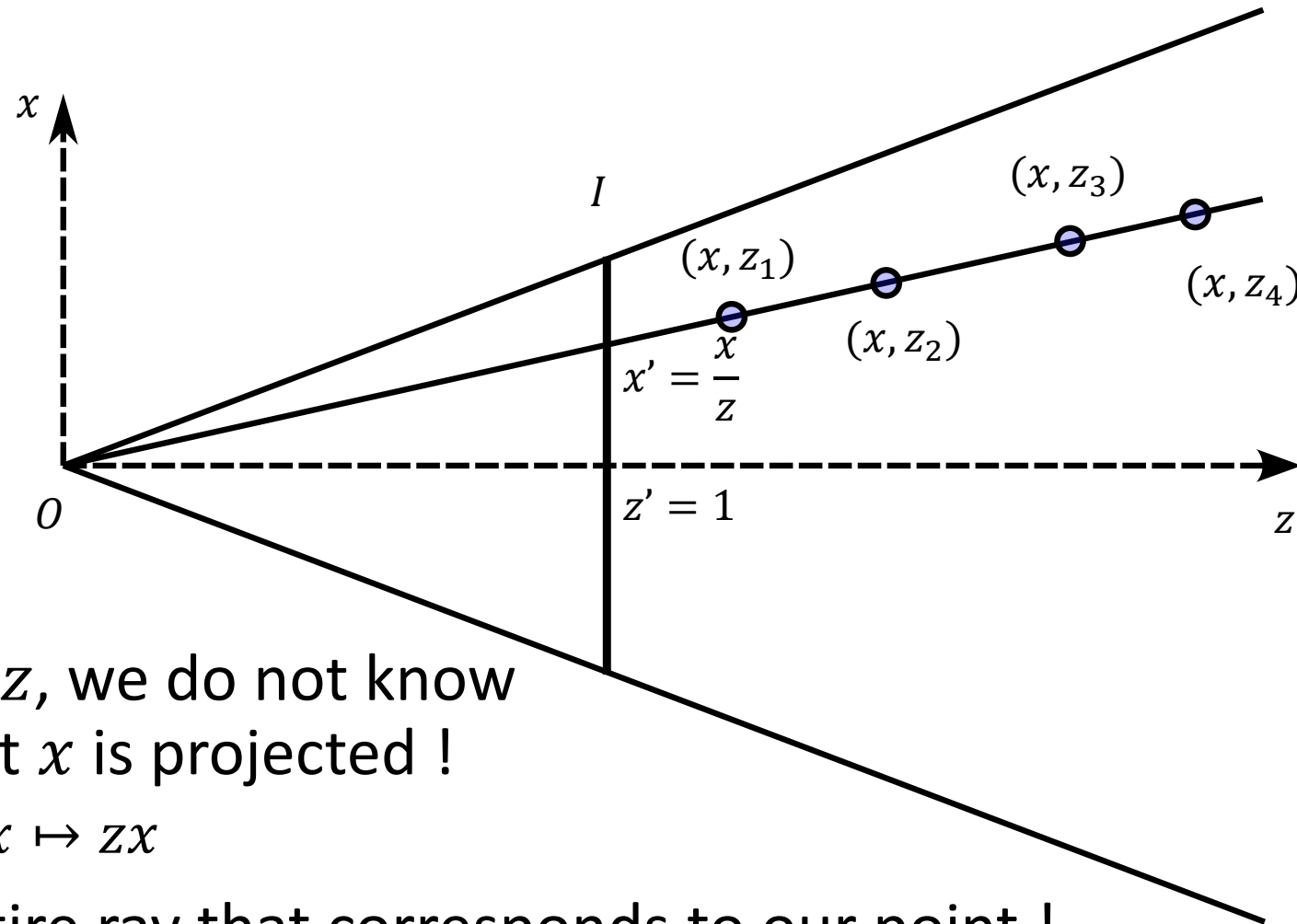
- That's very similar to the homogeneous coordinates !

- $p: \mathbb{RP} \rightarrow \mathbb{R}: (x, z) \mapsto \frac{x}{z}$

But.. What happens to the inverse transform?

The Pinhole Camera Model

- In 2D, objects are *projected* to the image plane (at distance $z' = 1$)



- If we loose the z , we do not know where the point x is projected !
 - $p^{-1}: \mathbb{R} \rightarrow \mathbb{RP}: x \mapsto zx$
- We have an entire ray that corresponds to our point !

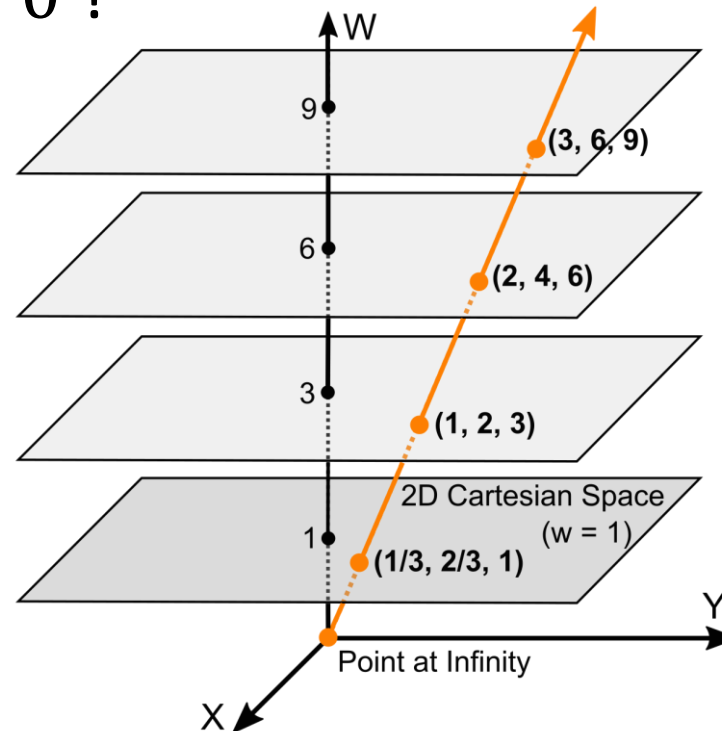
The Pinhole Camera Model

- In 2D, objects are *projected* to the image plane (at distance $z' = 1$)
- If we loose the z , we do not know where the point x is projected !
 - $p^{-1}: \mathbb{R} \rightarrow \mathbb{R}P: x \mapsto zx$
- We have an entire ray that corresponds to our point !
- Projections do **not** preserve the depth of the points !
- As previously, we need to keep the *depth* $k(z)$ in our matrices to be able to reconstruct the values

Projective Geometry - Interpretations

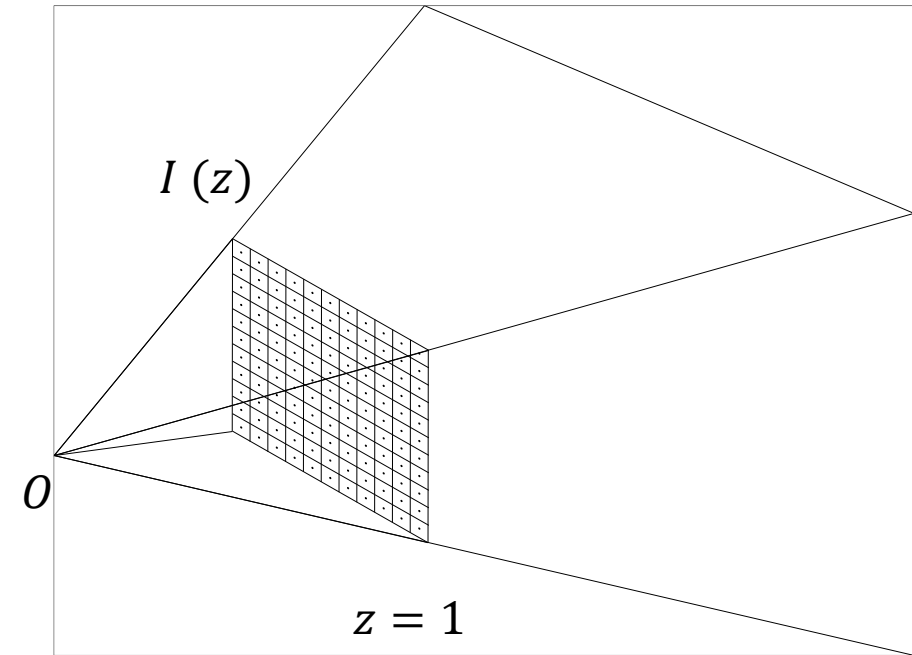
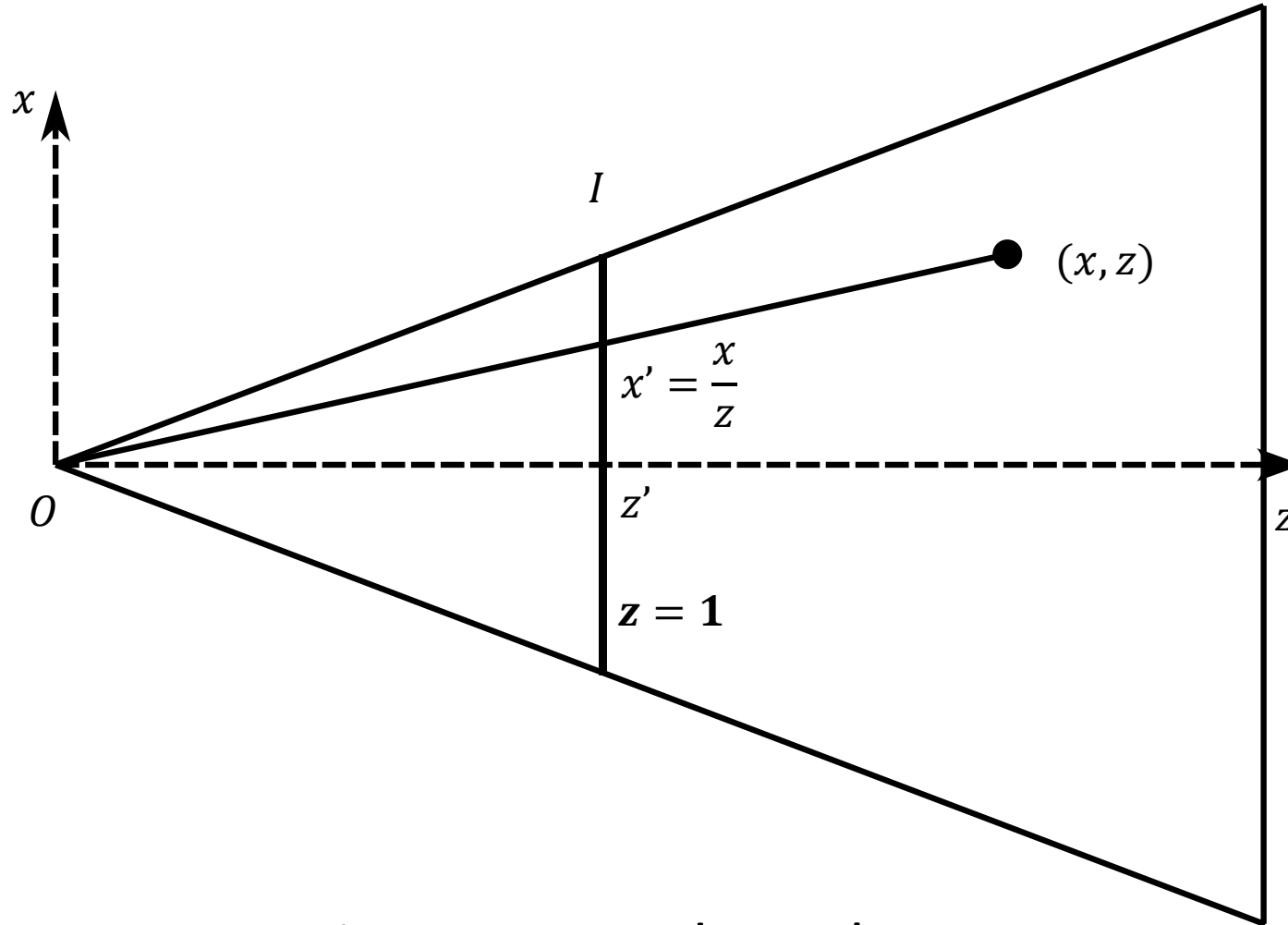
- Why is it called projective space ?
- **What does that $k = 1$ that we added mean ?**
- What happens for objects in $0 < k < 1$?
- What happens for objects with negative k ?
- What happens when $k = 0$?

(elements such as $(1,2,0)$)



What does that 1 that we added mean ?

- We setted our image plane to be at $z = 1$

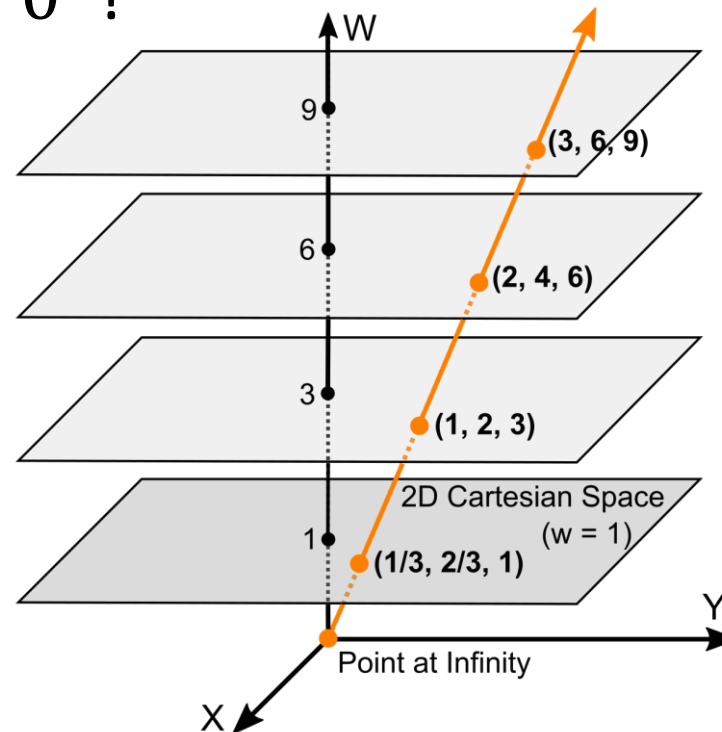


- In a camera it corresponds to the Image sensor

Projective Geometry - Interpretations

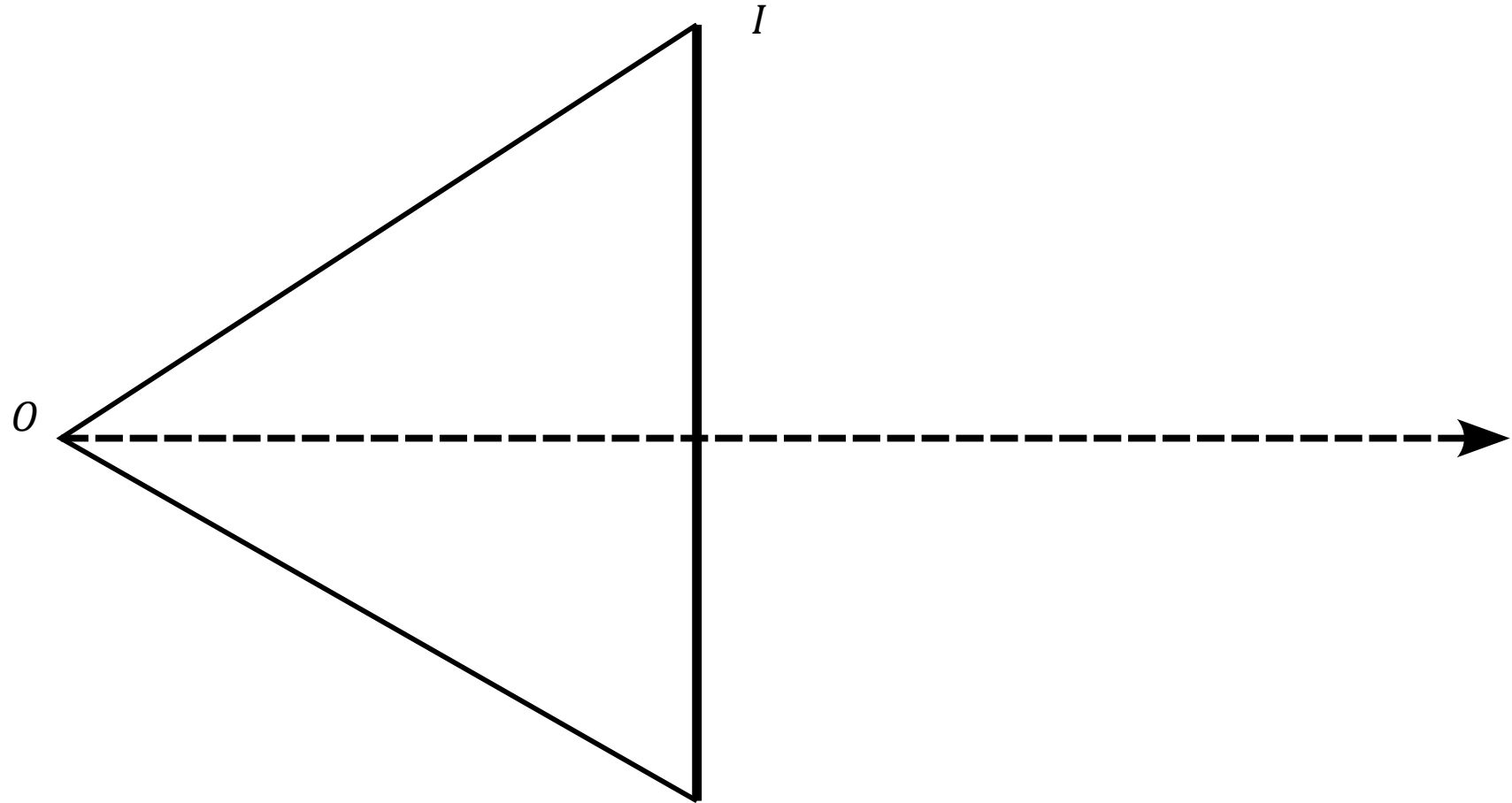
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- What does that $k = 1$ that we added mean ?
- **What happens for objects in $0 < k < 1$?**
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- What happens when $k = 0$?

(elements such as $(1,2,0)$)



What happens for objects with different k ?

- Suppose a camera with optical center O ($z = 0$) and image plane I ($z = 1$)



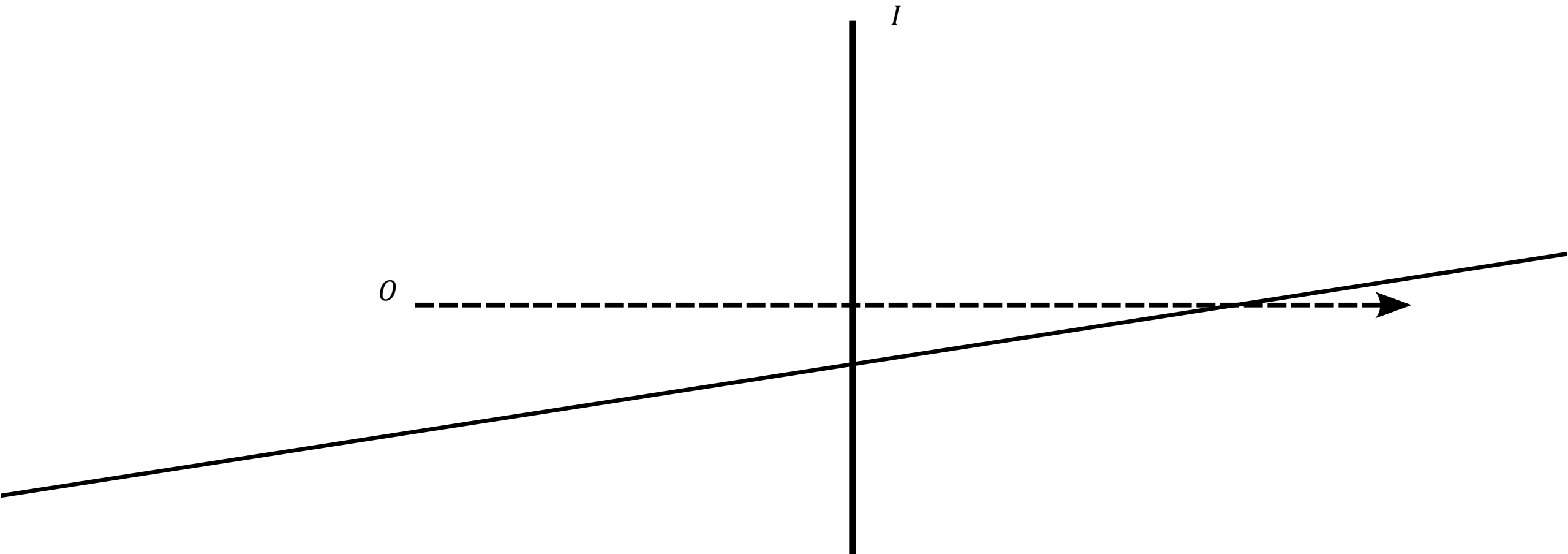
$$k < 0$$

$$0 < k < 1$$

$$k \geq 1$$

What happens for objects with different k ?

- Suppose a line goes from the front to after the optical center O



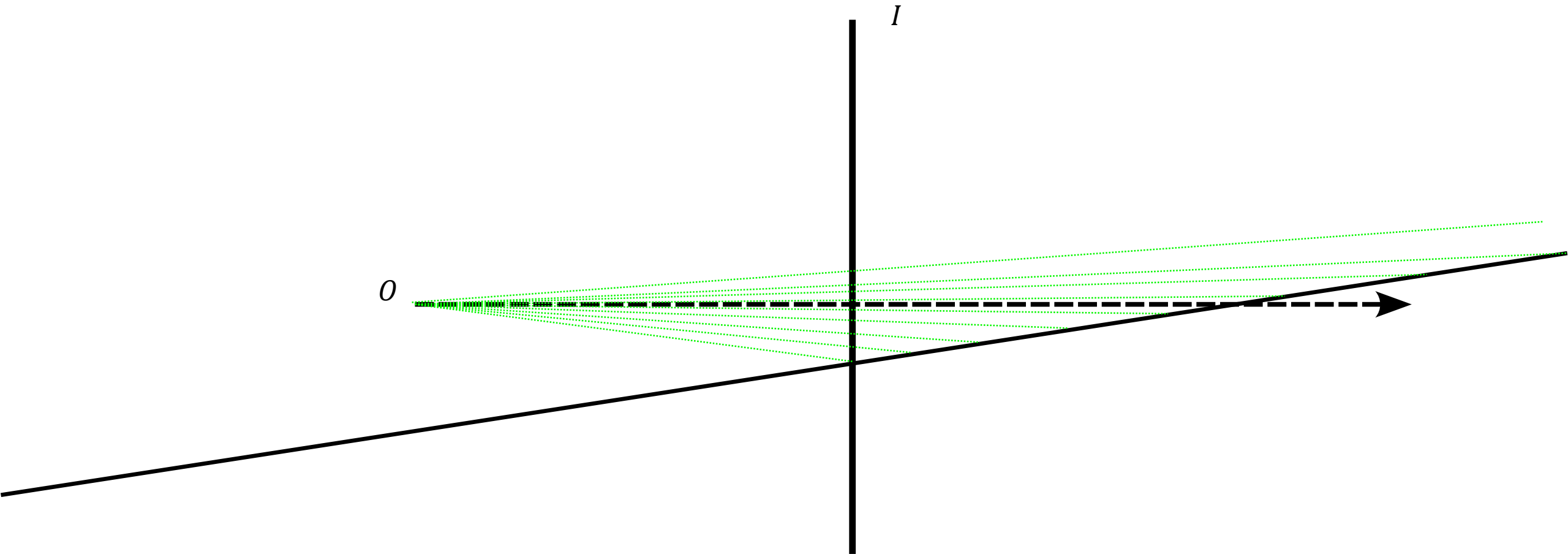
$$k < 0$$

$$0 < k < 1$$

$$k \geq 1$$

What happens for objects with different k ?

- Front parts of the line ($k \geq 1$) are projected to the image plane



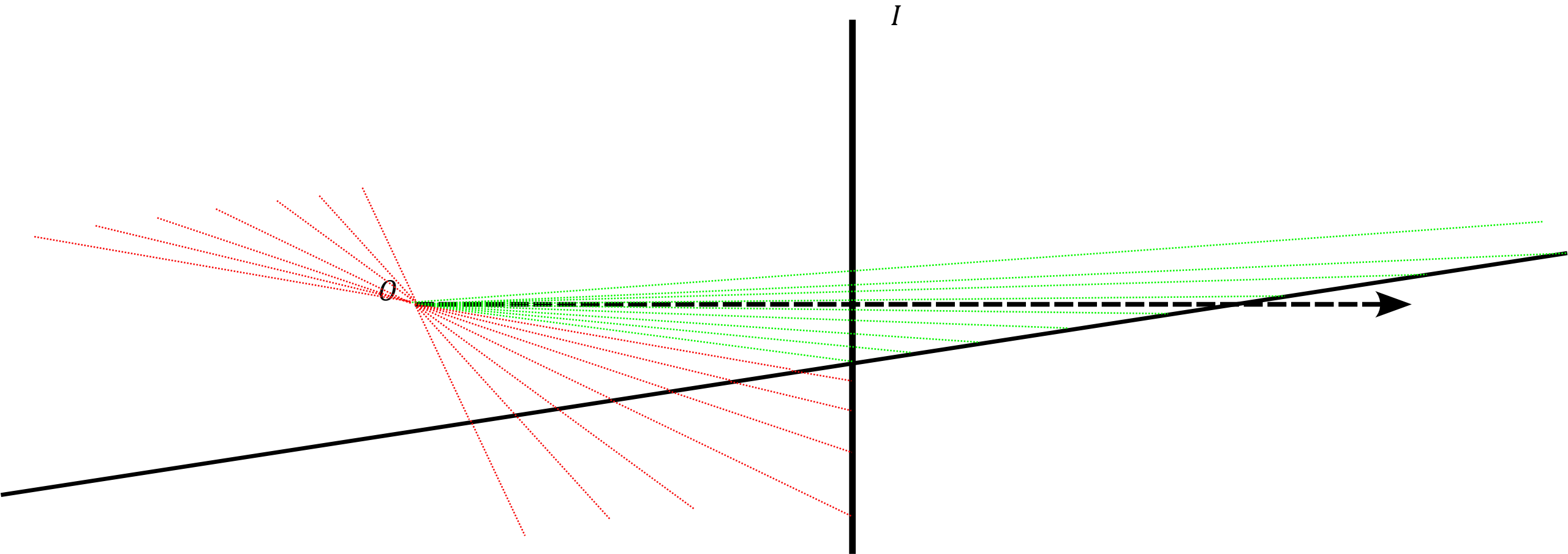
$k < 0$

$0 < k < 1$

$k \geq 1$

What happens for objects with different k ?

- Parts in $0 < k < 1$ of the object are projected to the image plane



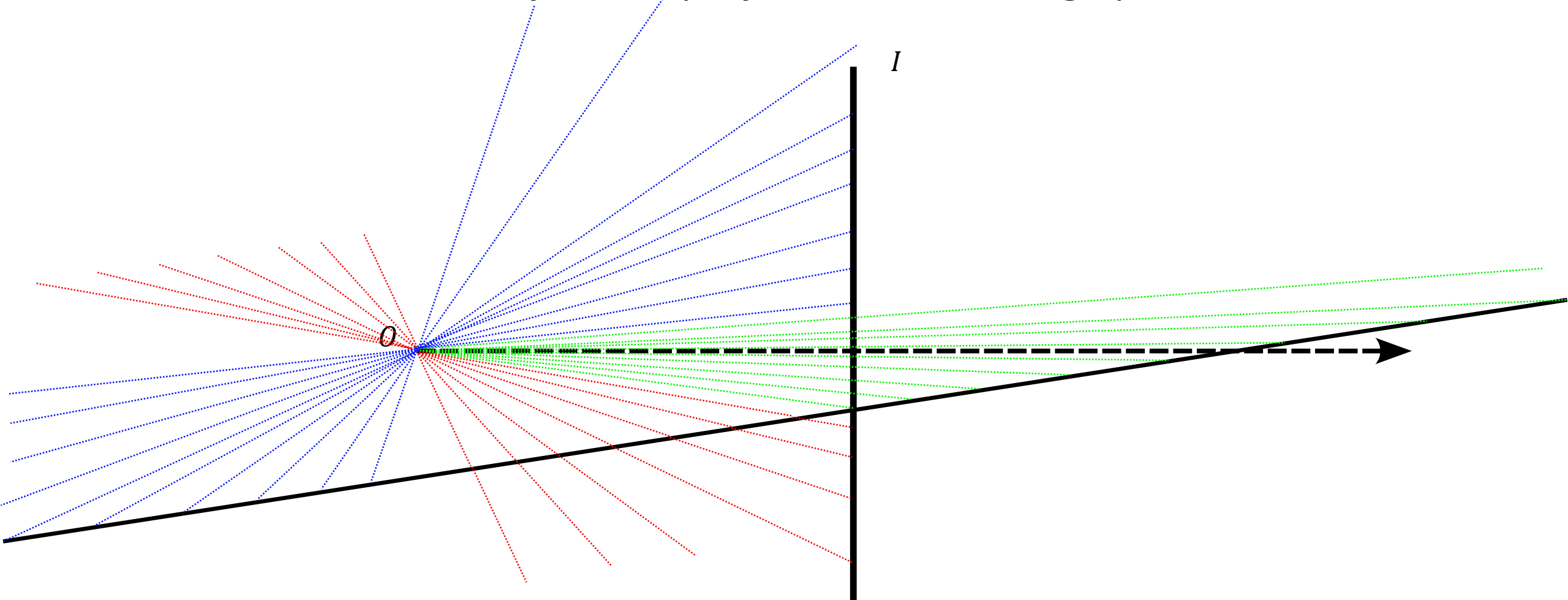
$k < 0$

$0 < k < 1$

$k \geq 1$

What happens for objects with different k ?

- Parts in $k < 0$ of the object are projected to the image plane



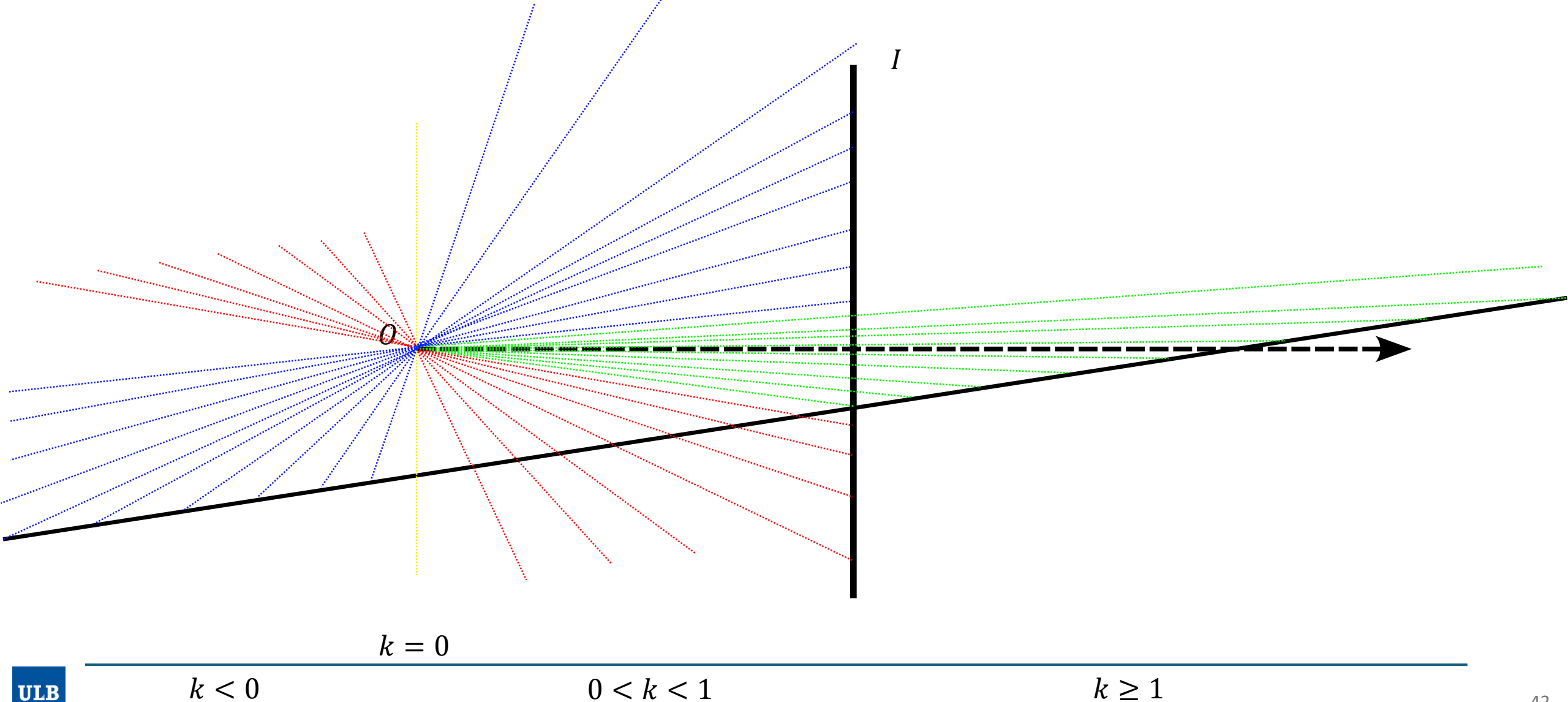
$k < 0$

$0 < k < 1$

$k \geq 1$

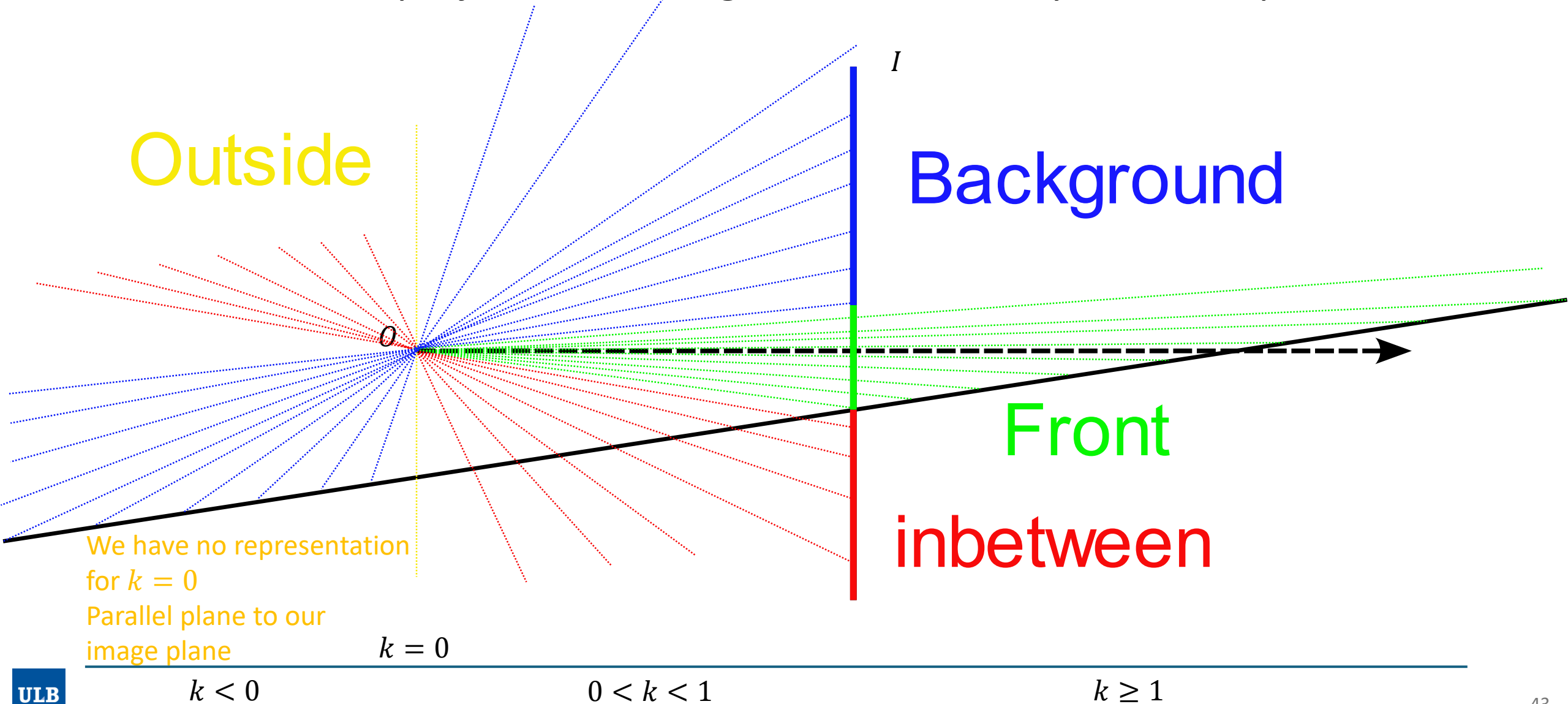
What happens for objects with different k ?

- At $k = 0$ There is no projection



What happens for objects with different k ?

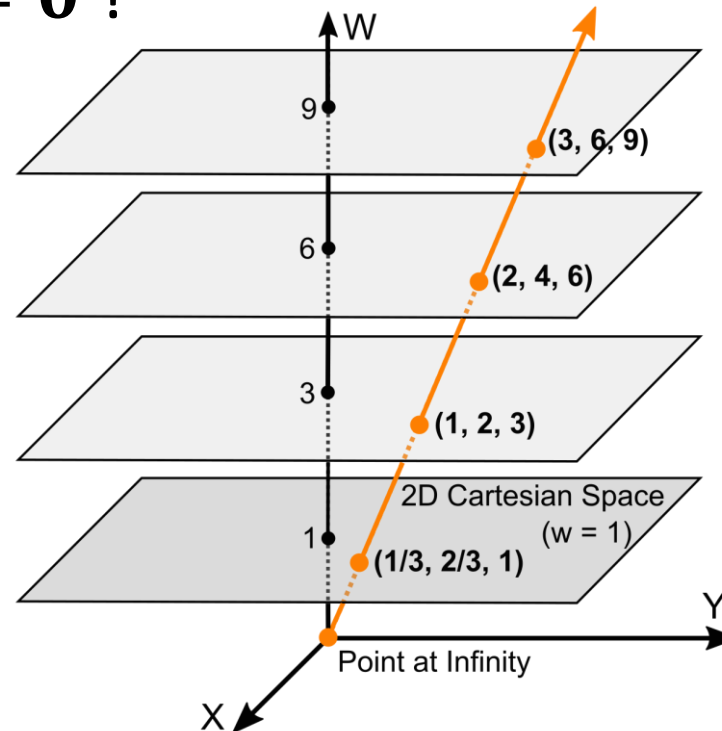
- We see that the projections through O send back rays to the top



Projective Geometry - Interpretations

- Why is it called projective space ?
- What does that $k = 1$ that we added mean ?
- What happens for objects in $0 < k < 1$ (near plane) ?
- What happens for objects with negative k ?
- **What happens when $k = 0$?**

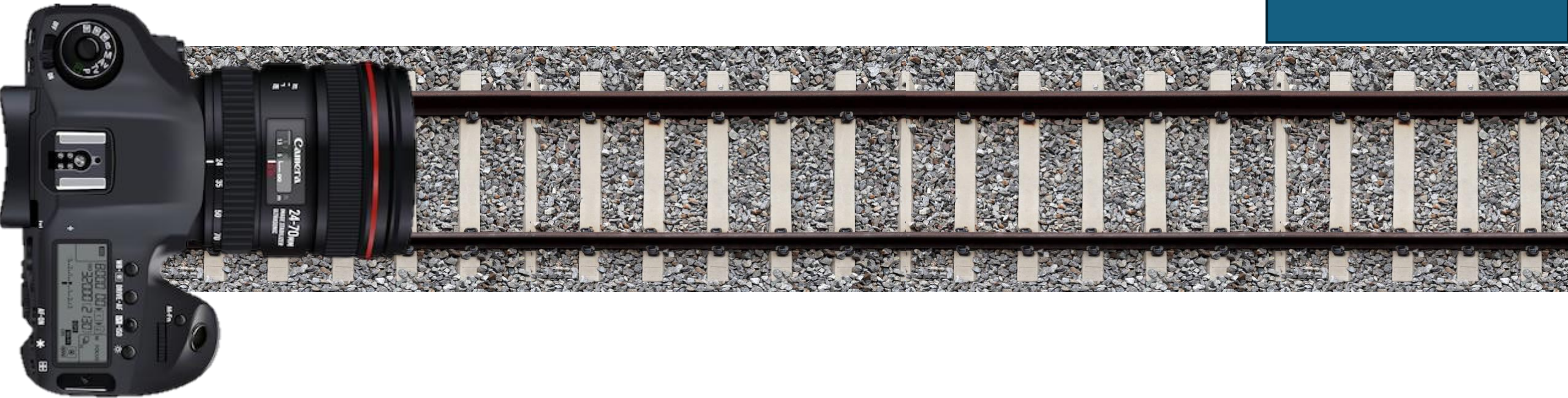
(elements such as $(1,2,0)$)



Projective geometry – Parallel lines

- Let's have a bit of intuition of how parallel lines are projected
- Suppose your camera is in front of a railway

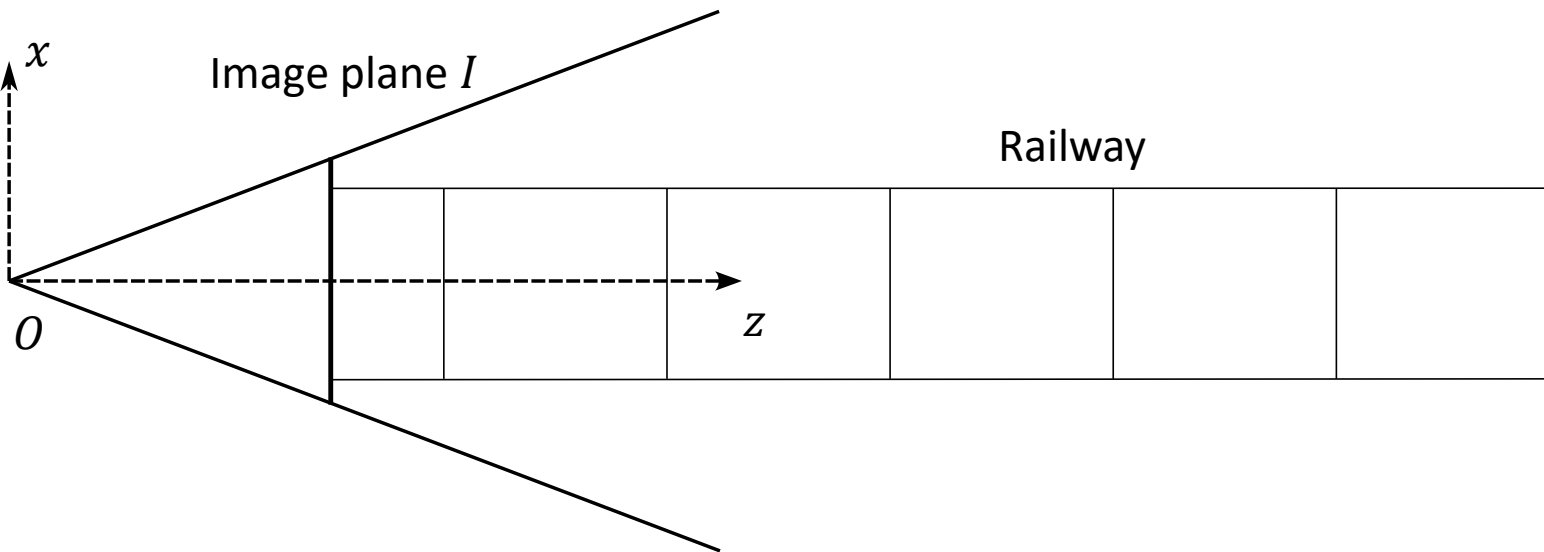
The Picture



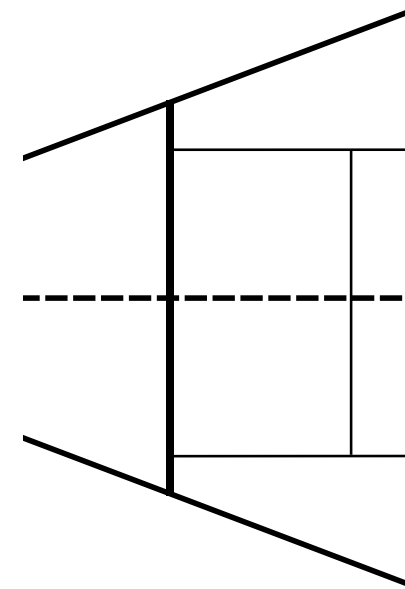
- What does it sees ?

Projective geometry – Parallel lines

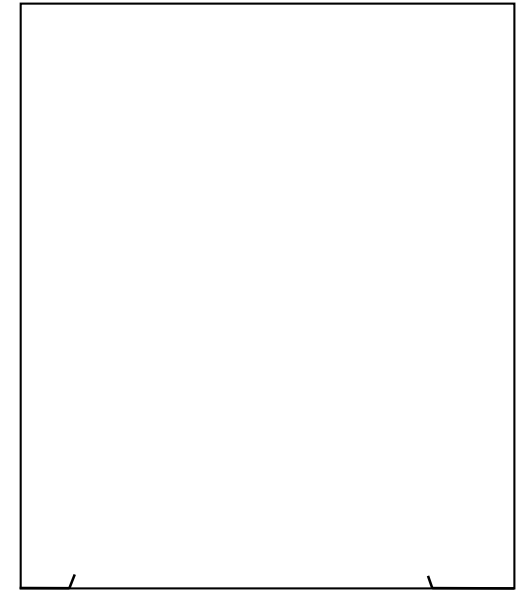
- Let's have a bit of intuition of how parallel lines are projected
- Suppose your camera is in front of a railway (in 2D)



Pinhole camera model



Zoom-in

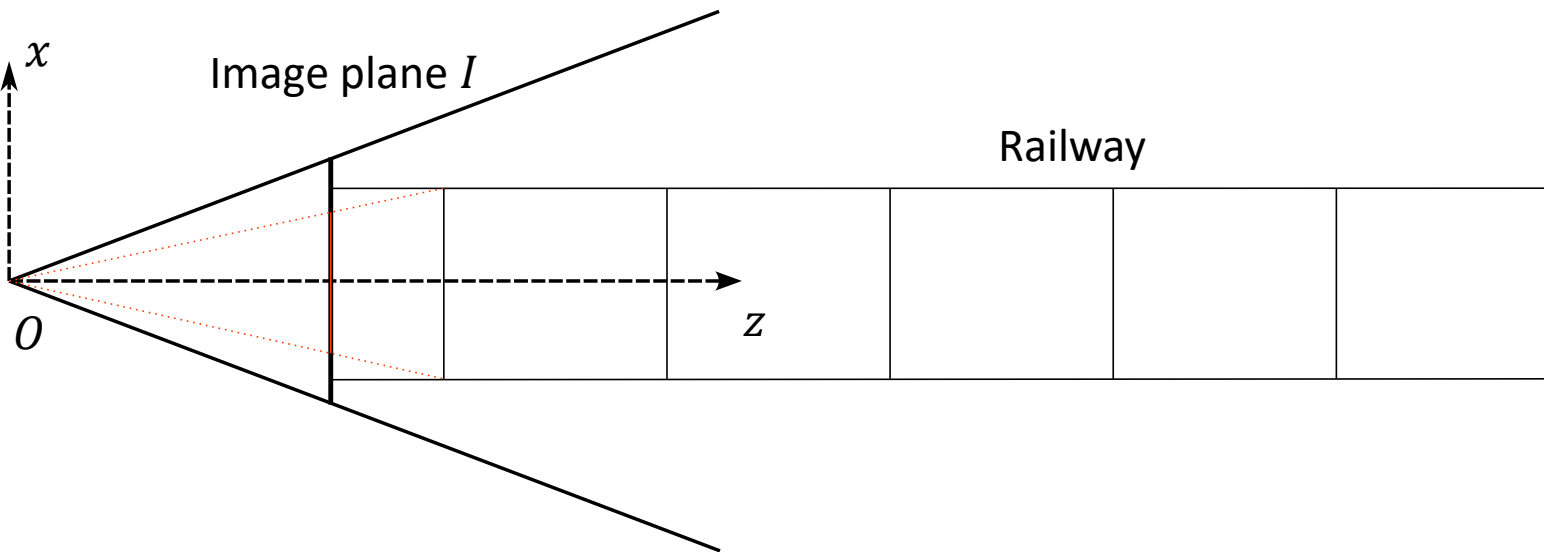


Picture/Image plane

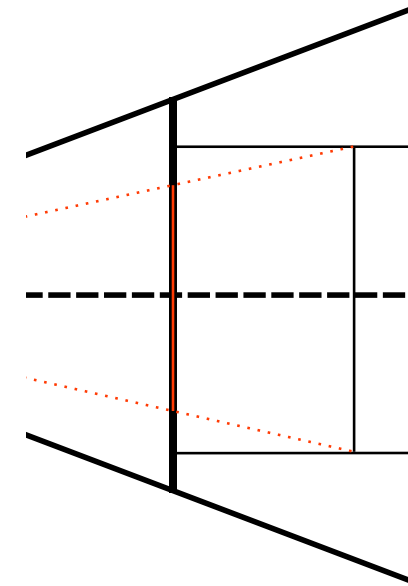
- First, we see the rail that *touches* the image plane

Projective geometry – Parallel lines

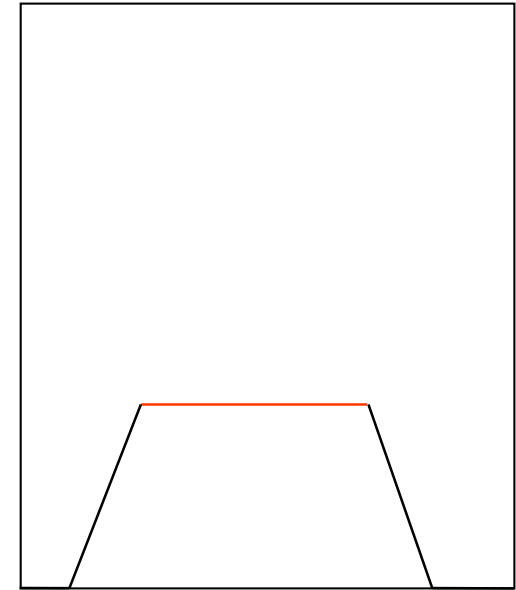
- Let's have a bit of intuition of how parallel lines are projected
- Suppose your camera is in front of a railway (in 2D)



Pinhole camera model



Zoom-in

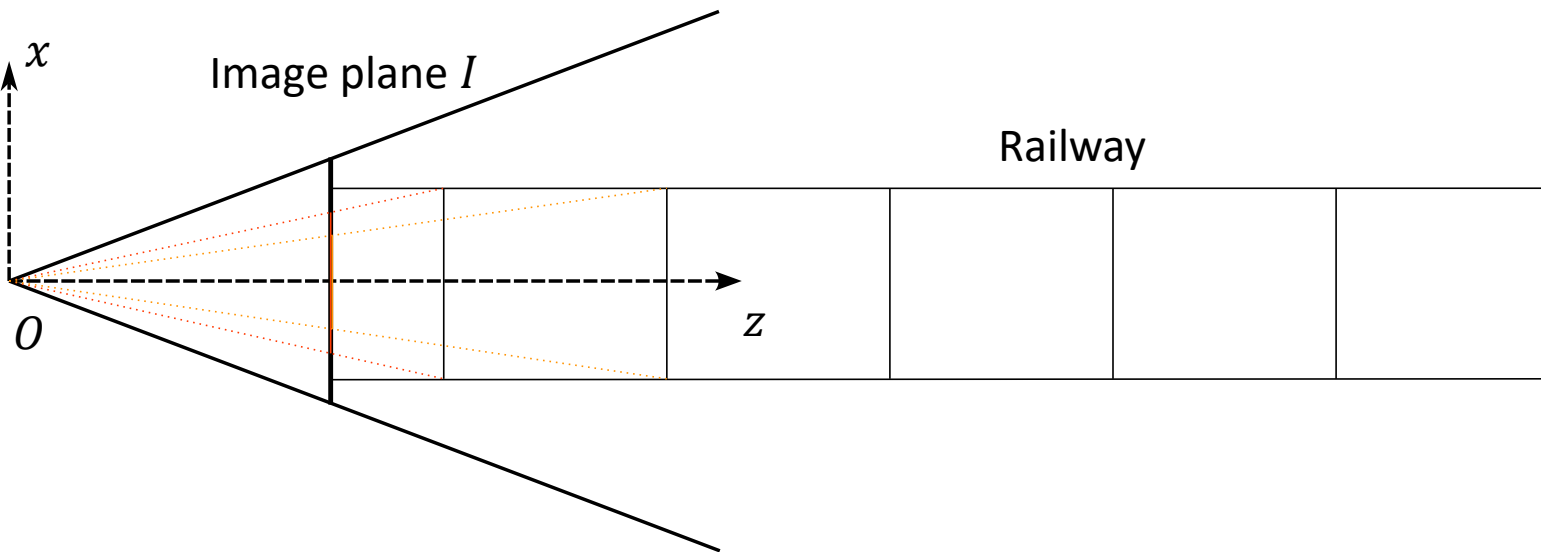


Picture/Image plane

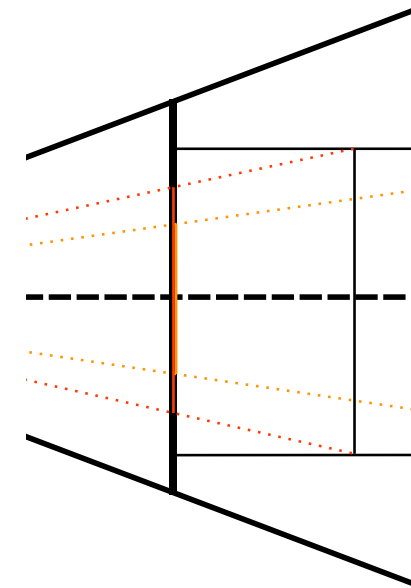
- As we project the first *stair* we see an image is forming

Projective geometry – Parallel lines

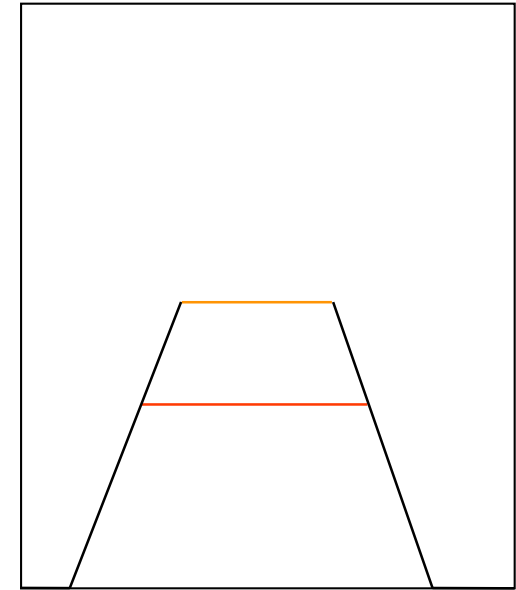
- Let's have a bit of intuition of how parallel lines are projected
- Suppose your camera is in front of a railway (in 2D)



Pinhole camera model



Zoom-in

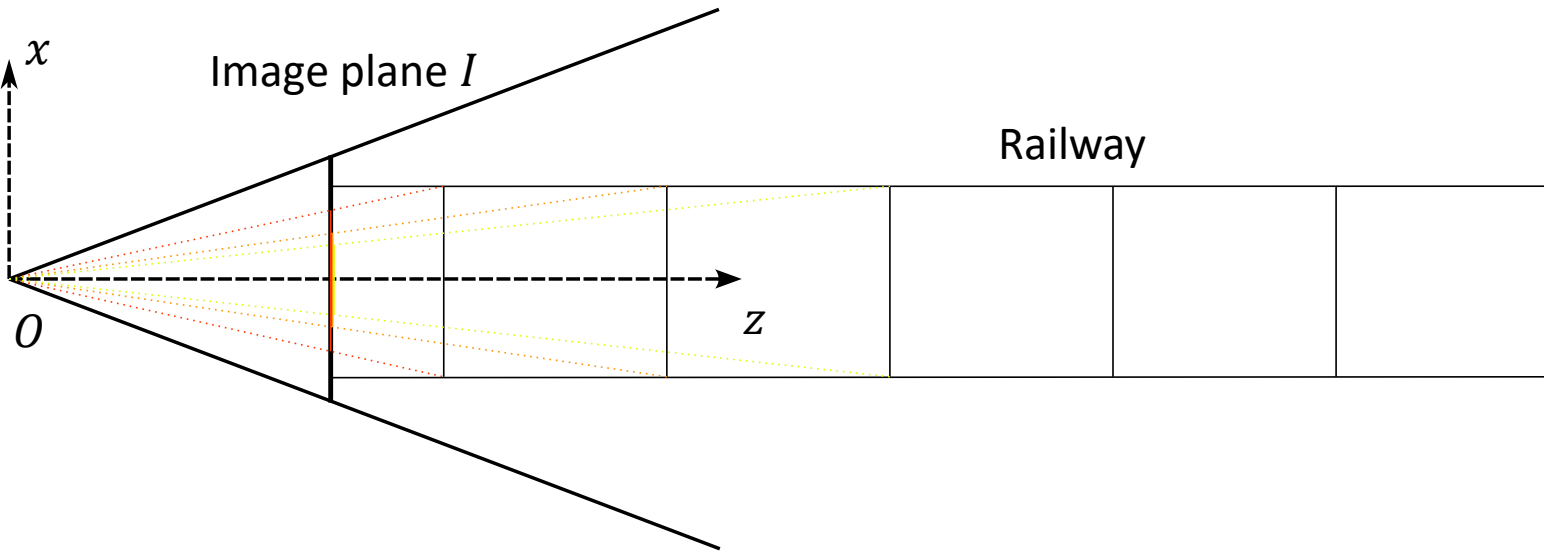


Picture/Image plane

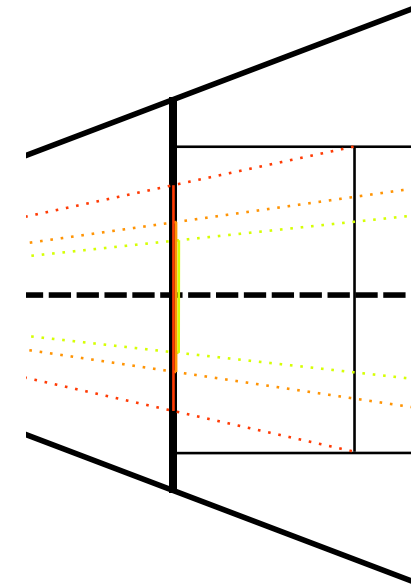
- We continue with the second, oh, it is smaller than the first, and *closer* to it.

Projective geometry – Parallel lines

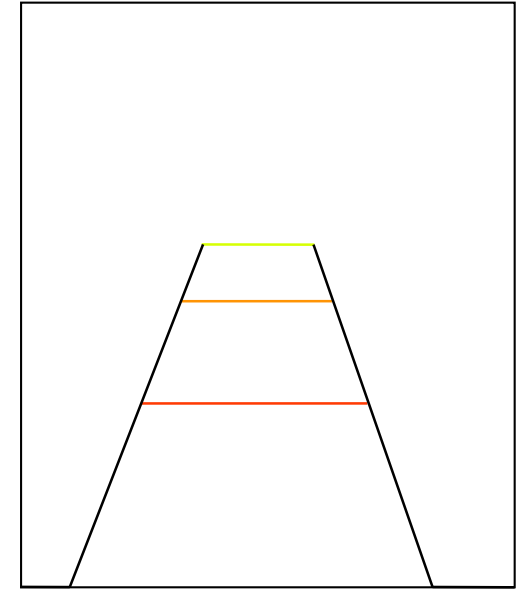
- Let's have a bit of intuition of how parallel lines are projected
- Suppose your camera is in front of a railway (in 2D)



Pinhole camera model



Zoom-in

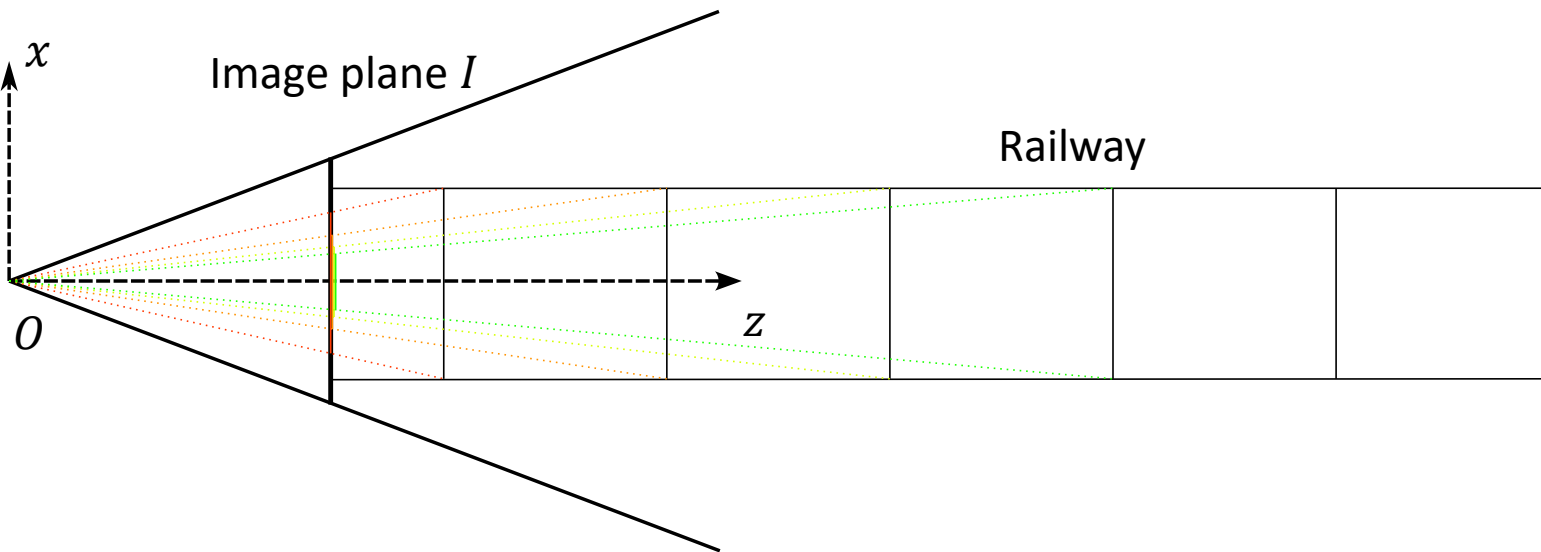


Picture/Image plane

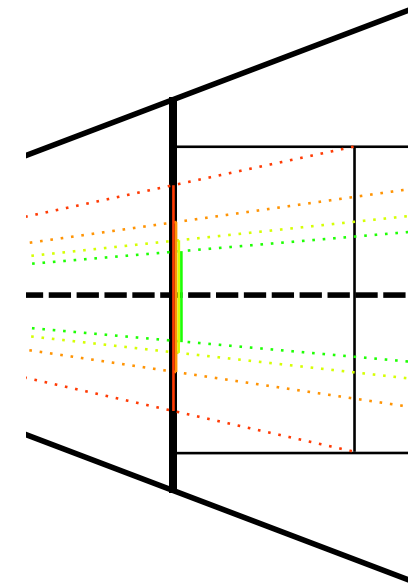
- This pattern continues

Projective geometry – Parallel lines

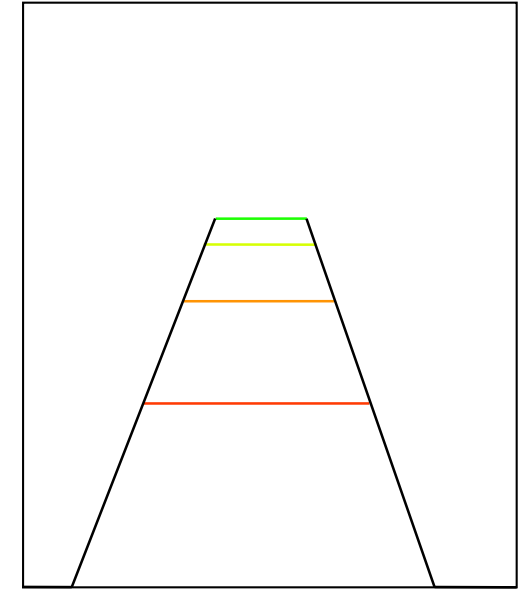
- Let's have a bit of intuition of how parallel lines are projected
- Suppose your camera is in front of a railway (in 2D)



Pinhole camera model



Zoom-in

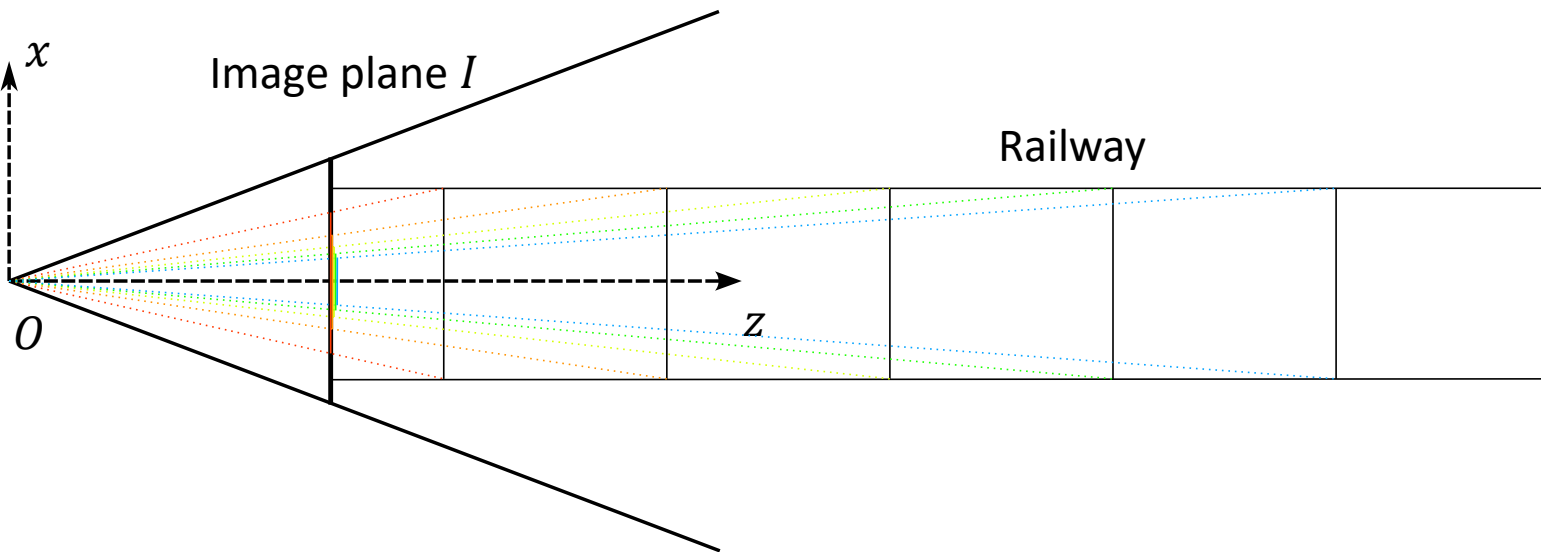


Picture/Image plane

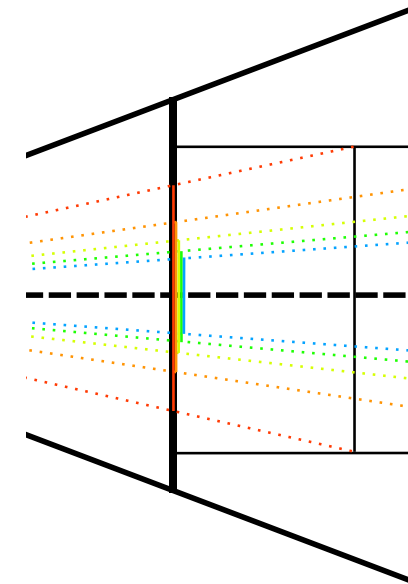
- And.. continues

Projective geometry – Parallel lines

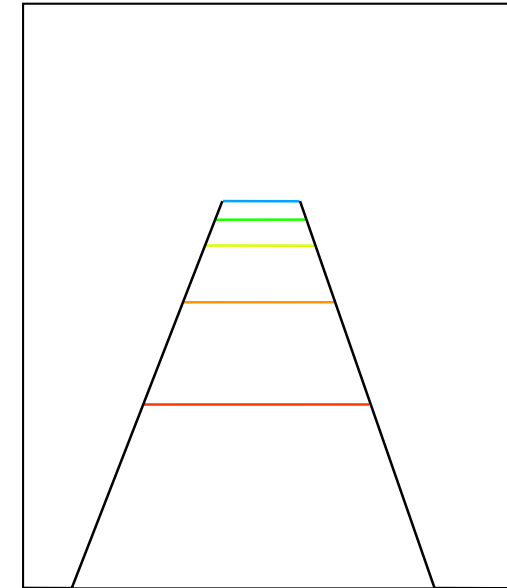
- Let's have a bit of intuition of how parallel lines are projected
- Suppose your camera is in front of a railway (in 2D)



Pinhole camera model



Zoom-in

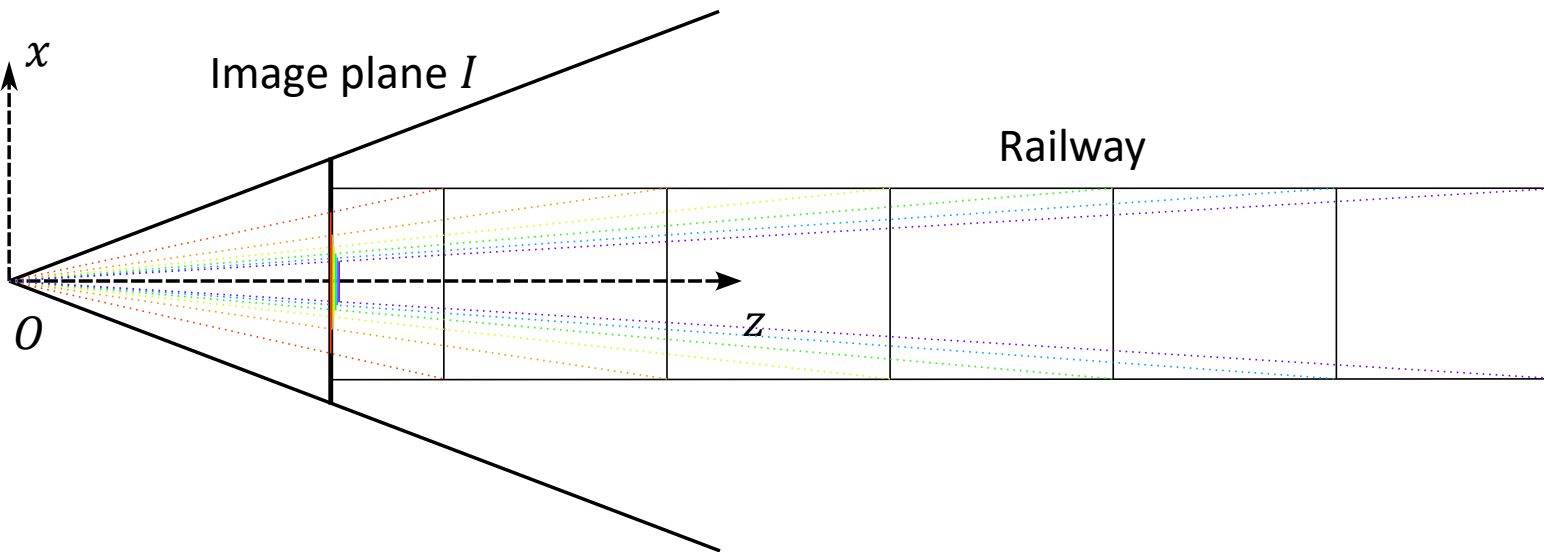


Picture/Image plane

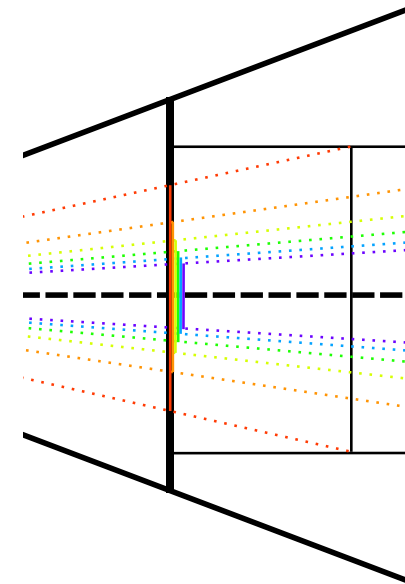
- And.. continues

Projective geometry – Parallel lines

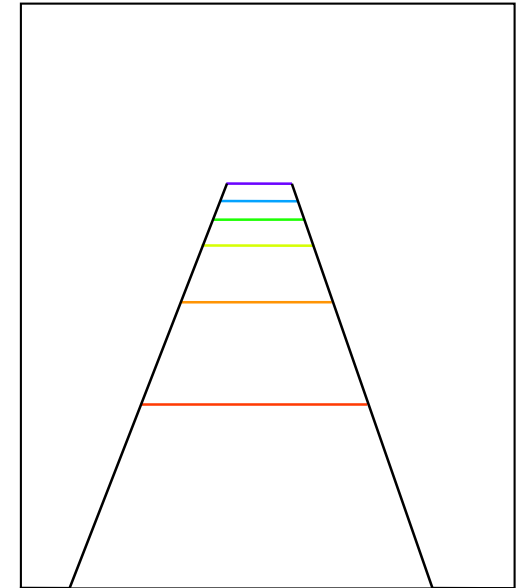
- Let's have a bit of intuition of how parallel lines are projected
- Suppose your camera is in front of a railway (in 2D)



Pinhole camera model



Zoom-in

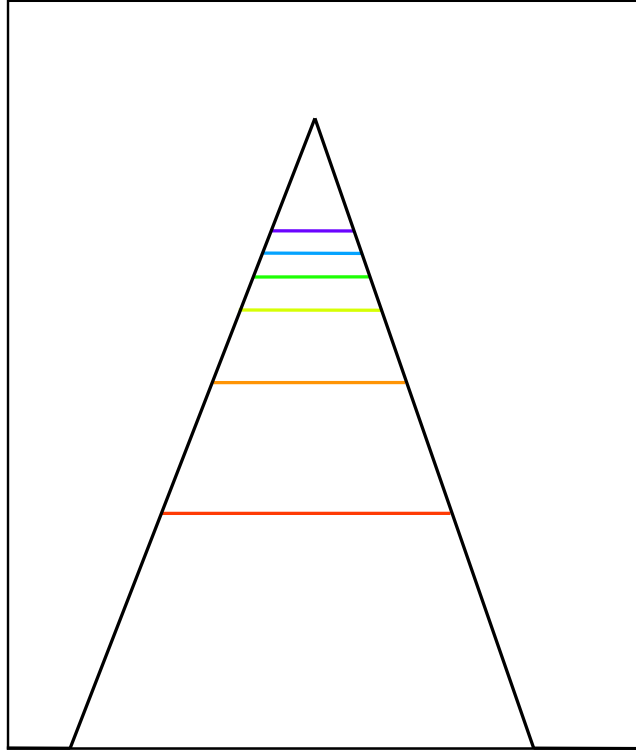


Picture/Image plane

- Hum.. Our drawing is not infinite.. But.. We can infer what will happens !

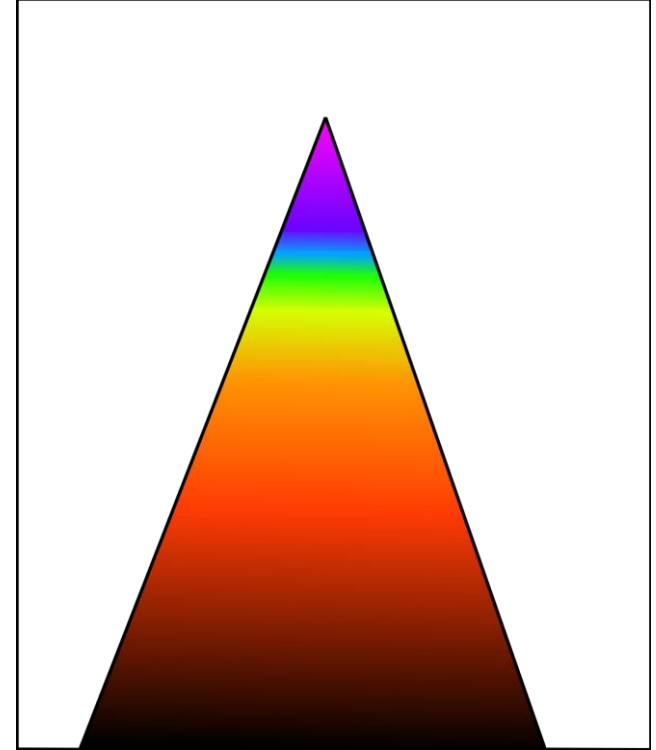
Projective geometry – Parallel lines

- So our parallel lines... Join.. ?



Picture/Image plane

Continuous version



Picture/Image plane

- Yes! Parallel lines join at infinity in projective geometry !

Projective geometry – Parallel lines

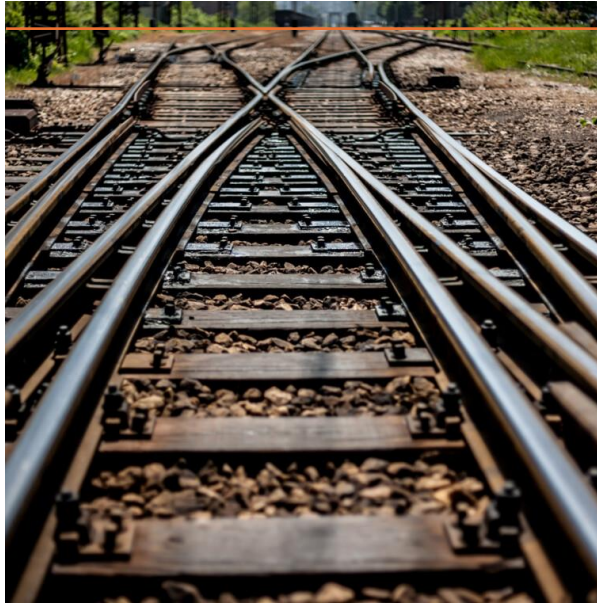
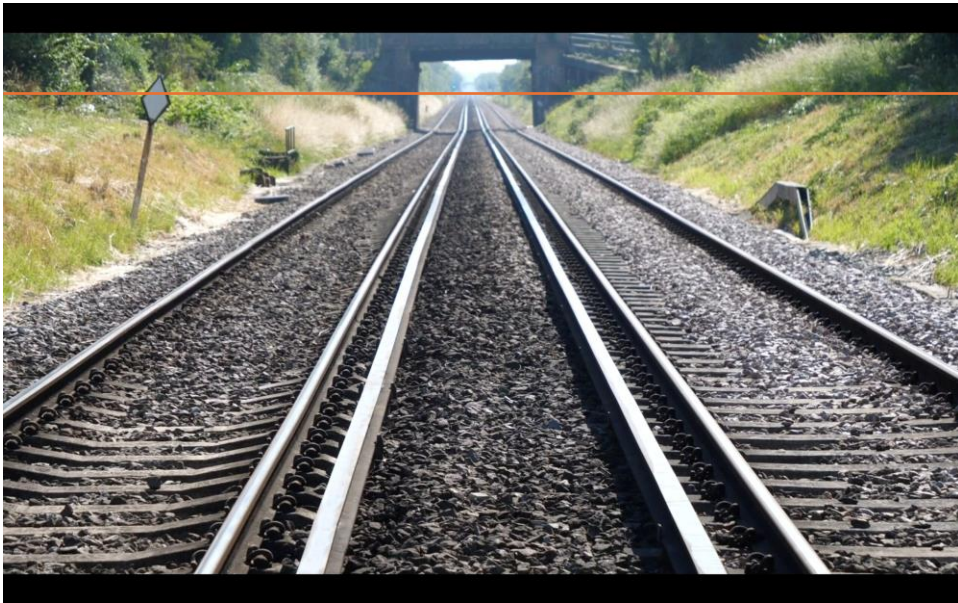
- So our parallel lines... Join.. ?
- Yes! Parallel lines join at infinity in projective geometry !



- Do the always join in the center of the picture ?

Projective geometry – Parallel lines

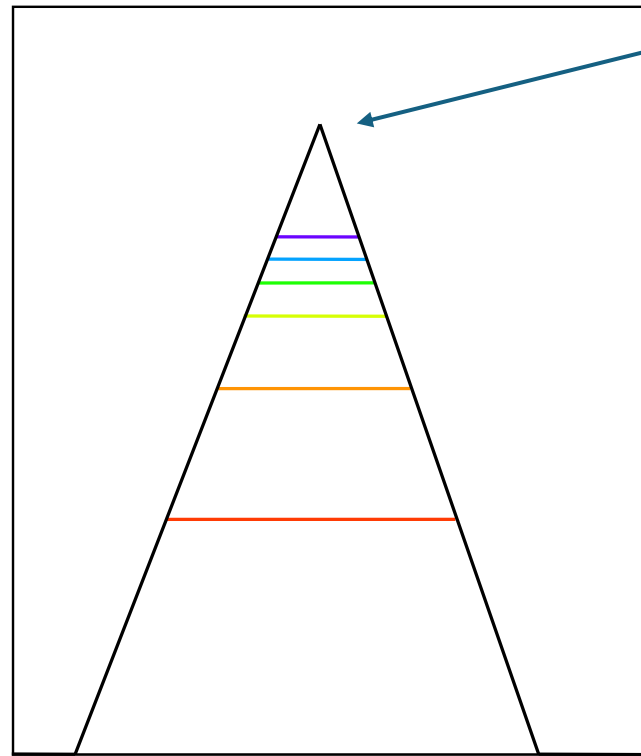
- Yes! Parallel lines join at infinity in projective geometry !
- Do they always join in the center of the picture ?



No.. Where is the point at which they join ?

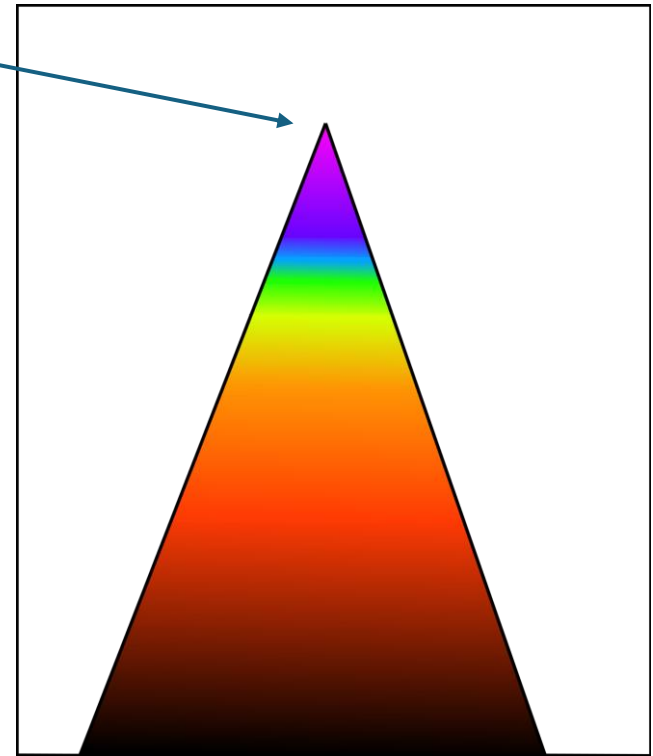
Projective geometry – Parallel lines

- What is the point at which they join ?



Picture/Image plane

Continuous version

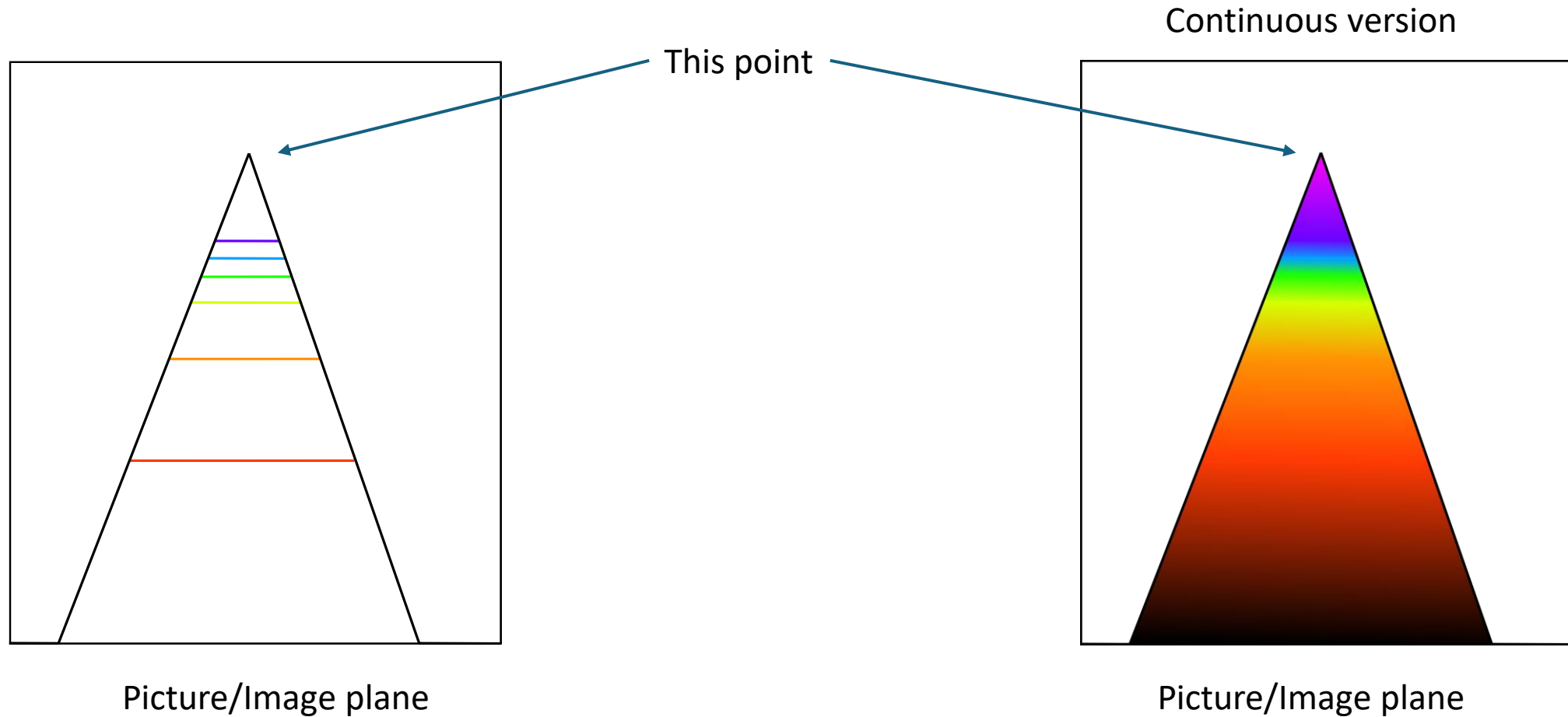


Picture/Image plane

This point

Projective geometry – Parallel lines

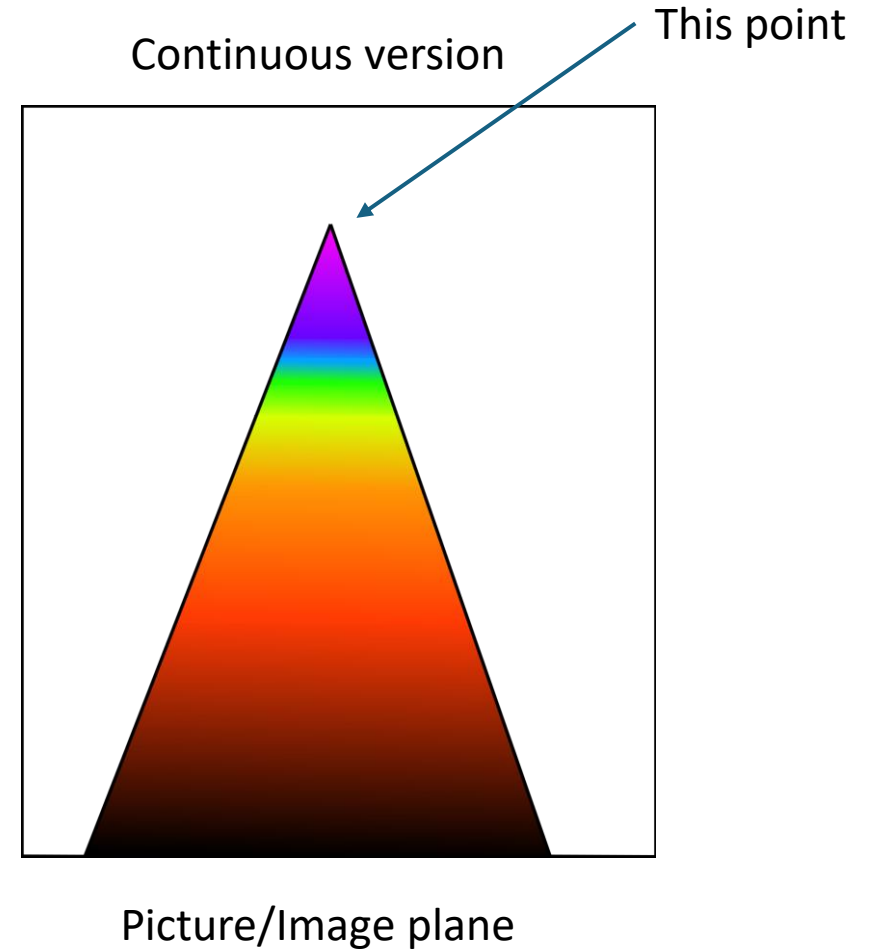
- What is the point at which they join ?



- So, $k \rightarrow \infty$ is like having a plane infinitely far away from the camera.

Projective geometry – Parallel lines

- It represent infinitely far away from the camera ! So $k \rightarrow \infty$
- Even if $k \rightarrow \infty$, (x, y) converge toward values !
- ∞ is not a physical location in space, but it's so far away, that the coordinates (x, y) stabilize in the image plane.
- We can think of (x, y) as the *direction* to which every parallel line in that direction converges to !
- So this point in image plane represent a direction, no more just a point.



- But.. We cannot encode it using projective geometry !

Projective geometry – Parallel lines

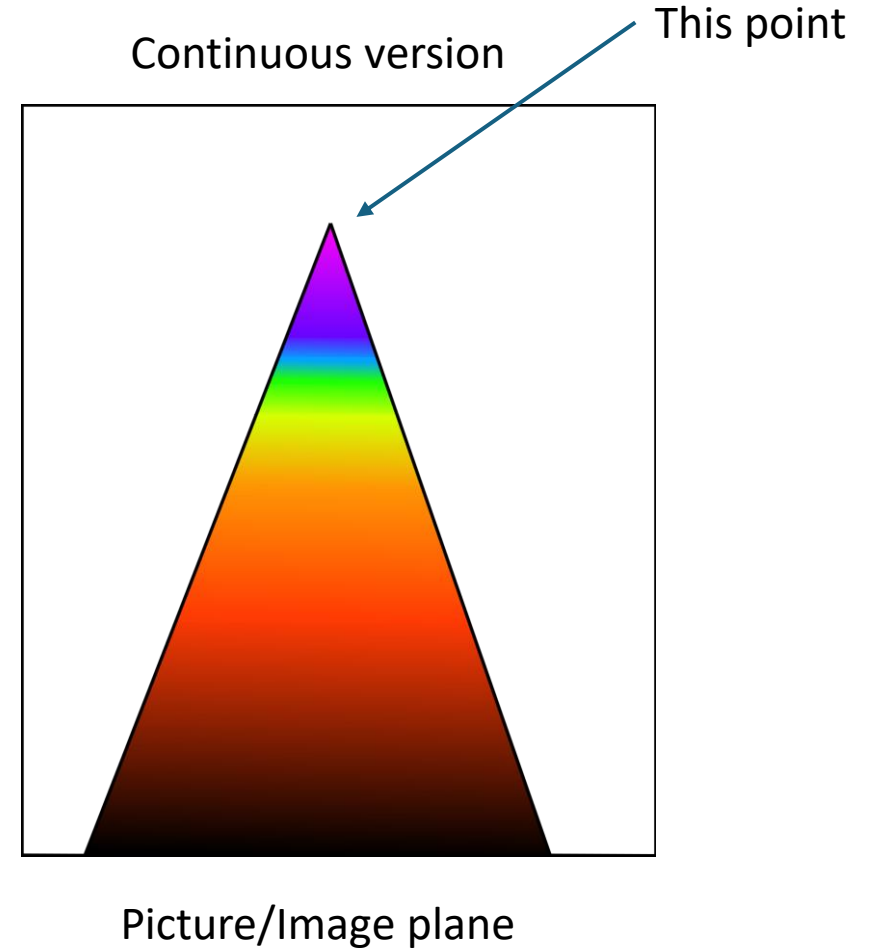
- But.. We cannot encode it using projective geometry !

- $p: \mathbb{RP} \rightarrow \mathbb{R}: (x, y, \infty) \mapsto \left(\frac{x}{\infty}, \frac{y}{\infty}\right) = (0,0)$

- This is not meaningful.
- We want to preserve that the point is a direction
- And infinities are not easy to handle.

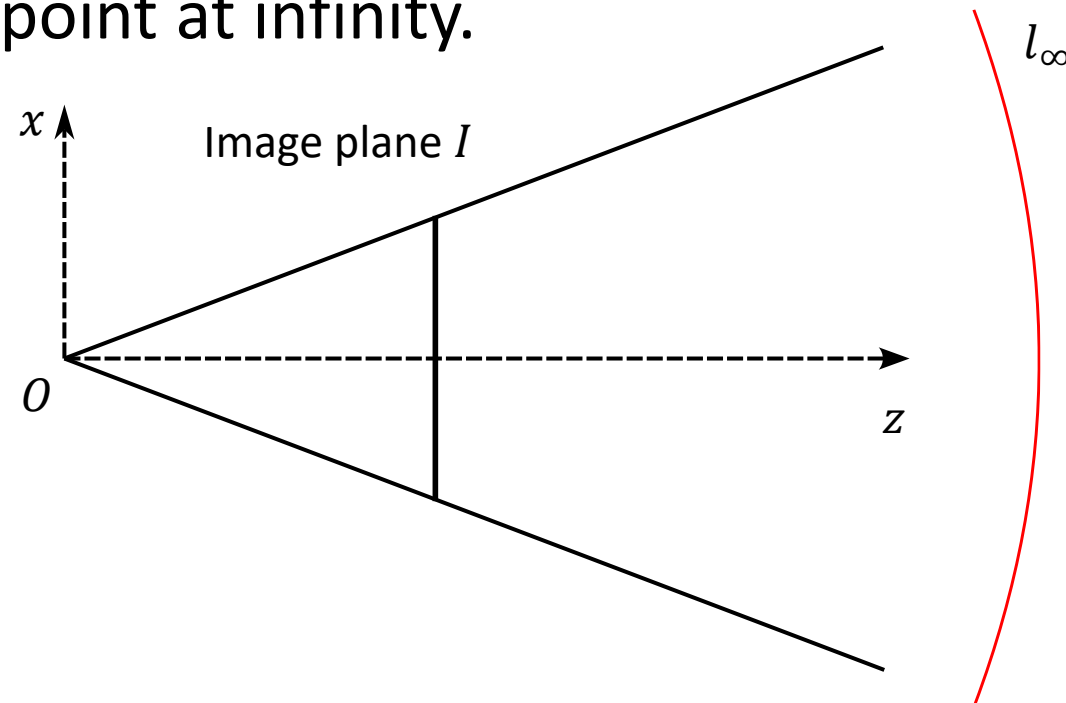
- Instead, this point does **not have a depth**, we encode it as $k = 0$

- So points at $(x, y, 0)$ are said to be at infinity.
- They have no defined depth, they are only directions



Projective Geometry – Parallel lines

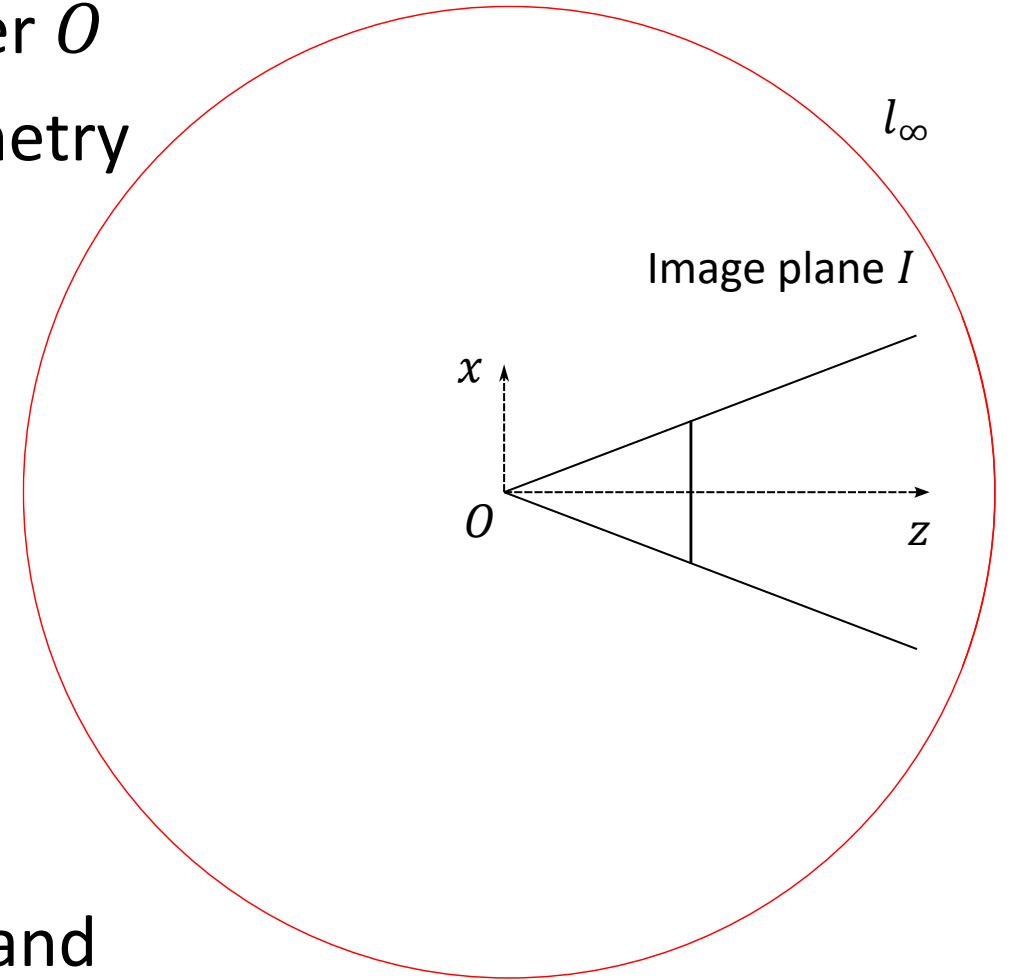
- This point represent **only** the lines that are parallel and goes toward that direction (x, y)
- **But**, any two parallel lines converge to a different point at infinity, with another *direction* point at infinity.



- We can visualize that as if there was an invisible line at infinity
- Where every parallel lines meet at some point on it

Projective Geometry – Parallel lines

- This is not limited to our field of view
- The line extends all around the optical center O
- It's only a representation of projective geometry
- As infinities have not to be at the same distance
- But its one of the best ways we have to visualize what is happening.
- This line is encoded with all points $(x, y, 0)$ and represent the direction of parallel lines.



Projective Geometry – $\mathbb{R}P^2$ – A line

- We gained intuition of how lines act in projective geometry.
- Concretely, a line

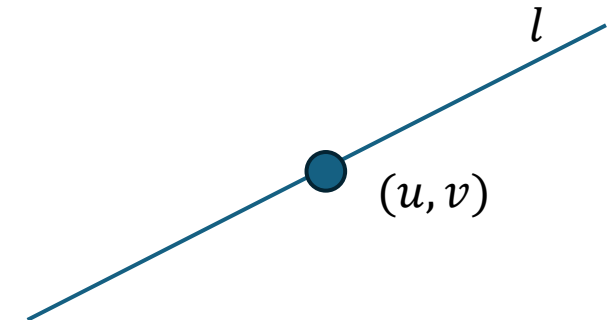
$$l \equiv ax + by + c = 0$$

- Can be written in vectorial form:

$$l \equiv (a, b, c) \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = 0$$

- To check if a point $p = (u, v)$ is on the line, we only need to check that

$$(a, b, c) \begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = 0$$



Projective Geometry – $\mathbb{R}P^2$ – Parallel lines

$$l \equiv (a, b, c) \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = 0$$

- Note that a line is invariant by scaling ! i.e.,

$$\forall k \neq 0, \quad (ka)x + (kb)y + kc = 0 \Leftrightarrow k(ax + by + c) = 0 \Leftrightarrow ax + by + c = 0$$

- By using the definition:

$$kl \equiv k(a, b, c) \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = (ka, kb, kc) \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = (a, b, c) \begin{pmatrix} kx \\ ky \\ k \end{pmatrix} = 0$$

- k moves the point $(x, y, 1)^T$ **along** the line !

Projective Geometry – $\mathbb{R}P^2$ – Duality

- Notice how a line: $l \equiv ax + by + c = 0$, a point: $p = (a, b)^T$
Rewritten in $\mathbb{R}P^2$ as $l \equiv (a, b, c)^T = 0$, and $p = (a, b, w)^T$
Ressemble each others.
- This is called the **duality** of lines and points in projective geometry.
- *Duality principle*: To any theorem of 2-dimensional projective geometry there corresponds a dual theorem, which may be derived by interchanging the roles of points and lines in the original theorem.
- So, in projective geometry, we only need to prove a theorem on lines **or** on points.

Projective Geometry – $\mathbb{R}P^2$ – Intersections

- What is the intersection of two lines? $l \equiv (a, b, c)^T$ and $l' \equiv (a', b', c')^T$
 - Pretty easy ! $p = l \times l'$ (yes.. Just a cross product!)

- It is true because we can check that the point lie on both lines:

$$l \cdot p = l' \cdot p \Leftrightarrow l \cdot (l \times l') = l' \cdot (l \times l') = 0$$

- Example: $l = (-1, 0, 1)^T$ and $l' = (0, -1, 1)^T$

$$p = \begin{vmatrix} e_1 & e_2 & e_3 \\ -1 & 0 & 1 \\ 0 & -1 & 1 \end{vmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \mapsto \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

We are in homogeneous coordinates,
the last element is w

Projective Geometry – $\mathbb{R}P^2$ – Intersections

- What is the intersection of two lines? $l \equiv (a, b, c)^T$ and $l' \equiv (a', b', c')^T$
 $p = l \times l'$
- What if the lines are parallel ?
- Example: $l = (-1, 0, 1)^T$ and $l' = (-1, 0, 2)^T$ ($l \equiv x = 1$ and $l' \equiv x = 2$)

$$p = \begin{vmatrix} e_1 & e_2 & e_3 \\ -1 & 0 & 1 \\ -1 & 0 & 2 \end{vmatrix} = \begin{pmatrix} 0 \\ 1 \\ \textcolor{red}{0} \end{pmatrix}$$

Thanks to projective geometry, if the lines do not intersect this point will be at infinity ! No more special cases 😊

We are in homogeneous coordinates,
the last element is w

Projective Geometry – $\mathbb{R}P^2$ – Intersections

- What is the line between two points? $p \equiv (a, b, c)^T$ and $p' \equiv (a', b', c')^T$
 - Easy again ! $l = p \times p'$
 - We use the same argument as before to prove it ! (or just the duality principle ☺)
- Example: $p = (-1, 0, 1)^T$ and $p' = (0, -1, 1)^T$

$$l = \begin{vmatrix} e_1 & e_2 & e_3 \\ -1 & 0 & 1 \\ 0 & -1 & 1 \end{vmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$l \equiv x + y + 1 = 0$$

Projective Geometry – Notice the beauty

- Notice the beauty of projective geometry ($\mathbb{RP}^2$).
- Euclidean Geometry (identified to $\mathbb{R}^2$):

Intersection two lines	Line from two points	Distance line point
$ \begin{aligned} l_1 &\equiv y_1 = ax + c, \\ l_2 &\equiv y_2 = bx + d \\ ax + c &= bx + d \\ ax - bx &= d - c \\ x &= \frac{d - c}{a - b} \\ y &= a \frac{d - c}{a - b} + c \\ p &= \left(\frac{d - c}{a - b}, a \frac{d - c}{a - b} + c \right) \end{aligned} $	$ \begin{aligned} p_1 &= (x_1, y_1), \\ p_2 &= (x_2, y_2) \\ slope &= \frac{y_2 - y_1}{x_2 - x_1} \\ l &\equiv y = slope \cdot x + b \\ \text{Solve for } b \\ y_1 &= \frac{y_2 - y_1}{x_2 - x_1} x_1 + b \\ l &\equiv y = \frac{y_2 - y_1}{x_2 - x_1} x + b \end{aligned} $	Very ugly equation from previous lesson

- Projective Geometry:

Intersection two lines	Line from two points	Distance line point
$p = l \times l'$	$l = p \times p'$	$d = l \cdot p$

Check it

Notice the duality between points and lines

Projective Geometry – Notice the beauty

- Even more beauty in $\mathbb{RP}^3$.
- Euclidean Geometry (identified to $\mathbb{R}^3$):

Distance point plane	A plane with 3 points	Line from two points	Intersection line plane	A plane from a point and a line
Ugly formulas $d = \frac{ ax + by + cz + d }{\sqrt{a^2 + b^2 + c^2}}$	Ugly formulas $n = (p_2 - p_1) \times (p_3 - p_1)$ $n_x x + n_y y + n_z z + k = 0$ Find k by substitution.	See in $\mathbb{R}^2$	Ugly formulas Substitute line in plane $((u + vt) - p) \cdot n = 0$ Solving for t Check conditions for intersection	Ugly formulas find two points on the line, use three points to find the plane with previous formula.

- Projective Geometry:

Distance point plane	Define a plane with 3 points	Line from two points	Intersection line plane	A plane from a point and a line
$d = p \cdot \pi$	$\pi = (p_1 \times p_2)p_3$	$l = p_1 \times p_2$	$p = l\pi$	$\pi = lp$

Matrix product here

Points and planes are represented by vectors
 Lines by matrices (see Plücker coordinates)

Projective Geometry – $\mathbb{R}P^2$ – Directions

- Let's have more mathematical intuition behind **directions**
- All the points at infinity are of the form
$$p_{\infty} = (a, b, 0)^T$$
- Do the set of all points at infinity define a geometric shape?

Projective Geometry – $\mathbb{R}P^2$ – Directions

- Do the set of all points at infinity define a geometric shape?

- It turn out that yes, they are all on a *line*, **the line at infinity** l_∞

$$l_\infty = (0,0,1)^T$$

- This is true because $l_\infty^T \cdot p = (0,0,1) \cdot (a,b,0)^T = 0$
 - So the points lie on that line.
 - (it is a strange line, as we saw, it circles back)
- What is the intersection of any lines to this line ?

Projective Geometry – $\mathbb{R}P^2$ – Directions

- What is the intersection of any lines to this line ?

- A line l intersect l_∞ at $p = l \times l_\infty = \begin{vmatrix} e_1 & e_2 & e_3 \\ a & b & c \\ 0 & 0 & 1 \end{vmatrix} = \begin{pmatrix} b \\ -a \\ 0 \end{pmatrix}$

- We see that this quantity is defined for any line l
- But what happens for another line l' parallel to l ?

Projective Geometry – $\mathbb{R}P^2$ – Directions

- But what happens for another line l' parallel to l ?

$$p = l \times l_\infty = (b, -a, 0)^T$$

- A line l' **parallel** to l intersect l_∞ at

$$p' = l' \times l_\infty = \begin{vmatrix} e_1 & e_2 & e_3 \\ a & b & c' \\ 0 & 0 & 1 \end{vmatrix} = \begin{pmatrix} b \\ -a \\ 0 \end{pmatrix}$$

- We see that their intersection give the same point !
- This intersection does not depend on the offset c .

- In non-homogeneous coordinates:

$$p = (b, -a)^T$$

Projective Geometry – $\mathbb{R}P^2$ – Directions

- So we have that for any two parallel lines, they intersect at $p = l \times l_\infty = (b, -a, 0)^T$
$$p = (b, -a)^T$$

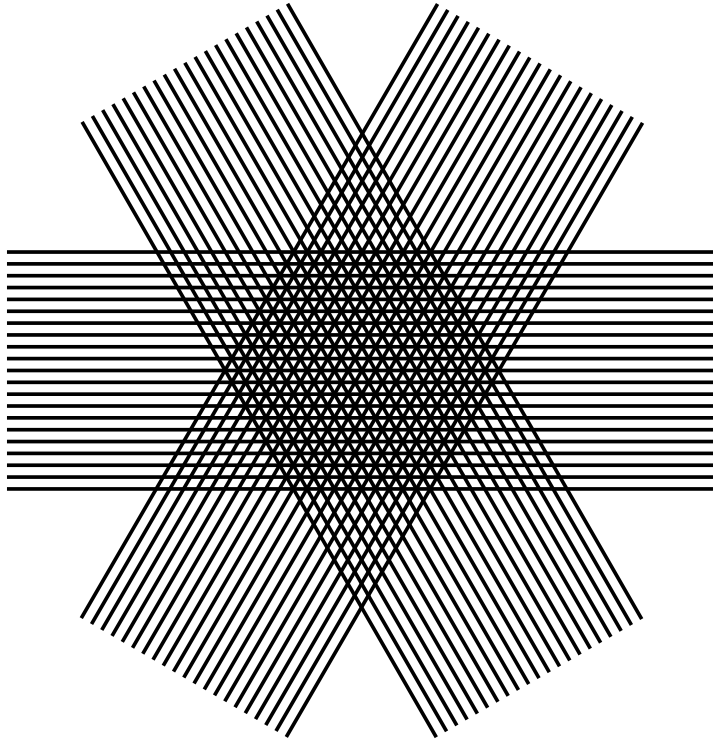
- What is the meaning of this point ?
- This vector is orthogonal to the line normal

$$(b, -a)^T \cdot (a, b) = ab - ab = 0$$

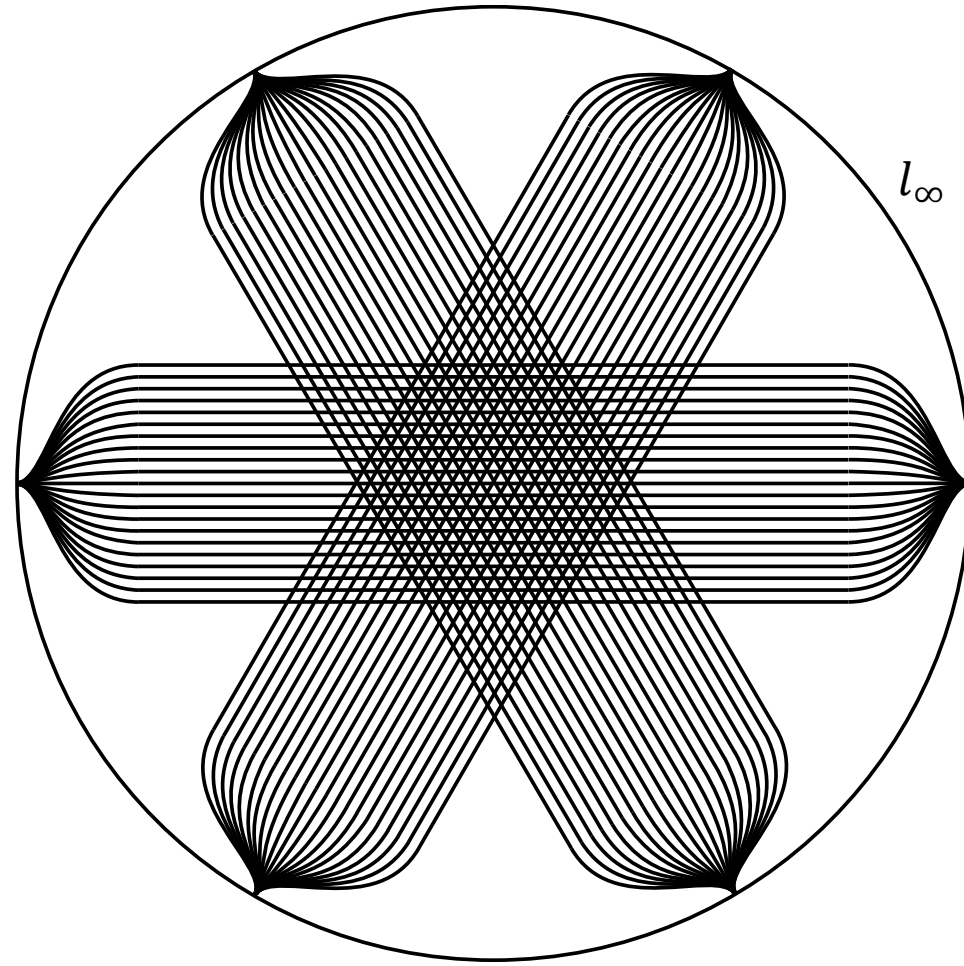
- Therefore it is tangent to the line, it represents the line *direction*.

Projective Geometry – Parallel lines

- Another view with parallel lines and directions.



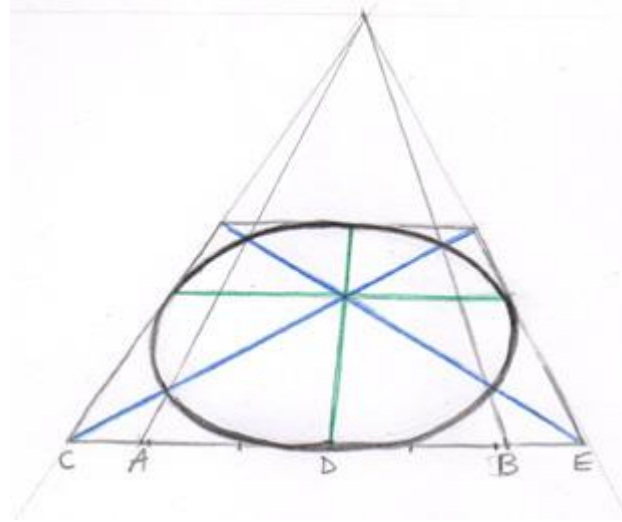
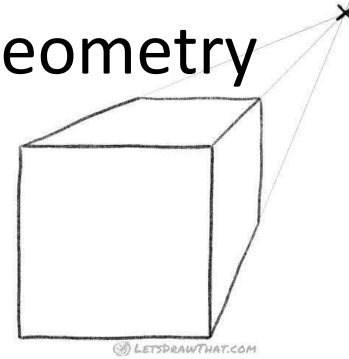
Euclidean Geometry



Projective Geometry

Projective Geometry – Summary

- Now, we are all familiar with projective geometry
- When we look at a picture:
 - Squares are not squares anymore
 - Circles are not circles anymore
 - Parallel lines join at infinity
- Transformations that map those planar objects into the picture is an example of a **perspective transformation**
- It allowed us to have a matrix representation of the rigid transform group !



Summary

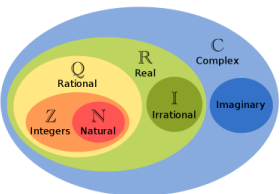
Add a law
 $+: G \times G \rightarrow G$

Add a law $\cdot: G \times G \rightarrow G$
 And $\cdot$ distribute $+$.

Objects = vectors
 Add 2 laws + inherit field ops
 $\oplus: V \times V \rightarrow V$
 $\odot: K \times V \rightarrow V$

Forget the origin $0 \in V$

Increase the dimension of space
 Add points at infinity

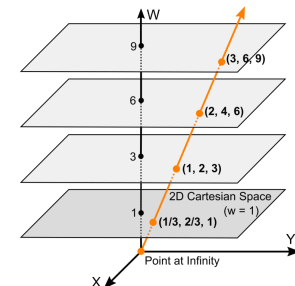
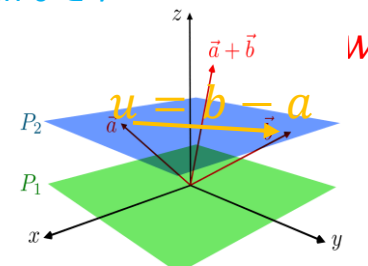
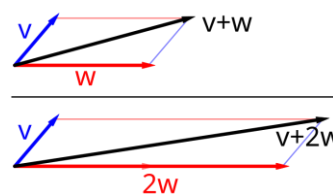


+	0	1	2	3
0	0	1	2	3
1	1	0	3	2
2	2	3	0	1
3	3	2	1	0

Addition Table

x	0	1	2	3
0	0	0	0	0
1	0	1	2	3
2	0	2	4	6
3	0	3	6	5

Multiplication Table



Sets

Groups $(G, \cdot)$

Field $(K, +, \cdot)$

Vector space $(V, \oplus, \odot)$

Affine spaces

Projective space

Elements = points

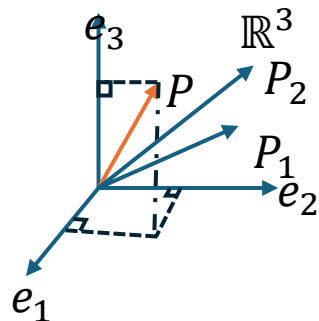
Add basis vectors
 Scalar product

Remove origin

$\mathbb{R}^3 \rightarrow \mathbb{RP}^3$

Euclidean Space $\mathbb{E}^3$
 A set of points

3D Space $(\mathbb{R}^3, +, \cdot)$



Dot product
 cross-product

$d(v_1, v_2)$

$\mu(v)$

$proj_Q(v)$

lines

Ops: point-line, line-line

Planes

Ops: line-plane, normal

Rigid bodies SE(3)

Ops: (R, t)

SE(3)
 Matrix form

We have only topology

We cannot measure

We are stuck to origin

We have no matrices

- The rotations in SE(3) are still a bit annoying, let's study them !

Problems with $R \in SO(3)$

- We defined the Special Orthogonal group $SO(3)$ as

$$SO(3) = \{R \in \mathbb{R}^{3 \times 3} \mid R^T R = I, \quad \det(R) = +1\}$$

- To rotate an object, we have special cases around the principal planes:

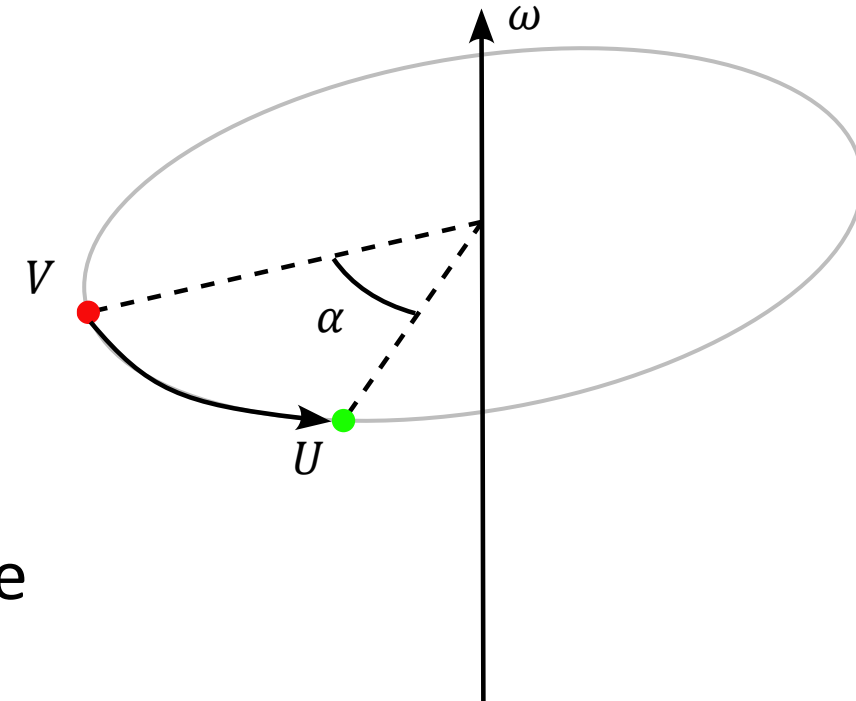
$$\bullet R_x = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha & \sin \alpha \\ 0 & -\sin \alpha & \cos \alpha \end{pmatrix}, R_y = \begin{pmatrix} \cos \beta & 0 & -\sin \beta \\ 0 & 1 & 0 \\ \sin \beta & 0 & \cos \beta \end{pmatrix}, R_z = \begin{pmatrix} \cos \gamma & \sin \gamma & 0 \\ -\sin \gamma & \cos \gamma & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- And the final rotation could be $R = R_z R_y R_x$
- However this is not practical ! (why?)
- Rotation matrices are non-commutatives, so the order of rotation matters
 - Convince yourself of that !

Rotation around an axis

- Instead, we would like to be able to rotate around an arbitrary axis
- Without taking into account the order of operations.

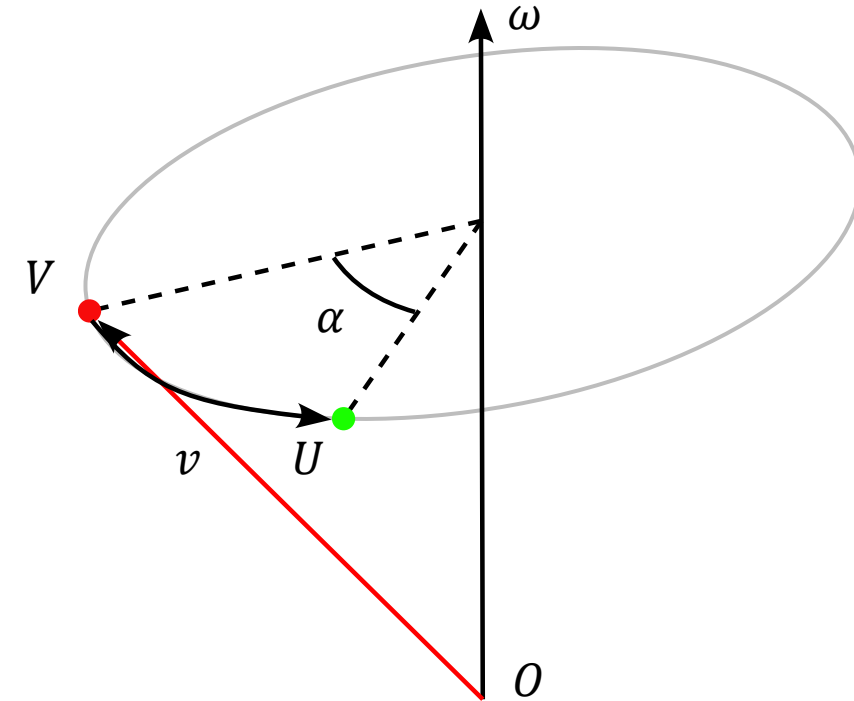
- Suppose a (normalized) axis ω and a point V that we rotate with an angle α



- We cannot directly rotate it, because it is not at the foot of ω
- How to find the corresponding rotation matrix R ?

Rodrigues Rotation Formula

- First of all, we define a vector v from the origin to the point V



Rodrigues Rotation Formula

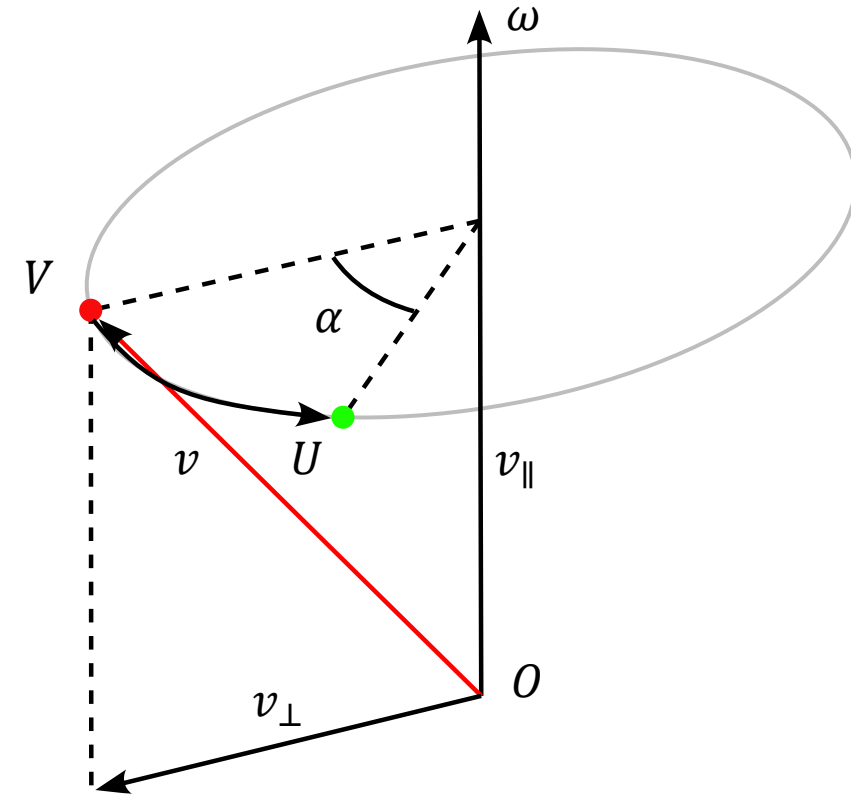
- Project v to ω :

$$v_{\parallel} = (\omega \cdot v)\omega$$

- Find the perpendicular vector

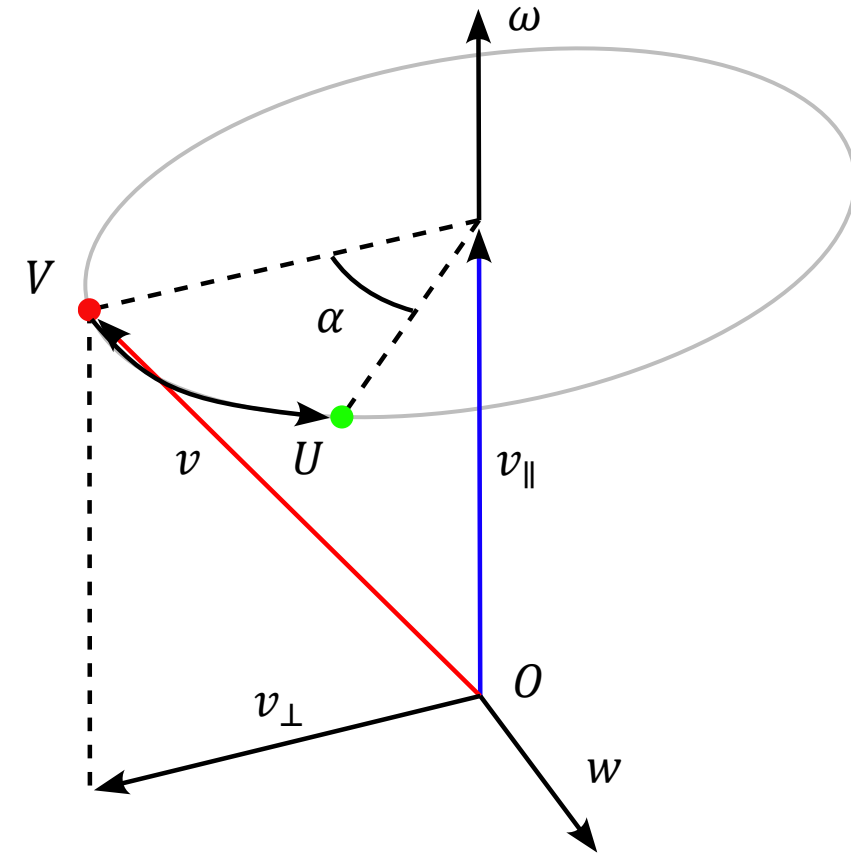
$$v_{\perp} = v - v_{\parallel}$$

- We have a plane π_{ω} with normal ω .
- To perform a 2D rotation in π_{ω} , we need a perpendicular axis to $v_{\perp}$.
- How to find it ?



Rodrigues Rotation Formula

- To perform a 2D rotation in the plane, we need a perpendicular axis to $v_{\perp}$.
- We can use the cross-product:
$$w = \omega \times v$$
- Now we can define the rotation of $v_{\perp}$ in π_{ω}
- How ?

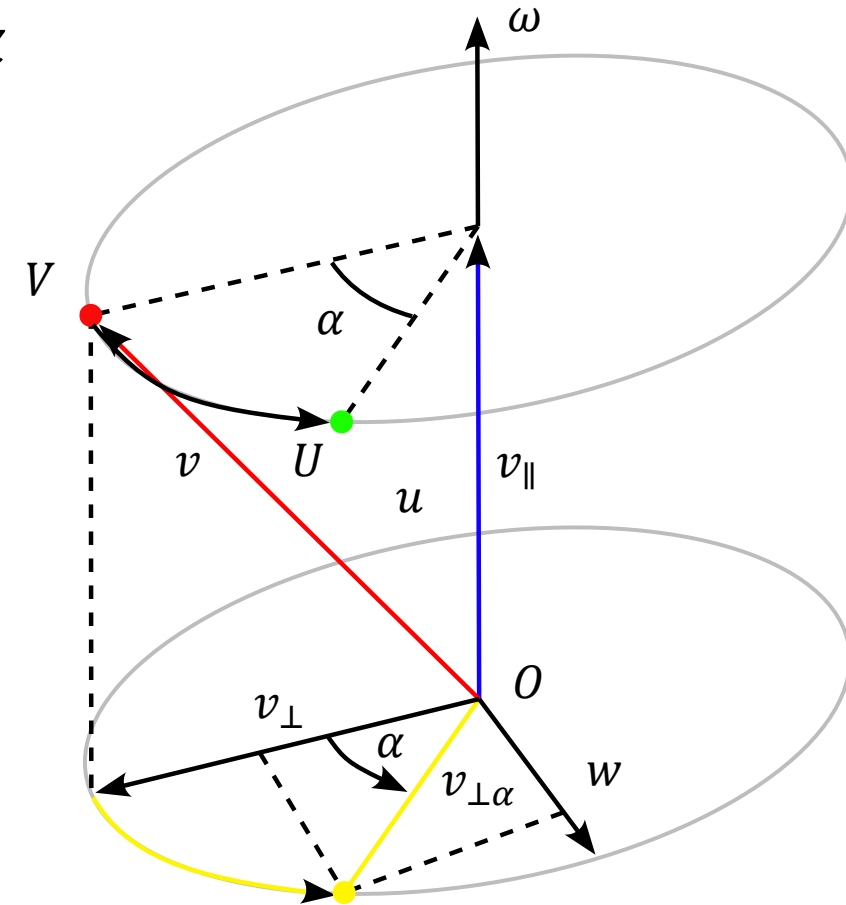


Rodrigues Rotation Formula

- Now we can define the rotation of $v_{\perp}$ in π_{ω}
- The rotation is

$$v_{\perp\alpha} = v_{\perp} \cos \alpha + w \sin \alpha$$

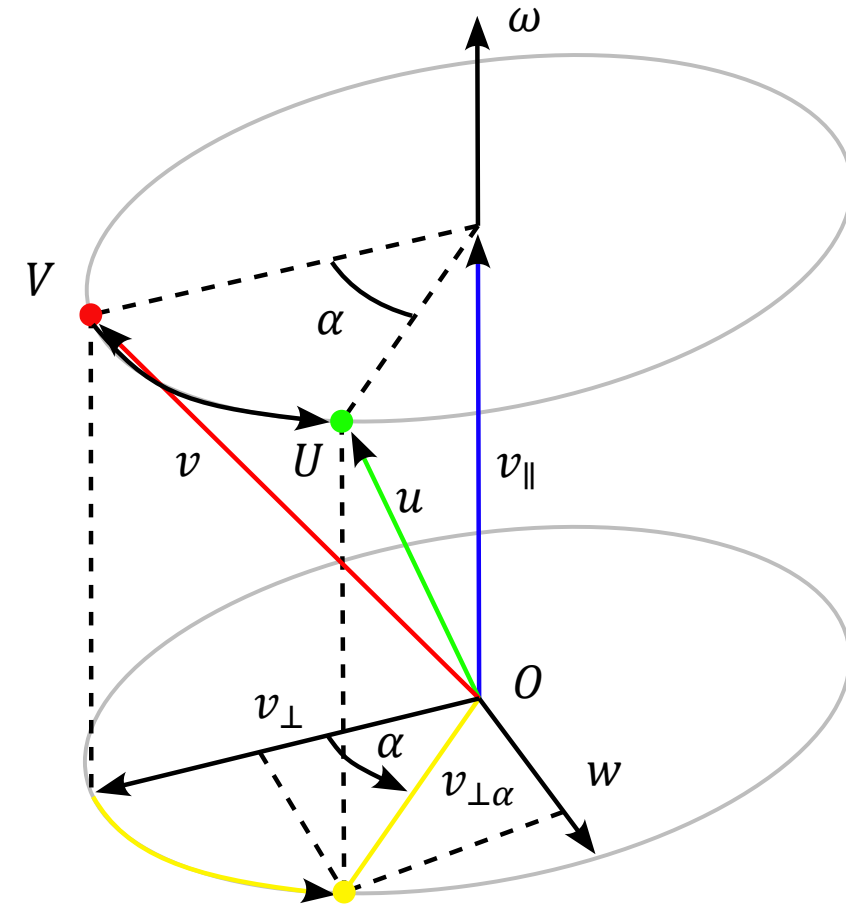
- We just need to get back to U



Rodrigues Rotation Formula

- We just need to get back to U
- Get back to the point U , using a vector

$$u = v_{\perp\alpha} + v_{\parallel}$$



Rodrigues Rotation Formula

- All the equations together:

- $v_{\parallel} = (\omega \cdot v)\omega$

- $v_{\perp} = v - v_{\parallel}$

- $w = \omega \times v$

- $v_{\perp\alpha} = v_{\perp} \cos \alpha + w \sin \alpha$

- $u = v_{\perp\alpha} + v_{\parallel}$

- If we substitute everything:

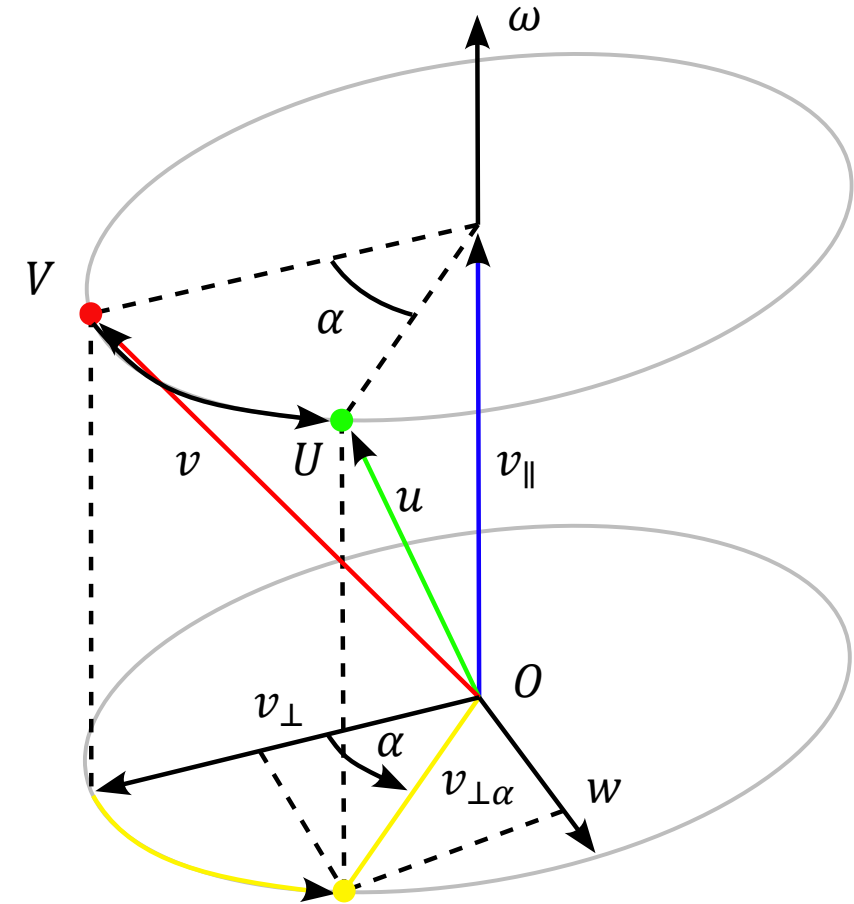
- $u = v_{\perp} \cos \alpha + w \sin \alpha + (\omega \cdot v)\omega$

- $u = (v - v_{\parallel}) \cos \alpha + (\omega \times v) \sin \alpha + (\omega \cdot v)\omega$

- $u = (v - (\omega \cdot v)\omega) \cos \alpha + (\omega \times v) \sin \alpha + (\omega \cdot v)\omega$

- $u = v \cos \alpha - (\omega \cdot v)\omega \cos \alpha + (\omega \times v) \sin \alpha + (\omega \cdot v)\omega$

- $u = v \cos \alpha + (\omega \cdot v)\omega (1 - \cos \alpha) + (\omega \times v) \sin \alpha$



Rodrigues Rotation Formula

- Now we have this:

$$u = v \cos \alpha + (\omega \cdot v) \omega (1 - \cos \alpha) + (\omega \times v) \sin \alpha$$

- We don't like dot and cross products in computer graphics:

- Convert the dot into a multiplication: $v_{\parallel} = (\omega \cdot v) \omega = \omega^T v \omega = \omega \omega^T v$

- Convert the cross-product into a multiplication: $w = \omega \times v = [\omega]_{\times} v$

$$u = v \cos \alpha + \omega \omega^T v (1 - \cos \alpha) + [\omega]_{\times} v \sin \alpha$$

- We still would like to extract a rotation matrix.

Rodrigues Rotation Formula

$$u = v \cos \alpha + \omega \omega^T v (1 - \cos \alpha) + [\omega]_{\times} v \sin \alpha$$

- To get the rotation matrix, we split this equation in the R and v components

$$u = (\cos \alpha + \omega \omega^T (1 - \cos \alpha) + [\omega]_{\times} \sin \alpha) v$$
$$u = R v$$

- So

$$R = I \cos \alpha + \omega \omega^T (1 - \cos \alpha) + [\omega]_{\times} \sin \alpha$$

- Can we still simplify ?

Rodrigues Rotation Formula

- Can we still simplify ?

$$R = I \cos \alpha + \omega^T \omega (1 - \cos \alpha) + [\omega]_{\times} \sin \alpha$$

- Yes! With a trick

- We have $\omega \omega^T = \begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix} (\omega_1 \quad \omega_2 \quad \omega_3) = \begin{pmatrix} \omega_1^2 & \omega_1 \omega_2 & \omega_1 \omega_3 \\ \omega_1 \omega_2 & \omega_2^2 & \omega_2 \omega_3 \\ \omega_1 \omega_3 & \omega_2 \omega_3 & \omega_3^2 \end{pmatrix}$

- And $[\omega]_{\times} = \begin{pmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{pmatrix} \rightarrow [\omega]_{\times}^2 = \begin{pmatrix} -\omega_3^2 - \omega_2^2 & \omega_1 \omega_2 & \omega_1 \omega_3 \\ \omega_1 \omega_2 & -\omega_3^2 - \omega_1^2 & \omega_2 \omega_3 \\ \omega_1 \omega_3 & \omega_2 \omega_3 & -\omega_2^2 - \omega_1^2 \end{pmatrix}$

- $\omega \omega^T$ and $[\omega]_{\times}^2$ look very similar !

- They differ only on their diagonals !

- How to make their diagonals better?

Rodrigues Rotation Formula

$$R = I \cos \alpha + \omega^T \omega (1 - \cos \alpha) + [\omega]_{\times} \sin \alpha$$

$$\omega \omega^T = \begin{pmatrix} \omega_1^2 & \omega_1 \omega_2 & \omega_1 \omega_3 \\ \omega_1 \omega_2 & \omega_2^2 & \omega_2 \omega_3 \\ \omega_1 \omega_3 & \omega_2 \omega_3 & \omega_3^2 \end{pmatrix}$$

$$[\omega]_{\times}^2 = \begin{pmatrix} -\omega_3^2 - \omega_2^2 & \omega_1 \omega_2 & \omega_1 \omega_3 \\ \omega_1 \omega_2 & -\omega_3^2 - \omega_1^2 & \omega_2 \omega_3 \\ \omega_1 \omega_3 & \omega_2 \omega_3 & -\omega_2^2 - \omega_1^2 \end{pmatrix}$$

- Diagonals?
- ω has unit length ! So $\omega_1^2 + \omega_2^2 + \omega_3^2 = 1 \Rightarrow \omega_1^2 = -\omega_2^2 - \omega_3^2 + 1$, so:

$$\omega \omega^T = \begin{pmatrix} \omega_1^2 & \omega_1 \omega_2 & \omega_1 \omega_3 \\ \omega_1 \omega_2 & \omega_2^2 & \omega_2 \omega_3 \\ \omega_1 \omega_3 & \omega_2 \omega_3 & \omega_3^2 \end{pmatrix} = \begin{pmatrix} -\omega_3^2 - \omega_2^2 & \omega_1 \omega_2 & \omega_1 \omega_3 \\ \omega_1 \omega_2 & -\omega_3^2 - \omega_1^2 & \omega_2 \omega_3 \\ \omega_1 \omega_3 & \omega_2 \omega_3 & -\omega_2^2 - \omega_1^2 \end{pmatrix} + I$$

Rodrigues Rotation Formula

$$R = I \cos \alpha + \omega^T \omega (1 - \cos \alpha) + [\omega]_{\times} \sin \alpha$$

$$\omega \omega^T = \begin{pmatrix} \omega_1^2 & \omega_1 \omega_2 & \omega_1 \omega_3 \\ \omega_1 \omega_2 & \omega_2^2 & \omega_2 \omega_3 \\ \omega_1 \omega_3 & \omega_2 \omega_3 & \omega_3^2 \end{pmatrix} = \begin{pmatrix} -\omega_3^2 - \omega_2^2 & \omega_1 \omega_2 & \omega_1 \omega_3 \\ \omega_1 \omega_2 & -\omega_3^2 - \omega_1^2 & \omega_2 \omega_3 \\ \omega_1 \omega_3 & \omega_2 \omega_3 & -\omega_2^2 - \omega_1^2 \end{pmatrix} + I$$

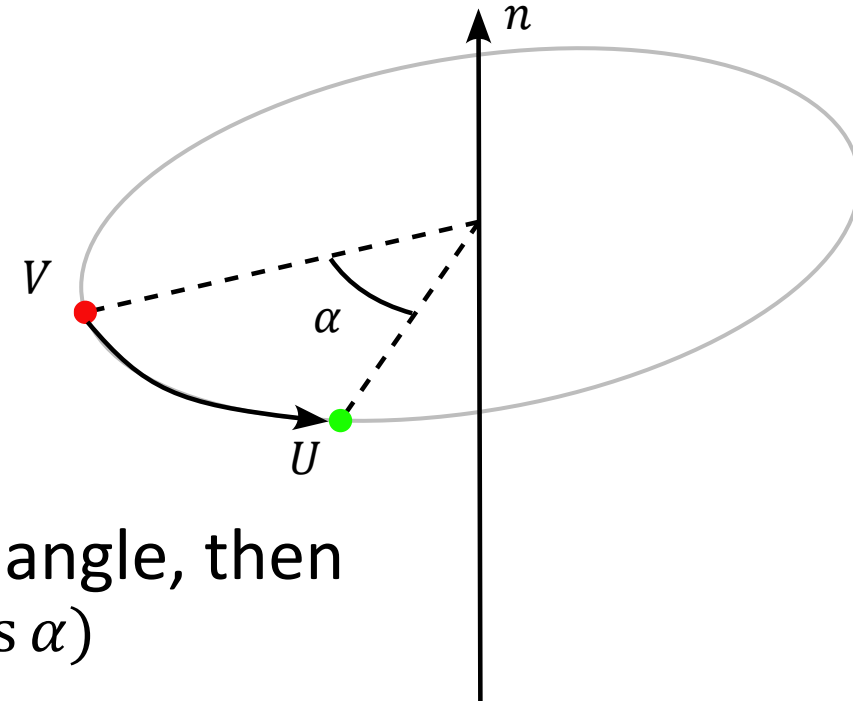
- This is $\omega \omega^T = [\omega]_{\times}^2 + I$
- Using this identity, we finally obtain:
 - $R = I \cos \alpha + \omega^T \omega (1 - \cos \alpha) + [\omega]_{\times} \sin \alpha$
 - $R = I \cos \alpha + ([\omega]_{\times}^2 + I) (1 - \cos \alpha) + [\omega]_{\times} \sin \alpha$
 - $R = I \cos \alpha + [\omega]_{\times}^2 - [\omega]_{\times}^2 \cos \alpha + I - I \cos \alpha + [\omega]_{\times} \sin \alpha$
 - $R = I + [\omega]_{\times}^2 - [\omega]_{\times}^2 \cos \alpha + [\omega]_{\times} \sin \alpha$
 - $R = I + [\omega]_{\times} \sin \alpha + [\omega]_{\times}^2 (1 - \cos \alpha)$

Rotation around an axis

- This matrix

$$R = I + [\omega]_{\times} \sin \alpha + [\omega]_{\times}^2 (1 - \cos \alpha)$$

- Is the Rodrigues rotation matrix, it allows us to rotate a point around an arbitrary axis !



- **Definition: Rodrigues rotation formula,**
let ω be a unitary rotation axis, and α the rotation angle, then

$$R = I + [\omega]_{\times} \sin \alpha + [\omega]_{\times}^2 (1 - \cos \alpha)$$

Other ways of rotating?

- We found several ways of obtaining $R \in SO(3)$:

- Rotation in each plane and composing

$$\bullet R_x = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha & \sin \alpha \\ 0 & -\sin \alpha & \cos \alpha \end{pmatrix}, R_y = \begin{pmatrix} \cos \alpha & 0 & -\sin \alpha \\ 0 & 1 & 0 \\ \sin \alpha & 0 & \cos \alpha \end{pmatrix}, R_z = \begin{pmatrix} \cos \alpha & \sin \alpha & 0 \\ -\sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- Rotating around an arbitrary axis

- $R = I + [\omega]_{\times} \sin \alpha + [\omega]_{\times}^2 (1 - \cos \alpha)$

- Are there other ways ?

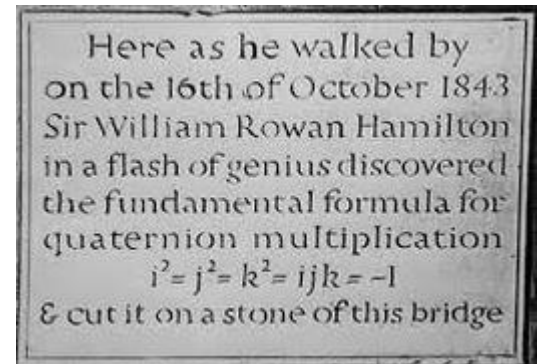
Complex Numbers $\mathbb{C}$

- A complex number is a number of the form $z = \Re(z) + \Im(z)i$
 - Where $\Re(z)$ is the real part of the number and $\Im(z)$ the imaginary part.
 - i has the property that $i^2 = -1$
- In 2D, complex numbers can be used to represent rotations
 - Multiplying by i corresponds to a 90 degrees anti-clockwise rotation.
- We want to generalize that to the 3D space.

Quaternions $\mathbb{Q}_8/\mathbb{H}$

- Quaternions were discovered by William Rowan Hamilton in 1843 as an extension of complex numbers into higher dimensions
- They possess more imaginary units: i, j, k with those rules:
 - $i^2 = j^2 = k^2 = ijk = -1$
- They provide a robust and compact way to represent 3D rotations
- They are represented as $q = a + bi + cj + dk$, $\forall a, b, c, d \in \mathbb{R}$
- They are often written as $q = [a, v]$ where $v = (b, c, d)$
- Using the rule we can derive that

$$\begin{array}{lll} i^2 = -1 & ij = k & ji = -k \\ j^2 = -1 & jk = i & kj = -i \\ k^2 = -1 & ki = j & ik = -j \end{array}$$



Dublin bridge

Quaternions $\mathbb{Q}_8/\mathbb{H}$ - Properties

$$\begin{array}{lll} i^2 = -1 & ij = k & ji = -k \\ j^2 = -1 & jk = i & kj = -i \\ k^2 = -1 & ki = j & ik = -j \end{array}$$

- Let $q = a + bi + cj + dk$
- The norm is defined as $|q| = \sqrt{a^2 + b^2 + c^2 + d^2}$
 - Unity quaternions $|q| = 1$ are essential for rotations
- We define its conjugate as $\bar{q} = a - bi - cj - dk$
- With the properties:
 - $q\bar{q} = \bar{q}q = q \cdot q = ||q||^2 \in \mathbb{R}$
 - $q^{-1} = \frac{\bar{q}}{||q||^2}$
 - Proof: $qq^{-1} = \frac{q\bar{q}}{||q||^2} = \frac{||q||^2}{||q||^2} = 1$

Quaternions $\mathbb{Q}_8/\mathbb{H}$ - Commutativity

- The multiplication of quaternions is **not** commutative !
 - Let $q_1 = a_1 + b_1i + c_1j + d_1k$, $q_2 = a_2 + b_2i + c_2j + d_2k$
 - $q_1q_2 = a_1a_2 + a_1b_2i + a_1c_2j + a_1d_2k + a_2b_1i + b_1b_2i^2 + b_1c_2ij + b_1d_2ik + a_2c_1j + b_2c_1ji + c_1c_2j^2 + c_1d_2jk + a_2d_1k + b_2d_1ki + c_2d_1kj + d_1d_2k^2$
- $$= a_1a_2 + a_1b_2i + a_1c_2j + a_1d_2k + a_2b_1i - b_1b_2 + b_1c_2k - b_1d_2j + a_2c_1j - b_2c_1k - c_1c_2 + c_1d_2i + a_2d_1k + b_2d_1j - c_2d_1i - d_1d_2$$
- $$= (a_1a_2 - b_1b_2 - c_1c_2 - d_1d_2) + (a_1b_2 + a_2b_1 + c_1d_2 - c_2d_1)i + (a_1c_2 - b_1d_2 + a_2c_1 + b_2d_1)j + (a_1d_2 + b_1c_2 - b_2c_1 + a_2d_1)k$$
- Test yourself with q_2q_1 and compare !

$i^2 = -1$	$ij = k$	$ji = -k$
$j^2 = -1$	$jk = i$	$kj = -i$
$k^2 = -1$	$ki = j$	$ik = -j$

Quaternions $\mathbb{Q}_8/\mathbb{H}$ and 3D rotations

- Rotating a vector: A vector v can be rotated using a unit quaternion q as:

$$v' = qvq^{-1}$$

Sandwich operation

- Where v is treated as a pure quaternion (i.e., $v = [0, v]$) Why ?
- Intuition: Quaternions live in 4D space, but their rotation effect happens in 3D. The quaternion axis and angle determine the rotation.
- **Relation to Axis-Angle Representation:** A quaternion can be derived from an axis n and angle θ as:

$$q = \cos \frac{\theta}{2} + \sin \frac{\theta}{2} (nbi + ncj + ndk)$$

Benefits of using quaternions

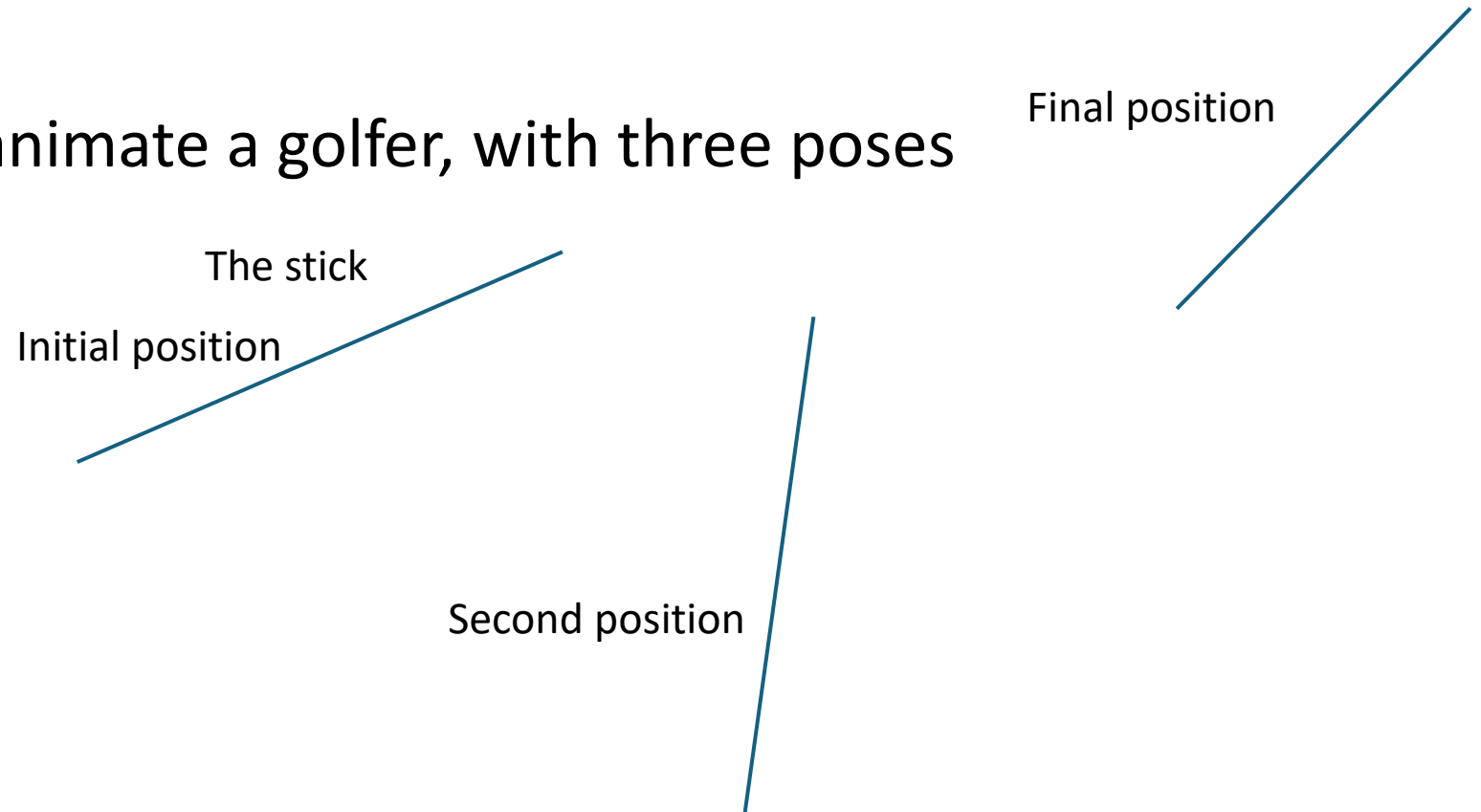
- There are several benefits of using quaternions instead of Euler angles or rotation matrices:
 - **Avoid the Gimbal lock** of Euler rotations, they maintain smooth rotations without singularities.
 - **Efficient**: Quaternion multiplication is cheaper than matrix multiplication!
Less elements that can drift, more compact representation.

When to use Rodrigues, quaternions, etc.. ?

- An issue we didn't raise previously:
 - Interpolating Euler angles, Rotation matrices, Rodriguez, etc.. Give **incorrect** rotation interpolations !
- Why ?

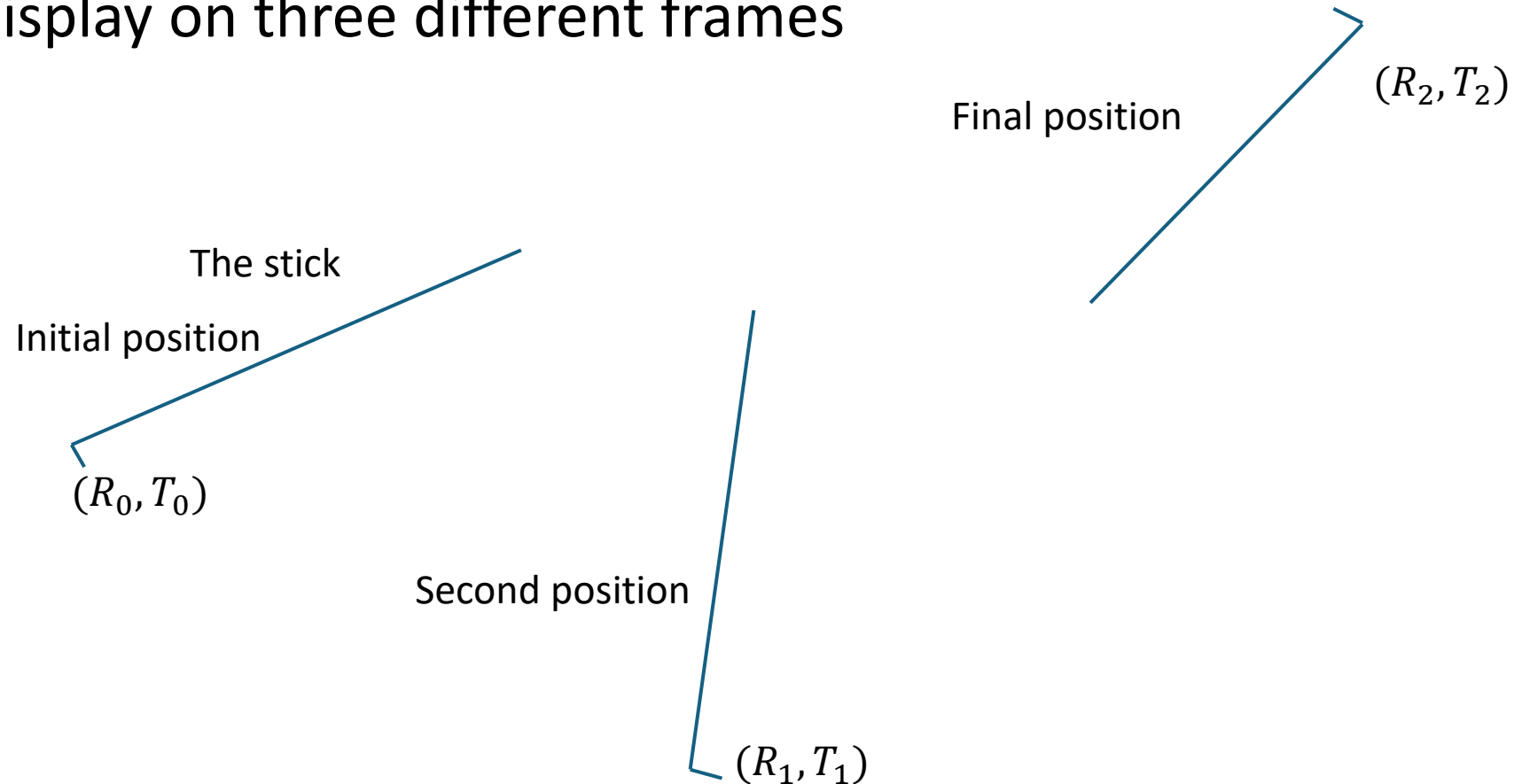
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- Suppose you want to animate a golfer, with three poses



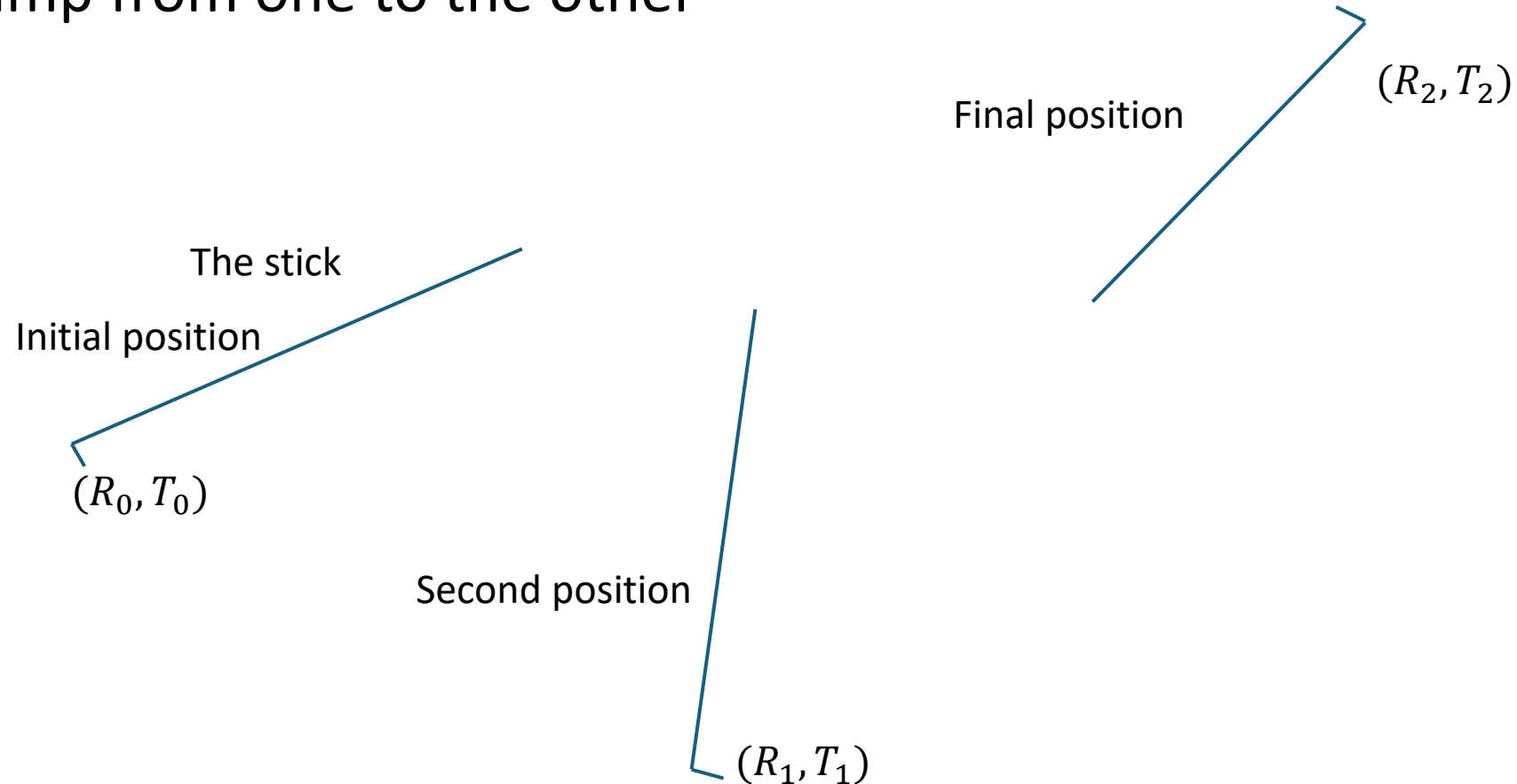
When to use Rodrigues, quaternions, etc.. ?

- Suppose you want to animate a golfer, with three poses
- you have rigid body transformations at the three positions
- What happens if you display on three different frames the stick position ?



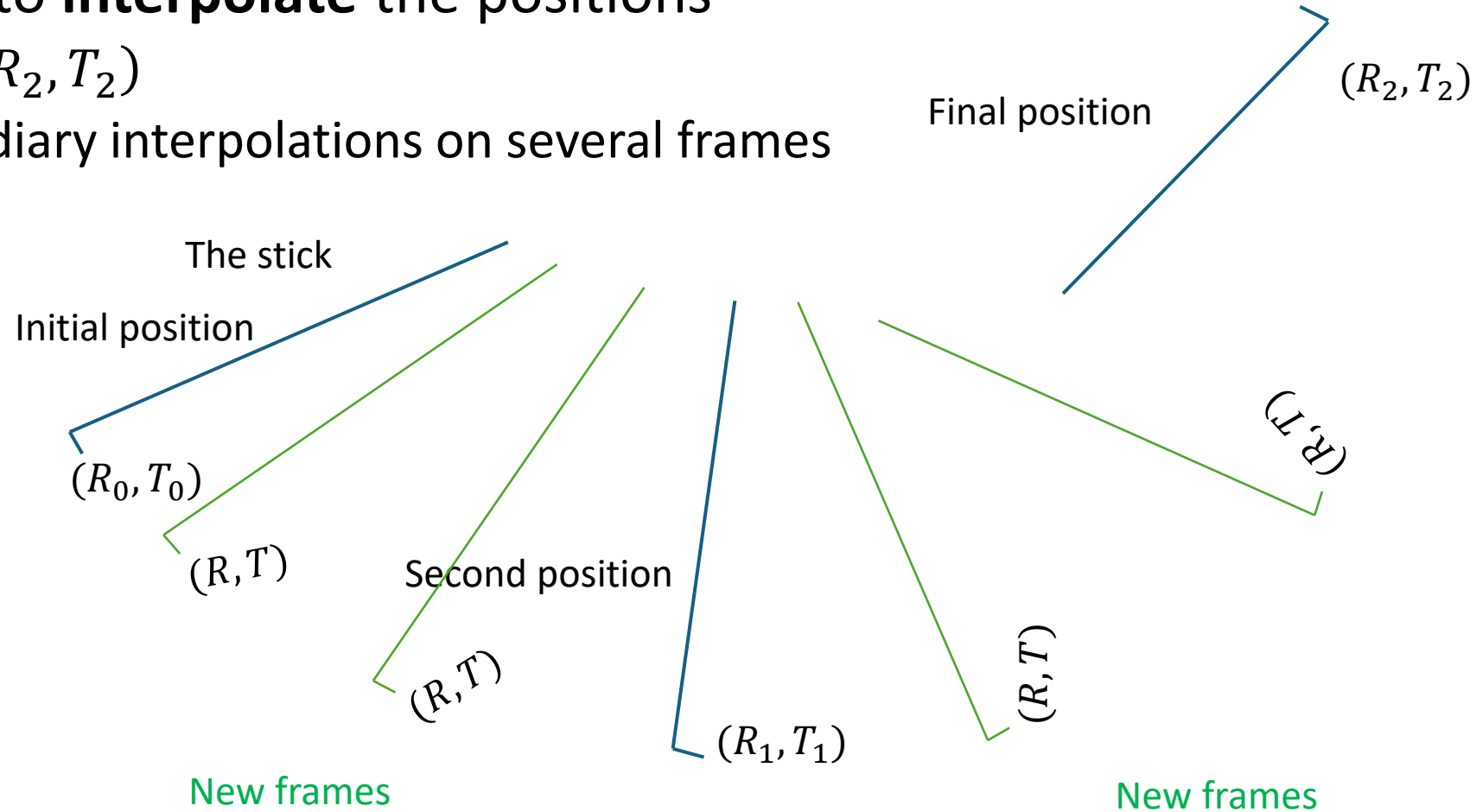
When to use Rodrigues, quaternions, etc.. ?

- What happens if you display on three different frames the stick position ?
- We will see the stick jump from one to the other
 - Not a nice animation



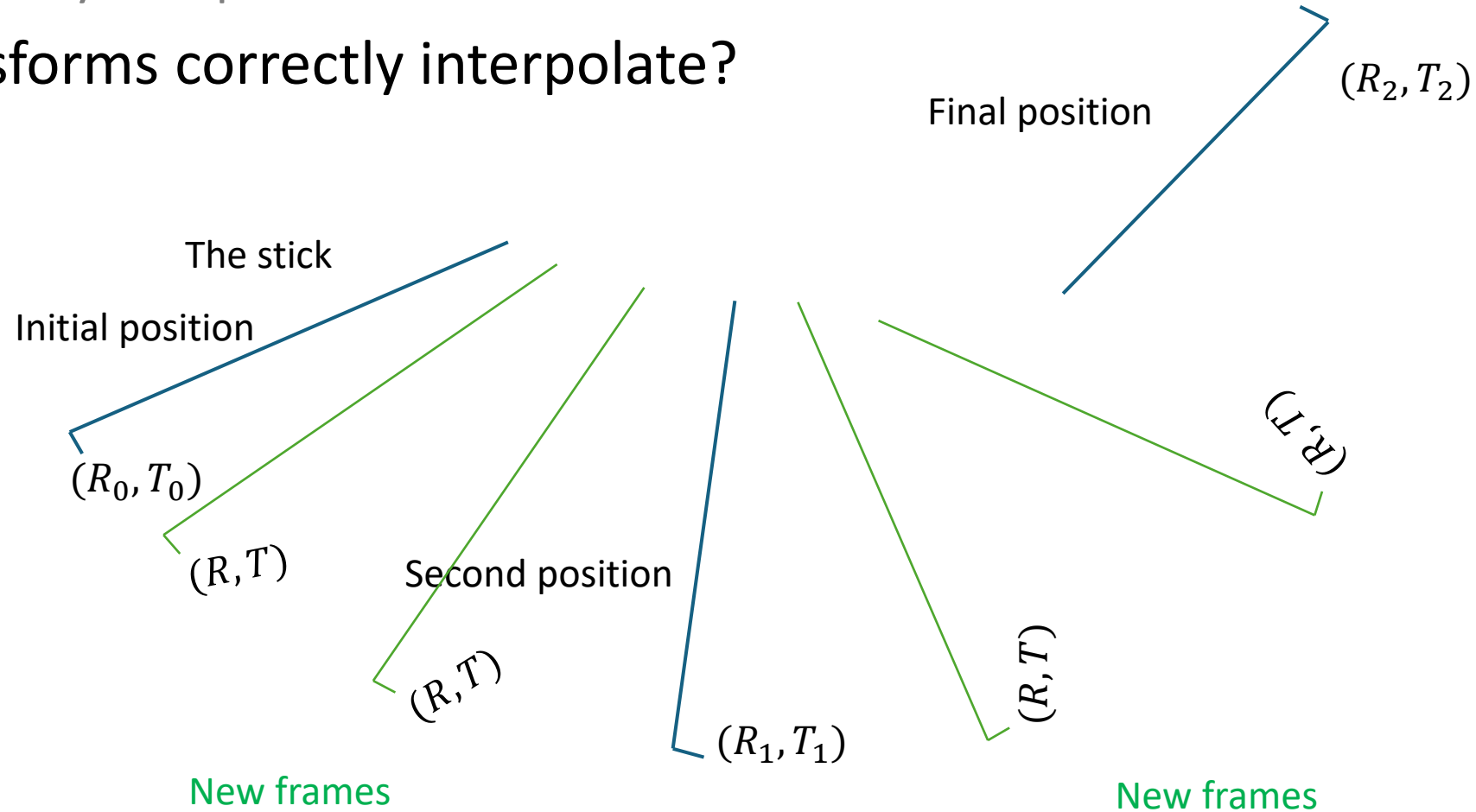
When to use Rodrigues, quaternions, etc.. ?

- We will see the stick jump from one to the other
 - Not a nice animation
- Instead we would like to **interpolate** the positions
 - $(R_0, T_0) \mapsto (R_1, T_1) \mapsto (R_2, T_2)$
 - And spread the intermediary interpolations on several frames



When to use Rodrigues, quaternions, etc.. ?

- Instead we would like to **interpolate** the positions
 - $(R_0, T_0) \mapsto (R_1, T_1) \mapsto (R_2, T_2)$
 - And spread the intermediary interpolations on several frames
- But do rigid body transforms correctly interpolate?



When to use Rodrigues, quaternions, etc.. ?

- But do rigid body transforms correctly interpolate?
- Let's try for the translation
 - $T_0 = (2, 4, 6)^T, T_1 = (8, 10, 12)^T$
 - $T = tT_0 + (1 - t)T_1 \quad t \in [0, 1]$

When to use Rodrigues, quaternions, etc.. ?

- But do rigid body transforms correctly interpolate?
- Let's try for the translation
 - $T_0 = (2, 4, 6)^T, T_1 = (8, 10, 12)^T$
 - $T = tT_0 + (1 - t)T_1 \quad t \in [0, 1]$
 - $t = \frac{1}{2} \Rightarrow T = \frac{1}{2}(2, 4, 6)^T + \frac{1}{2}(8, 10, 12)^T = (1, 2, 3) + (4, 5, 6) = (5, 7, 9)$
- This seems to work okay, 5 is in the middle of $[2, 8]$, 7 in the middle of $[4, 10]$ and 9 in the middle of $[6, 12]$
- But what about rotation?

When to use Rodrigues, quaternions, etc.. ?

- But what about rotation?
- Suppose you have two matrices

$$R_0 = \begin{pmatrix} r_{11} & r_{12} \\ r_{21} & r_{22} \end{pmatrix}, \quad R_1 = \begin{pmatrix} r'_{11} & r'_{12} \\ r'_{21} & r'_{22} \end{pmatrix}$$

Such that $R_0 R_0^T = I$ and $\det(R_0) = \det(R_1) = +1$ (rotations)

- What can we try ?

When to use Rodrigues, quaternions, etc.. ?

- But what about rotation?

- Suppose you have two matrices

$$R_0 = \begin{pmatrix} r_{11} & r_{12} \\ r_{21} & r_{22} \end{pmatrix}, \quad R_1 = \begin{pmatrix} r'_{11} & r'_{12} \\ r'_{21} & r'_{22} \end{pmatrix}$$

Such that $R_0 R_0^T = I$ and $\det(R_0) = \det(R_1) = +1$ (rotations)

- What can we try ?

- Let's interpolate the values, take as an example r_{11}

$$r_{11}^* = r_{11}t + r'_{11}(1 - t), \quad t \in [0, 1]$$

- Our new matrix is of the form

$$R^* = \begin{pmatrix} r_{11}t + r'_{11}(1 - t) & r_{12}t + r'_{12}(1 - t) \\ r_{21}t + r'_{21}(1 - t) & r_{22}t + r'_{22}(1 - t) \end{pmatrix}$$

- Is this still a rotation?

When to use Rodrigues, quaternions, etc.. ?

$$R^* = \begin{pmatrix} r_{11}t + r'_{11}(1-t) & r_{12}t + r'_{12}(1-t) \\ r_{21}t + r'_{21}(1-t) & r_{22}t + r'_{22}(1-t) \end{pmatrix}$$

- Is this still a rotation?

- Let's check

$$\begin{aligned} R^* R^{*T} &= \begin{pmatrix} r_{11}t + r'_{11}(1-t) & r_{12}t + r'_{12}(1-t) \\ r_{21}t + r'_{21}(1-t) & r_{22}t + r'_{22}(1-t) \end{pmatrix} \begin{pmatrix} r_{11}t + r'_{11}(1-t) & r_{21}t + r'_{21}(1-t) \\ r_{12}t + r'_{12}(1-t) & r_{22}t + r'_{22}(1-t) \end{pmatrix} \\ &= \begin{pmatrix} r_{11}^2 t^2 + \dots & \dots \\ \dots & \dots \end{pmatrix} \end{aligned}$$

- We can directly stop here !
- We see that it will not resemble to an identity matrix !
- Our linear interpolation is not anymore a rotation matrix ☹
 - Rodrigues uses matrices too ☹

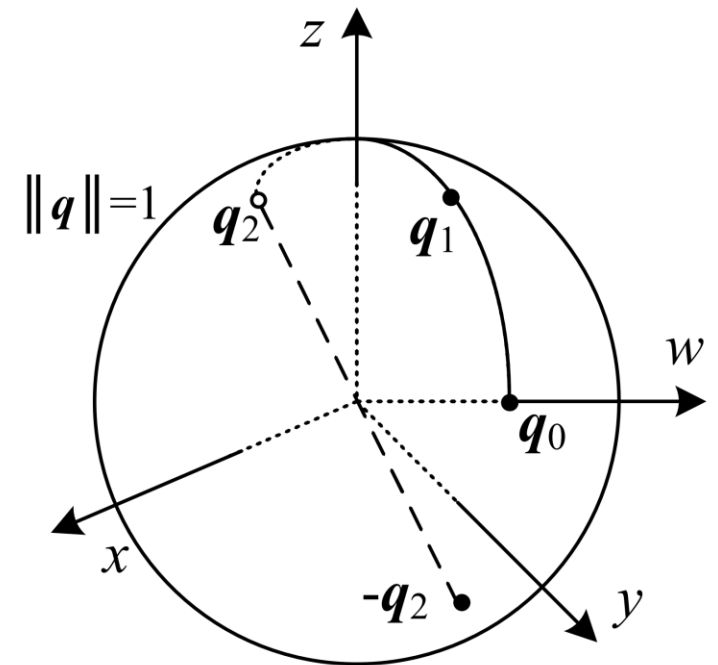
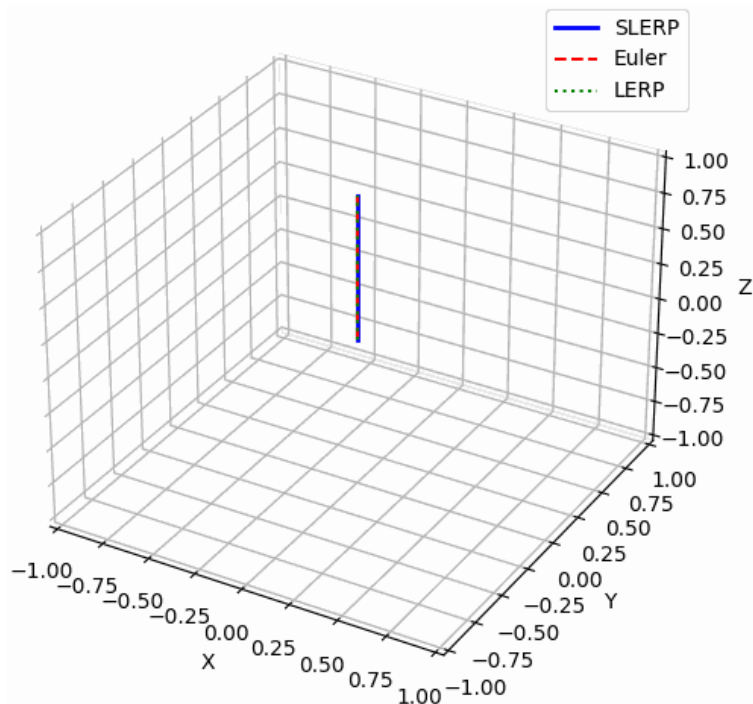
When to use Rodrigues, quaternions, etc.. ?

- An issue we didn't raise previously:
 - Interpolating Euler angles, Rotation matrices, Rodriguez, etc.. Give **incorrect** rotation interpolations !
- But in computer graphics, we need to interpolate between rotations a lot in animations !
- Quaternions interpolate correctly !
 - It is called the **spherical linear interpolation (SLERP)**

$$\text{slerp}(q_0, q_1, t) = q_0 (q_0^{-1} q_1)^t$$

When to use Rodrigues, quaternions, etc.. ?

- An issue we didn't raise previously:
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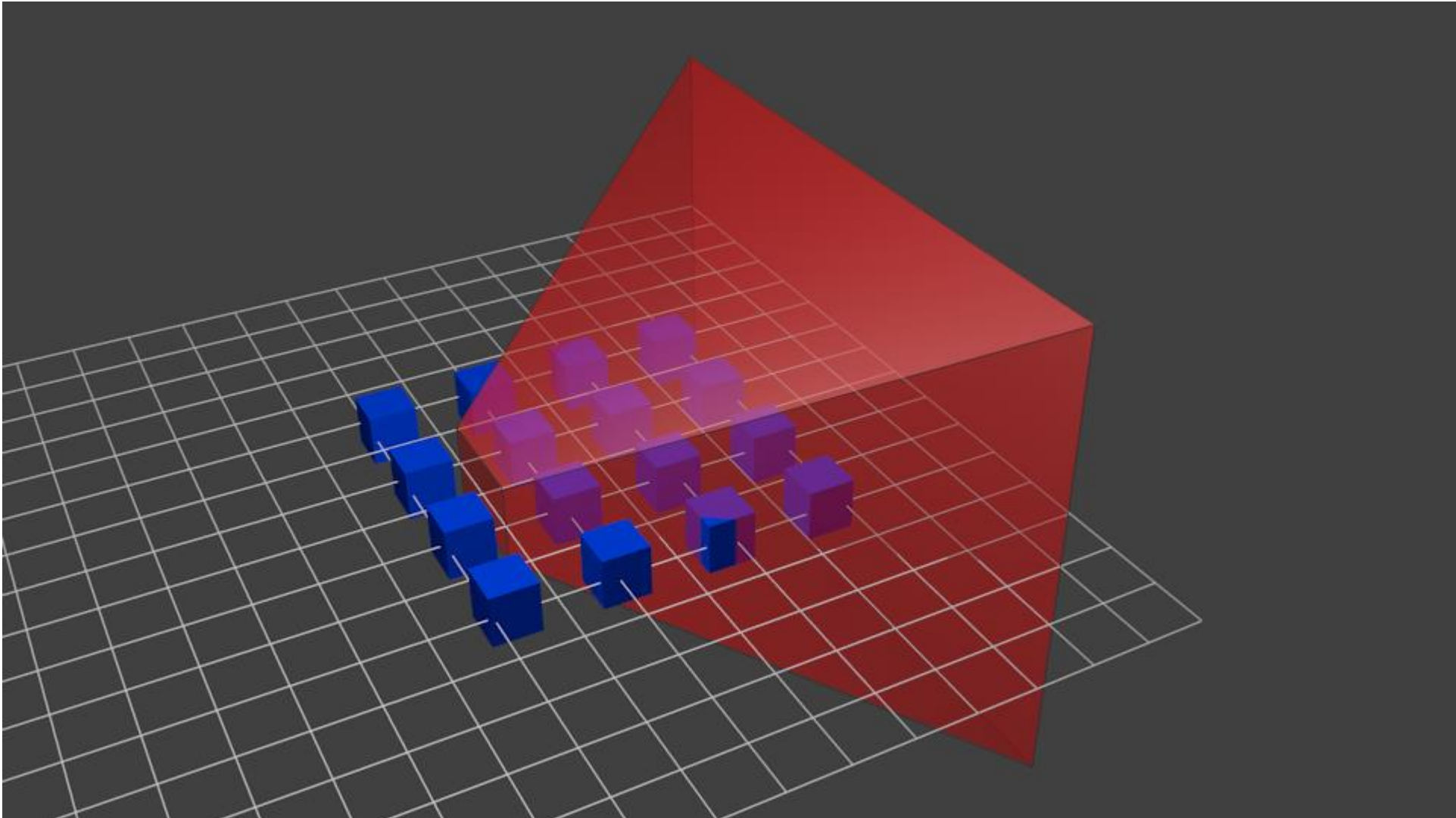


What I Expect for the exam

- After this lesson I expect that:
 - You have an intuitive feeling of projective geometry
 - You have an understanding of the rotations in computer graphics
 - And their links
 - Know when to apply which rotation
 - Know the proof for Rodrigues
- I DO NOT EXPECT
 - That you are able to re-derive all the formulas for quaternions, but still you should be able to use them and have an intuition for them

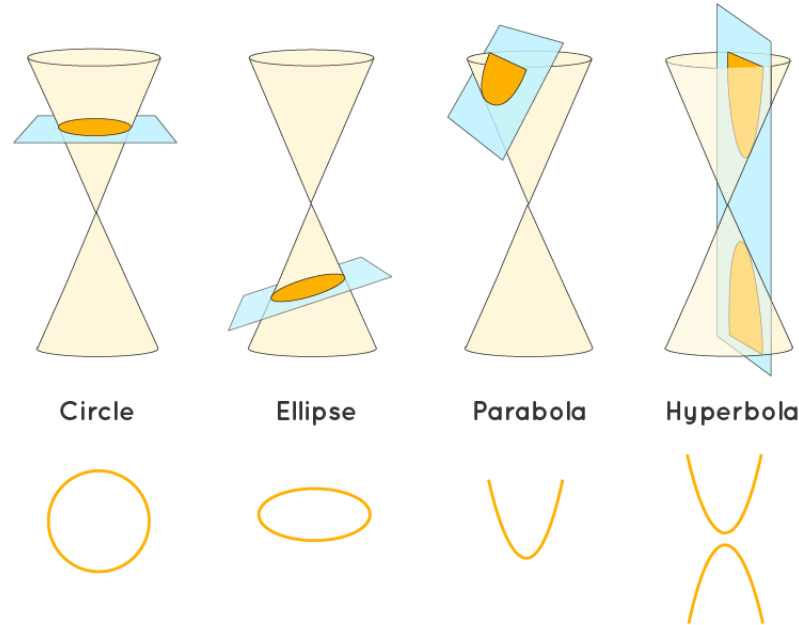
Next time

- We finally apply all this theory to create the perspective matrix !



Appendix: $\mathbb{R}P^2$ – Conics

- Conics (hyperbola, ellipse, parabola, degenerate cases) are conic sections by a plane



- Their equation in inhomogeneous coordinates is
$$ax^2 + bxy + cy^2 + dx + ey + f = 0$$
- We can homogenize it by replacing $x \mapsto x_1/x_3$ and $y \mapsto x_2/x_3$
$$ax_1^2 + bx_1x_2 + cx_2^2 + dx_1x_3 + ex_2 + fx_3^2 = 0$$

Appendix: $\mathbb{R}P^2$ – Conics

- We can homogenize it by replacing $x \mapsto x_1/x_3$ and $y \mapsto x_2/x_3$
 $ax_1^2 + bx_1x_2 + cx_2^2 + dx_1x_3 + ex_2 + fx_3^2 = 0$

- Or in matrix form: $x^T C x = 0$

$$C = \begin{pmatrix} a & b/2 & d/2 \\ b/2 & c & e/2 \\ d/2 & e/2 & f \end{pmatrix}$$

- The conic has 6 parameters $c = (a, b, c, d, e, f)^T$

- Note that this matrix is symmetric: $C^T = C$

Appendix: $\mathbb{R}P^2$ – Conics

- Suppose we have 5 points $p_i = (x_i, y_i)$ and we want to find the conic which passes through them.

- This gives us 5 constraints as we know that the points must be on the conic

$$\forall p_i: (ax_i^2 + bx_iy_i + cy_i^2 + dx_i + ey_i + f)c = 0$$

- Stacking them together we have

$$\begin{pmatrix} x_1^2 & x_1y_1 & y_1^2 & x_1 & y_1 & 1 \\ x_2^2 & x_2y_2 & y_2^2 & x_2 & y_2 & 1 \\ x_3^2 & x_3y_3 & y_3^2 & x_3 & y_3 & 1 \\ x_4^2 & x_4y_4 & y_4^2 & x_4 & y_4 & 1 \\ x_5^2 & x_5y_5 & y_5^2 & x_5 & y_5 & 1 \end{pmatrix} c = 0$$

- We can solve this system to find the coefficients of our conic c .

Appendix: $\mathbb{R}P^2$ – Conics

- A line l tangent to a conic c at a point p has a nice form:

$$l = Cp$$

- Proof:

- The line $l = Cp$ passes through p as $l^T p = (Cp)^T p = p^T C^T p = p^T Cp = 0$

- If l has a single point of contact with the conic, it is tangent to it, and we are done.

- If l meet the conic again at a point p' we have

- $p'^T Cp' = 0$ (p' is on the conic)

- $p^T Cp' = l^T p' = 0$ (the point p' is on the line $l = Cp$)

- As the points lie on the same line, we can interpolate between them $p + \alpha p'$ and stay on the line.

- If we evaluate all intermediary points between p and p' we have

- $(p + \alpha p')^T C(p + \alpha p')$

- $= p^T Cp + p^T C\alpha p' + \alpha p'^T Cp + \alpha p'^T C\alpha p$

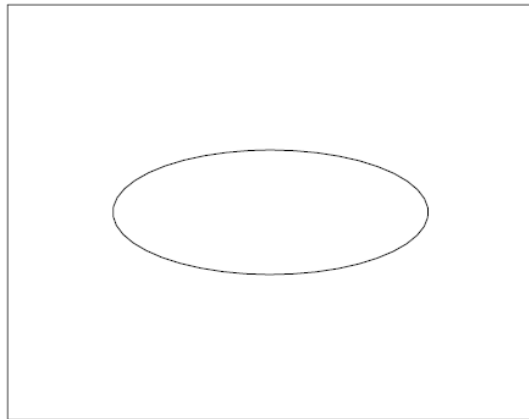
- $= p^T Cp + \alpha p^T Cp' + \alpha p'^T Cp + \alpha^2 p'^T Cp'$

- $= 0 + 0 + 0 + 0 = 0$

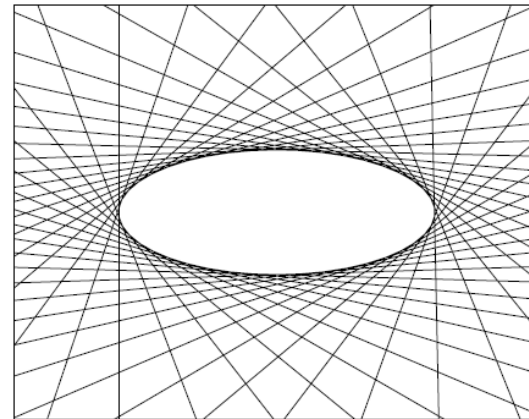
- So all the points of the line lie on the conic! We have a degenerated conic.

Appendix: $\mathbb{R}P^2$ – Conics

- The conics as defined are **point** conics, they determine the coordinates of points on the plane.
- By the duality principle, there exists **line** conics ! (or **dual** conics)
- Let C^* be the matrix for the line conic
 - To find it we can use $l = Cp \Leftrightarrow p = C^{-1}l \Rightarrow (C^{-1}l)^T C (C^{-1}l) = l^T C^{-1}l = 0$
 - So $C^* = C^{-1}$ (actually it is the adjoint matrix, but for non-singular matrices $C^* = C^{-1}$)
- A line l tangent to the conic C^* satisfies $l^T C^* l = 0$
- Dual conics are also known as conic envelopes



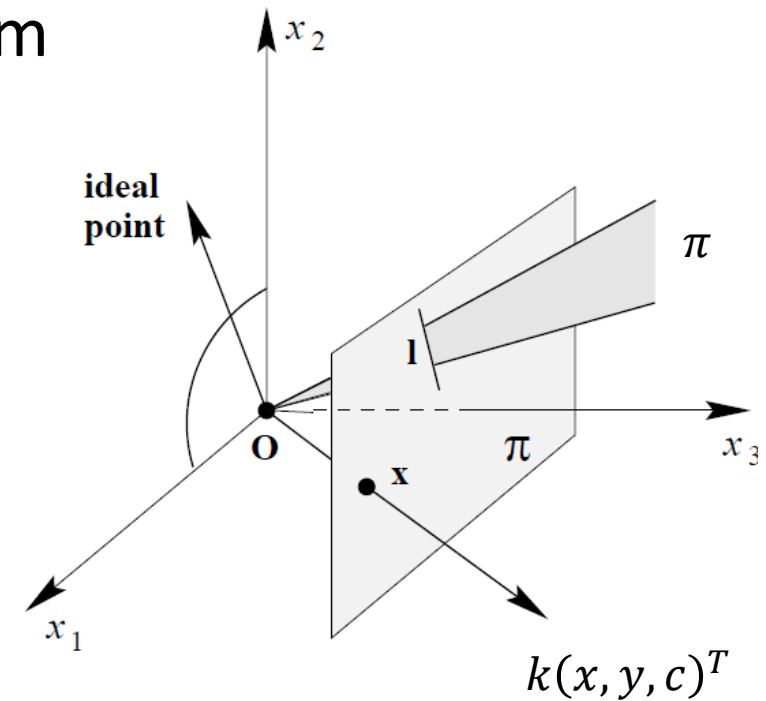
a



b

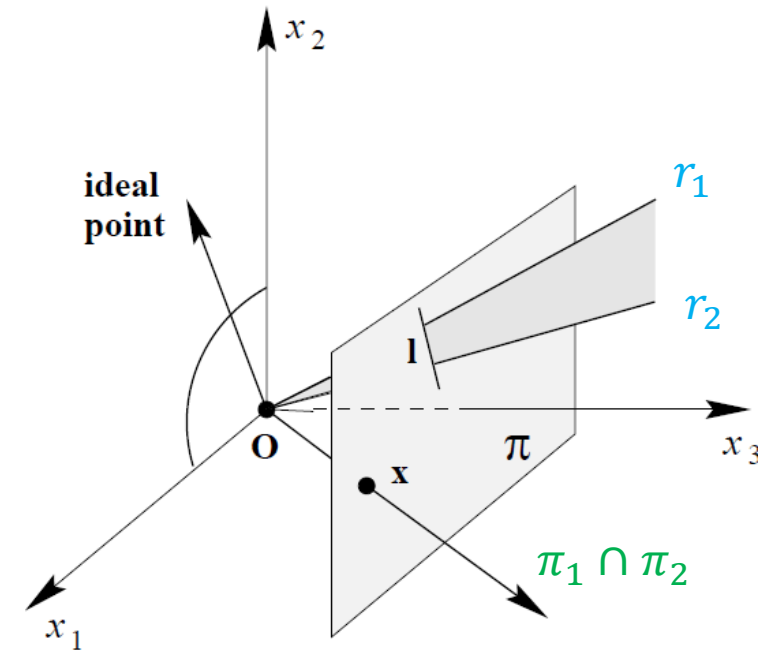
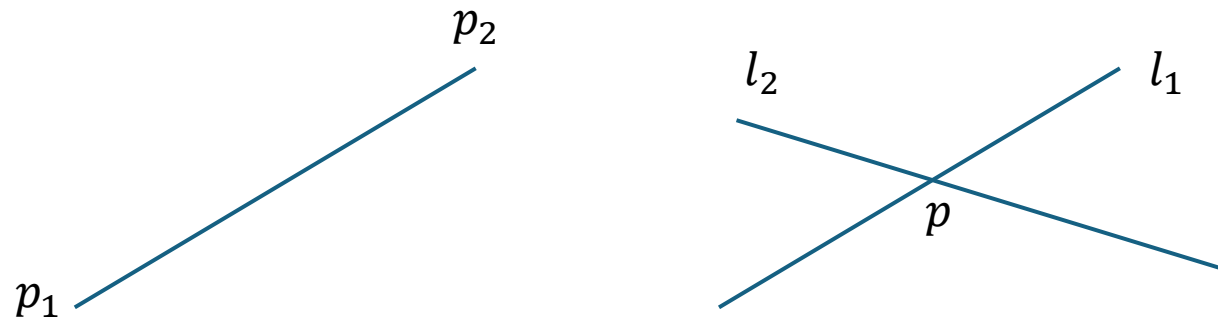
Appendix: $\mathbb{R}P^2$ – Model

- A useful way to think about $\mathbb{R}P^2$ is as a set of rays in $\mathbb{R}^3$
- The set of all vectors $k(x, y, c)^T \in \mathbb{R}^3$ as k varies form a ray through the origin.
 - This ray can be represented as a single point in $\mathbb{R}P^2$
- Similarly, planes passing through the origin in $\mathbb{R}^3$ represent lines in $\mathbb{R}P^2$



Appendix: $\mathbb{R}P^2$ – Model

- A useful way to think about $\mathbb{R}P^2$ is as a set of rays through the origin in $\mathbb{R}^3$
- Note that:
 - Two non-identical rays lie on exactly one plane
 - Two planes intersect in one ray
- This is the analogue in $\mathbb{R}^2$ of:
 - two distinct points uniquely define a line
 - two lines always intersect in one point.



Appendix: $\mathbb{R}P^3$

- In $\mathbb{R}P^3$ points and planes are dual.

- A plane can be written as

$$\pi_1 x + \pi_2 y + \pi_3 z + \pi_4 = 0$$

- In homogeneous coord., we map $x \mapsto x_1/x_4$, $y \mapsto x_2/x_4$, $z \mapsto x_3/x_4$:

$$\pi_1 x_1 + \pi_2 x_2 + \pi_3 x_3 + \pi_4 x_4 = 0$$

- Or more concisely using $\pi = (\pi_1, \pi_2, \pi_3, \pi_4)^T$
 $\pi x = 0$

- As with lines, this allows us to check if a point x is on the plane.

Appendix: $\mathbb{R}P^3$

- In $\mathbb{R}P^3$ there are numerous geometric relations between points, lines and planes:
 - A plane is defined uniquely by the join of three points, or the join of a line and point, in general position.
 - Two distinct planes intersect in a unique line.
 - Three distinct planes intersect in a unique point.
 - We can define quadrics
- We will not cover those, their development is similar to what we did in $\mathbb{R}P^2$
- You are encouraged to work them out !
 - But.. Google out how to define lines, it is a bit different ;) (e.g.: Plücker line coordinates)
 - Use the reference book: Hartley, Multiple view geometry in computer vision

Appendix: From Rigid-Body to Projective

- We saw that a 2D rigid body has the form

$$g = \begin{pmatrix} R & T \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} r_{11} & r_{12} & t_1 \\ r_{21} & r_{22} & t_2 \\ 0 & 0 & 1 \end{pmatrix}$$

- This mapping preserves the length and areas of objects
- By adding a scaling, we have

$$S = \begin{pmatrix} sR & T \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} sr_{11} & sr_{12} & t_1 \\ sr_{21} & sr_{22} & t_2 \\ 0 & 0 & 1 \end{pmatrix}$$

- This is called a **similarity** transformation
- it preserves ratio of lengths
- But we can continue to change our matrix

Appendix: From Rigid-Body to Projective

- By having any real number in place of sR

$$a = \begin{pmatrix} A & T \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & t_1 \\ a_{21} & a_{22} & t_2 \\ 0 & 0 & 1 \end{pmatrix}$$

- We obtain **affine** transformations
- This mapping preserves parallelism, ratios of areas, ratio of lengths on colinear or parallel lines, linear combination of vectors, the line at infinity l_∞

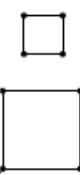
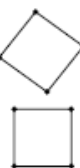
Appendix: From Rigid-Body to Projective

- Finally replacing all values we have

$$h = \begin{pmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{pmatrix}$$

- We obtain **projective** transformations
- This mapping preserves concurrency, colinearity

Appendix: From Rigid-Body to Projective

Group	Matrix	Distortion	Invariant properties
Projective 8 dof	$\begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix}$		Concurrency, collinearity, order of contact : intersection (1 pt contact); tangency (2 pt contact); inflections (3 pt contact with line); tangent discontinuities and cusps. cross ratio (ratio of ratio of lengths).
Affine 6 dof	$\begin{bmatrix} a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$		Parallelism, ratio of areas, ratio of lengths on collinear or parallel lines (e.g. midpoints), linear combinations of vectors (e.g. centroids). The line at infinity, l_∞ .
Similarity 4 dof	$\begin{bmatrix} sr_{11} & sr_{12} & t_x \\ sr_{21} & sr_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$		Ratio of lengths, angle. The circular points, I, J (see section 2.7.3).
Euclidean 3 dof	$\begin{bmatrix} r_{11} & r_{12} & t_x \\ r_{21} & r_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$		Length, area

Appendix: Converting Euler angles and Quaternions

- We can convert between Euler angles and quaternions easily:
- Euler angles to quaternions:
 - $a = \cos \frac{\theta}{2} \cos \frac{\varphi}{2} \cos \frac{\gamma}{2} + \sin \frac{\theta}{2} \sin \frac{\varphi}{2} \sin \frac{\gamma}{2}$
 - $b = \sin \frac{\theta}{2} \cos \frac{\varphi}{2} \cos \frac{\gamma}{2} - \cos \frac{\theta}{2} \sin \frac{\varphi}{2} \sin \frac{\gamma}{2}$
 - $c = \cos \frac{\theta}{2} \sin \frac{\varphi}{2} \cos \frac{\gamma}{2} + \sin \frac{\theta}{2} \cos \frac{\varphi}{2} \sin \frac{\gamma}{2}$
 - $d = \cos \frac{\theta}{2} \cos \frac{\varphi}{2} \sin \frac{\gamma}{2} - \sin \frac{\theta}{2} \sin \frac{\varphi}{2} \cos \frac{\gamma}{2}$
- Quaternions to Euler angles (assumes $R = R_z R_y R_x$ rotation)
 - $\theta = \arctan 2 \left(2(ab + cd), 1 - 2(b^2 + c^2) \right)$
 - $\varphi = \arcsin(2(ac - db))$
 - $\gamma = \arctan 2 \left(2(ad + bc), 1 - 2(c^2 + d^2) \right)$

Appendix: Converting Rotation matrices and Quat.

- We can convert between Rotation matrices and quaternions easily:

- Rotation matrices to quaternions:

- Trace: $T = r_{11} + r_{22} + r_{33}$

- If $T > 0$

$$a = \frac{\sqrt{1+T}}{2}$$

$$b = \frac{r_{32}-r_{23}}{4a}$$

$$c = \frac{r_{13}-r_{31}}{4a}$$

$$d = \frac{r_{21}-r_{12}}{4a}$$

- If $T \leq 0$

- If $r_{11} \geq r_{22}$ and $r_{11} \geq r_{33}$

- $b = \frac{\sqrt{1+r_{11}-r_{22}-r_{33}}}{2}$

- $a = \frac{r_{32}-r_{23}}{4b}$

- $d = \frac{r_{13}-r_{31}}{4b}$

- $c = \frac{r_{21}-r_{12}}{4b}$

- Similarly, compute for c or d if their diagonal elements are the largest.

- Quaternions to Euler angles (assumes $R = R_z R_y R_x$ rotation)

- $\theta = \arctan 2 \left(2(ab + cd), 1 - 2(b^2 + c^2) \right)$

- $\varphi = \arcsin(2(ac - db))$

- $\gamma = \arctan 2 \left(2(ad + bc), 1 - 2(c^2 + d^2) \right)$

Appendix: Converting Rotation matrices and Quat.

- We can convert between Rotation matrices and quaternions easily:
- Quaternions to Euler angles (assumes $R = R_z R_y R_x$ rotation)

$$R = \begin{pmatrix} 1 - 2(c^2 + d^2) & 2(bc - ad) & 2(bd + ac) \\ 2(bc + ad) & 1 - 2(b^2 + d^2) & 2(cd - ab) \\ 2(bd - ac) & 2(cd + ab) & 1 - 2(b^2 + c^2) \end{pmatrix}$$

- I copied and adapted those matrices a little fast from wikipedia, if someone want to cross-check them, feel free, otherwise, refer to wikipedia.